Working with HydroServer: Automated Data Ingestion, Visualization, and Quality Control

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Session 6a. Working with HydroServer
Tuesday, September 15, 2026
13:45 – 15:15

CENTER FOR GEOSPATIAL SOLUTIONS

WMO

NOAA

Utah Water Research Laboratory
UtahStateUniversity

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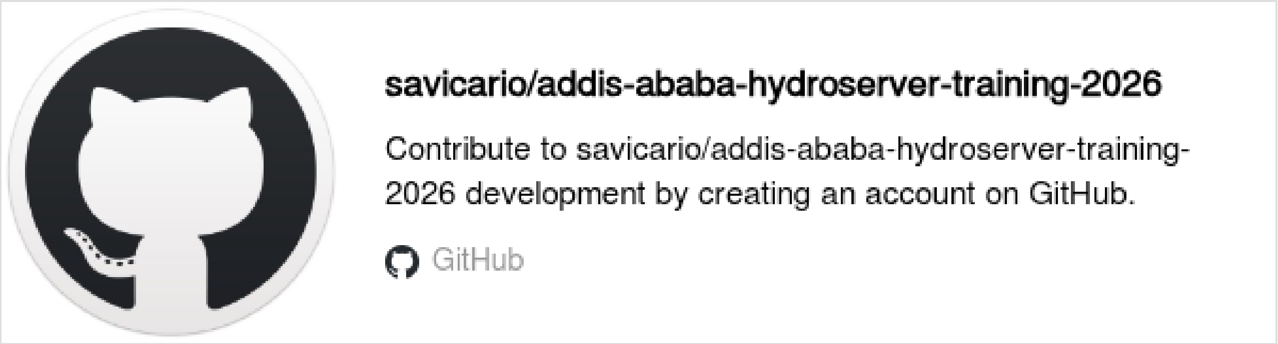
What We've Done So Far

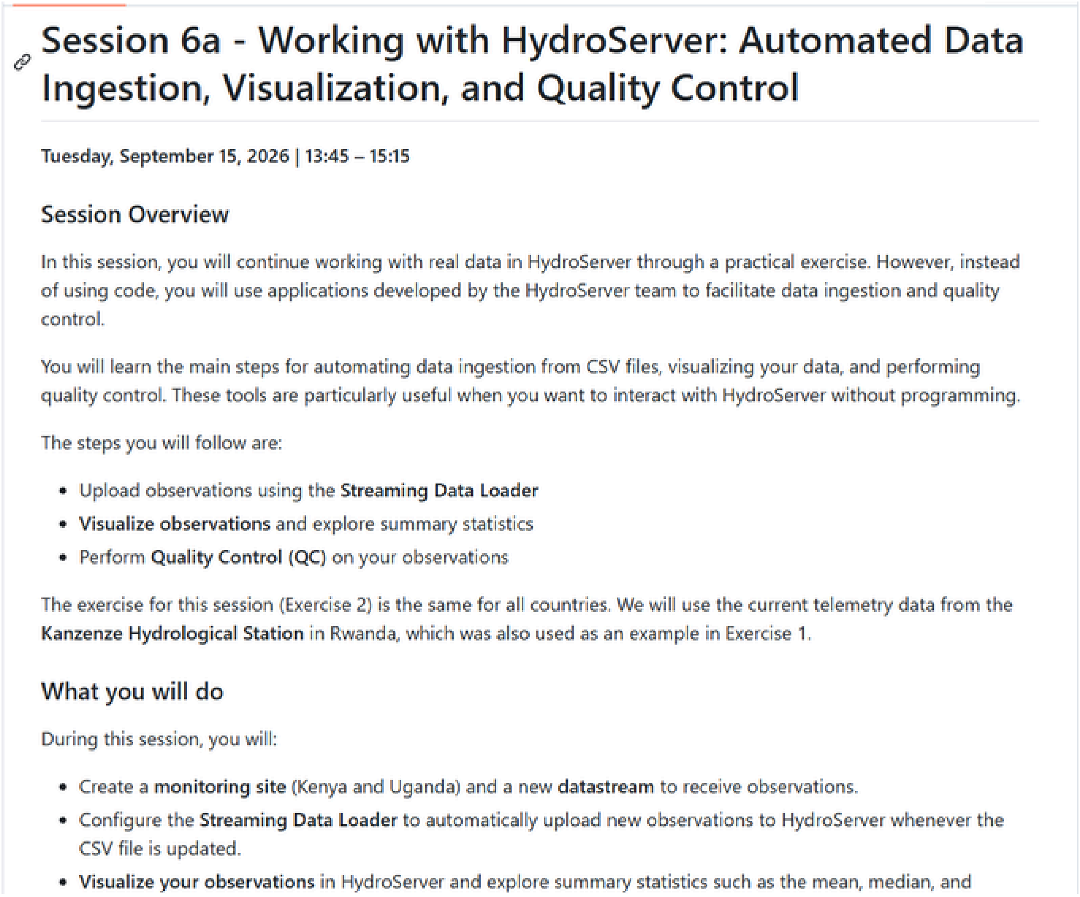
In the previous session, you have:

- Created a **Workspace**
- Created a **Monitoring Site**
- Created a **Datastream**
- **Loaded Historical Observations**

Now, you are ready to explore **automated data-loading methods**, **visualize your data in HydroServer**, and **perform data quality control**!

Please go to:





Session 6a - Working with HydroServer: Automated Data Ingestion, Visualization, and Quality Control

Tuesday, September 15, 2026 | 13:45 – 15:15

Session Overview

In this session, you will continue working with real data in HydroServer through a practical exercise. However, instead of using code, you will use applications developed by the HydroServer team to facilitate data ingestion and quality control.

You will learn the main steps for automating data ingestion from CSV files, visualizing your data, and performing quality control. These tools are particularly useful when you want to interact with HydroServer without programming.

The steps you will follow are:

- Upload observations using the **Streaming Data Loader**
- **Visualize observations** and explore summary statistics
- Perform **Quality Control (QC)** on your observations

The exercise for this session (Exercise 2) is the same for all countries. We will use the current telemetry data from the **Kanzenze Hydrological Station** in Rwanda, which was also used as an example in Exercise 1.

What you will do

During this session, you will:

- Create a **monitoring site** (Kenya and Uganda) and a new **datastream** to receive observations.
- Configure the **Streaming Data Loader** to automatically upload new observations to HydroServer whenever the CSV file is updated.
- **Visualize your observations** in HydroServer and explore summary statistics such as the mean, median, and

and take a couple of minutes to read through what we're going to do in this **Session 6a**

Let’s start with automated data ingestion!

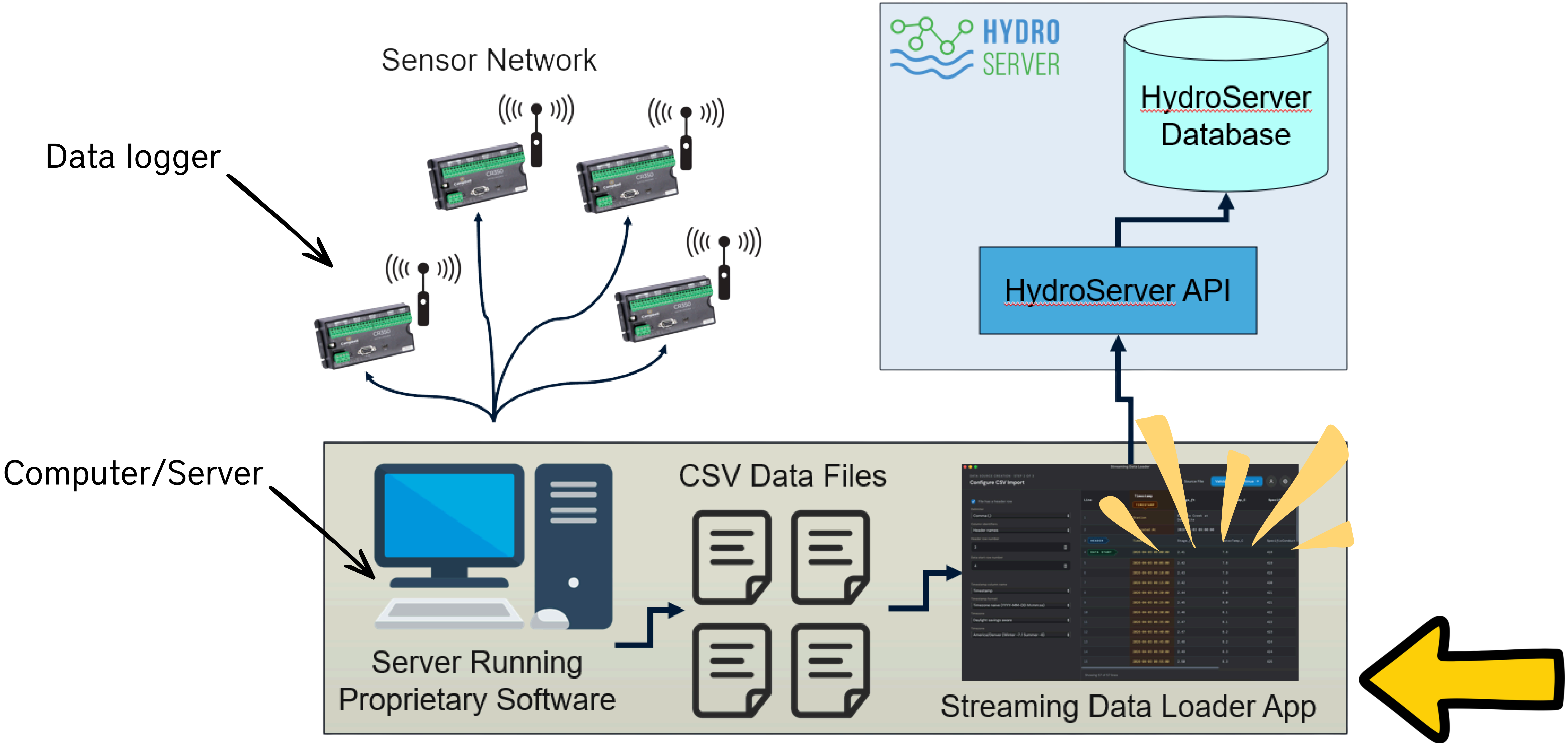
Automated Data Ingestion:

Streaming Data Loader App

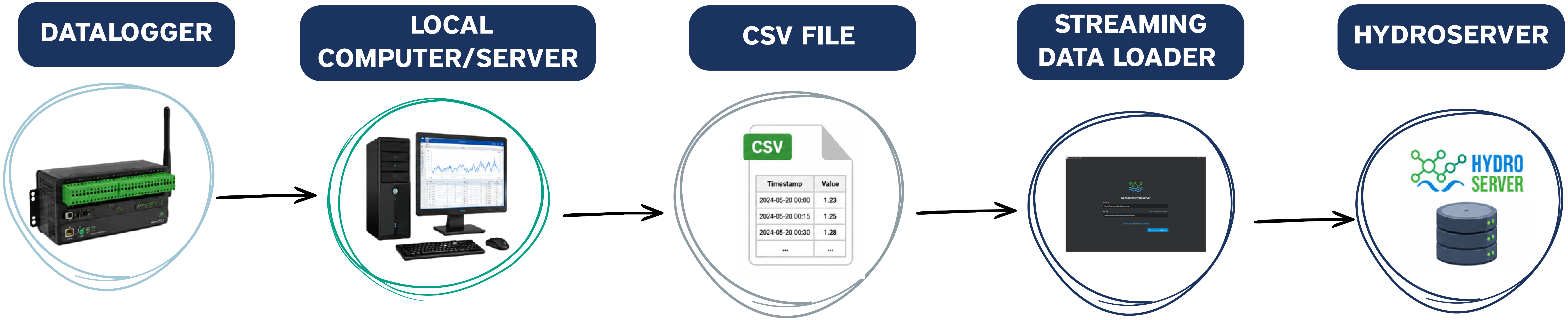
What is the Streaming Data Loader App?

The **Streaming Data Loader App** is a HydroServer application designed to automatically transfer time-series observations from local files into HydroServer.

Sensor data are exported from the source system as CSV files into your computer/server and then uploaded to HydroServer using the Streaming Data Loader App, which sends the observations to HydroServer through the API.



Streaming Data Loader Workflow



Streaming Data Loader App

- **Connects a local time-series file, such as a CSV, to a HydroServer datastream.**
- **Runs as a scheduled job** that periodically checks the source CSV file for new observations.
- **Tracks which observations have already been uploaded**, so it only loads new data rather than re-uploading the entire file.
- When new rows are added to the CSV, they are automatically read, processed, and sent to the corresponding HydroServer datastream.
- **Particularly useful for dataloggers and proprietary sensor software that continuously append new measurements to local CSV files.**

Now, we are going to try the Streaming Data Loader App by completing **Exercise 2**.

Exercise 2:

Loading Real-Time Sensor Data into HydroServer

For this exercise, we will use real-time stage data from the Kanzenze Hydrological Station in Rwanda.

Kanzenze Hydrological Station

REMEMBER

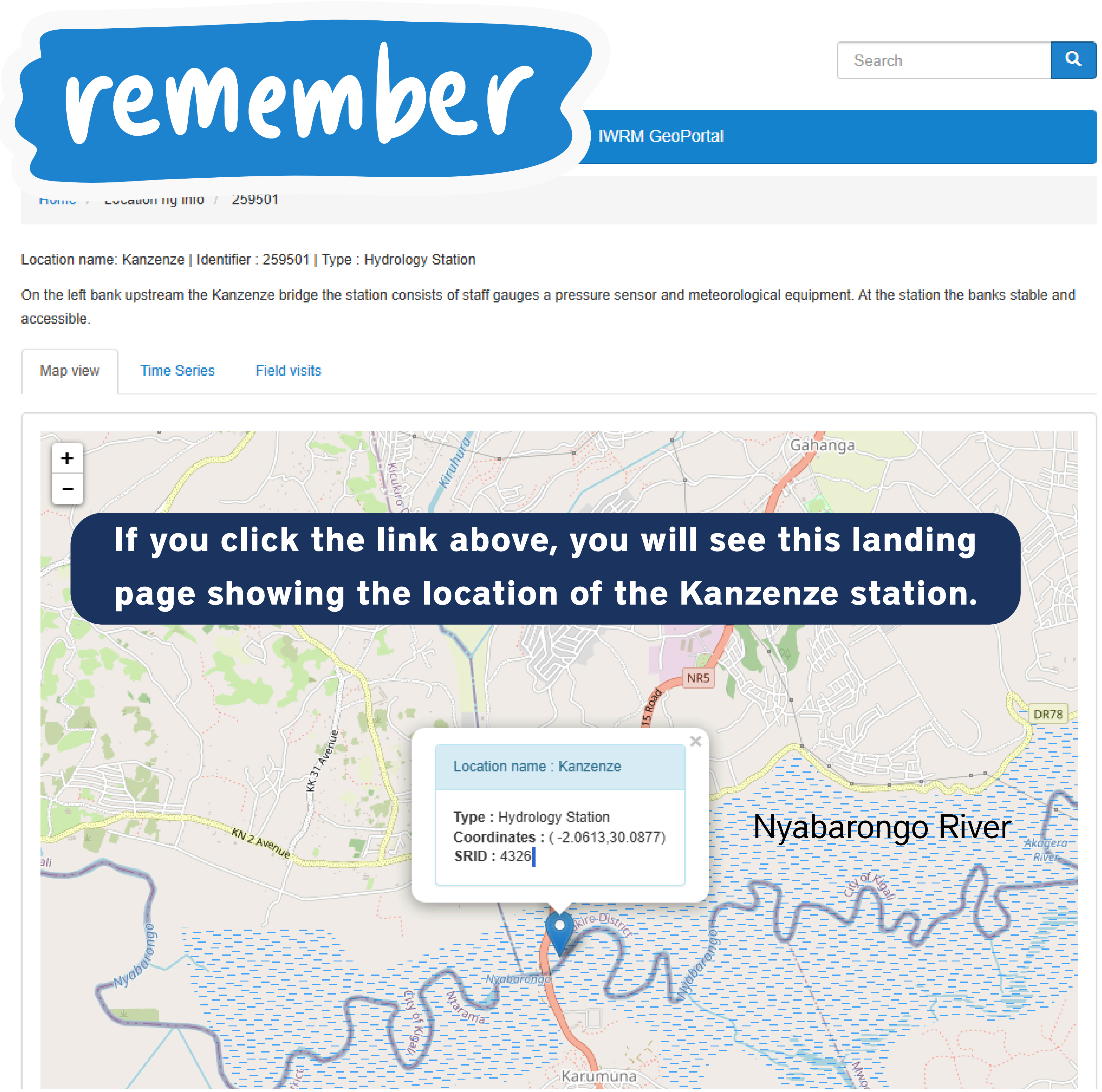
Rwanda receives equipment for six HydroMet stations

We selected **Kanzenze** because it is an **existing Rwandan hydrological monitoring site** that was **upgraded and equipped** with a modern telemetry system through the **Nile Basin Initiative (NBI) HydroMet Project**.

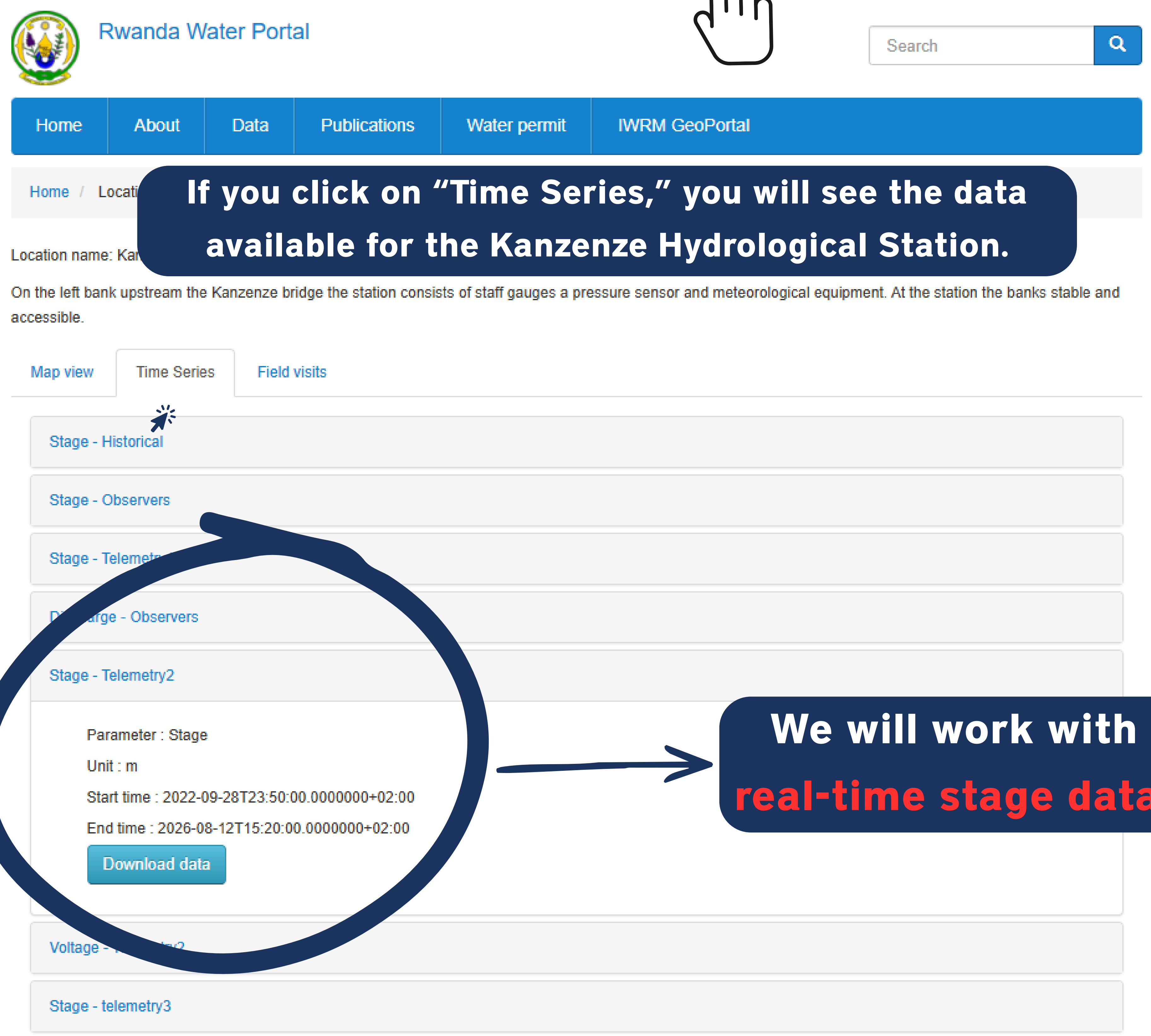
Kanzenze Hydrological Station

Link to the station data in the Rwanda Water Portal

remember



If you click the link above, you will see this landing page showing the location of the Kanzenze station.



If you click on "Time Series," you will see the data available for the Kanzenze Hydrological Station.

We will work with **real-time stage data!**

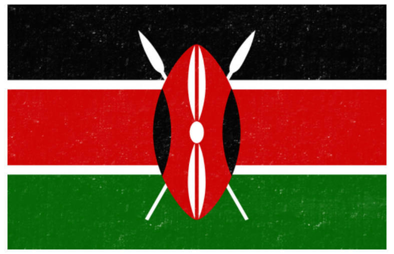
Kanzenze Hydrological Station Data

This table summarizes all the available data for the Kanzenze station. We are going to focus on the real-time stage data.

Time series	Variable	Unit	Period
Stage – Historical	Stage	m	1971-03-07 - 2015-06-08
Stage – Observers	Stage	m	2016-03-04 - 2017-02-28
Discharge – Observers	Discharge	m³/s	2016-03-04 - 2017-02-28
Stage – Telemetry1	Stage	m	2018-02-27 - 2019-09-22
Stage – Telemetry2	Stage	m	2022-09-28 - Present
Voltage – Telemetry2	Voltage	V	2022-09-28 - Present

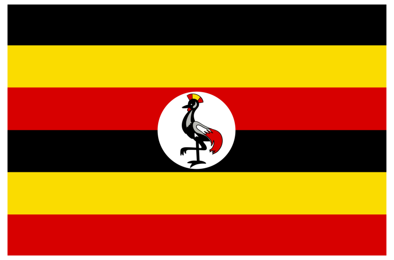
Access Exercise 2

The second exercise can be accessed through the following link for each country:



Kenya: Access Exercise 2

Rwanda: Access Exercise 2



Uganda: Access Exercise 2

Ethiopia: Access Exercise 2

You may be asked to sign in to your Google account.

Exercise 2 – Setting Up HydroServer for Automated Data Ingestion

This Python code (Exercise 2) performs the same initial setup steps as in Exercise 1:

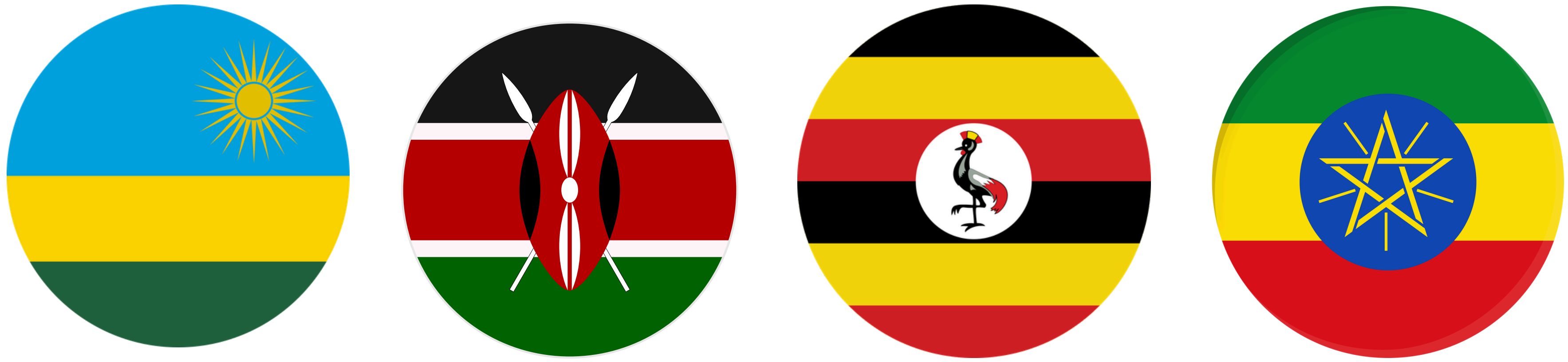
- Creates a Monitoring Site for the Kanzenze Hydrological Station (if needed).
- Creates a Datastream for the real-time stage data.

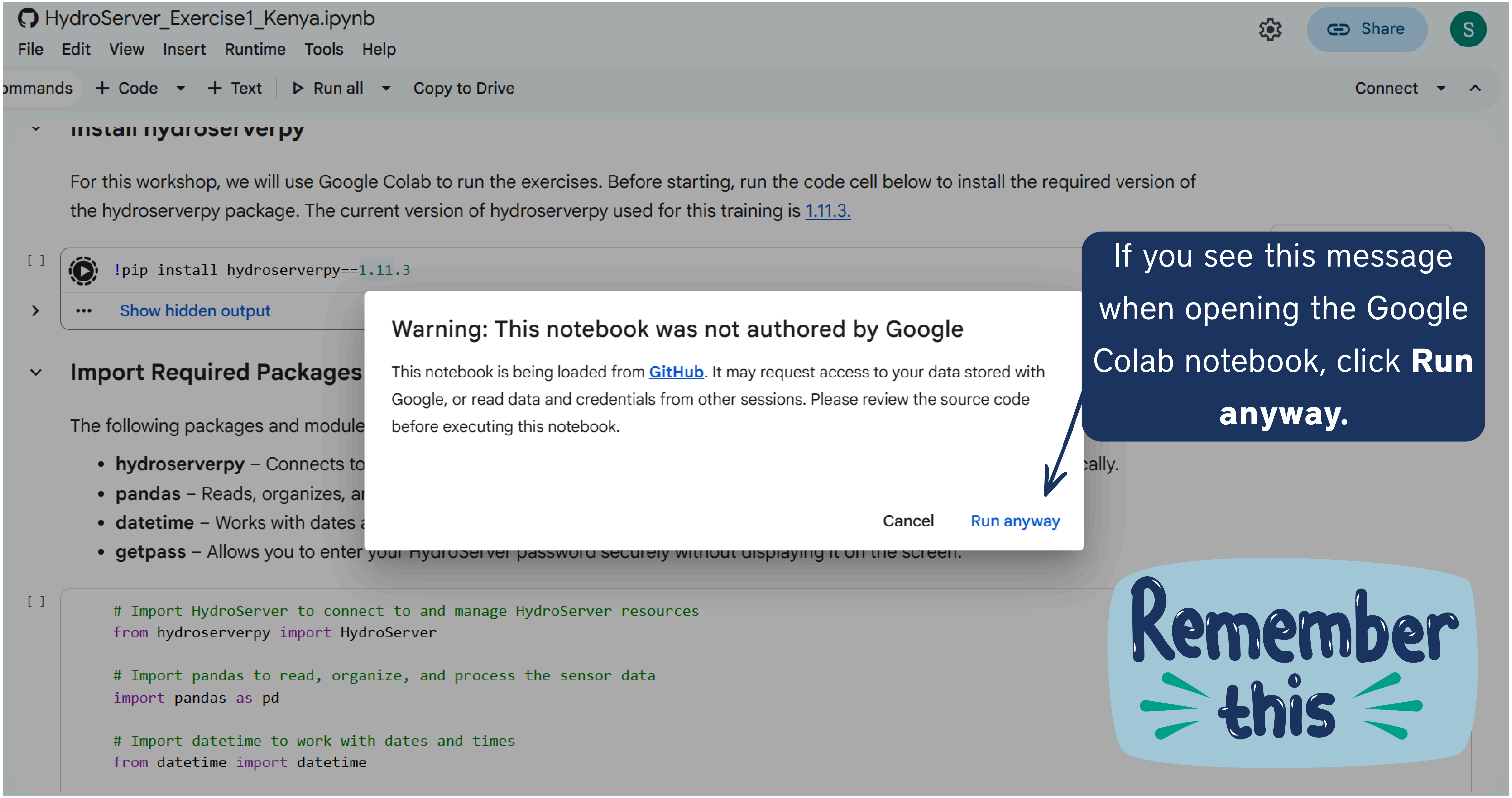
These steps can also be completed manually using the HydroServer Web Dashboard.

These steps must be completed before uploading data using the Streaming Data Loader App.

The setup required for Exercise 2 depends on the monitoring site created by each country in Exercise 1:

- **Rwanda and Ethiopia** already created the Kanzenze Hydrological Station monitoring site in Exercise 1, so they only need to create the new datastream for Exercise 2.
- **Kenya and Uganda** worked with their own monitoring stations in Exercise 1. Therefore, they first need to create the Kanzenze Hydrological Station monitoring site and then create the new datastream for Exercise 2.




HydroServer_Exercise1_Kenya.ipynb
File Edit View Insert Runtime Tools Help
Share
ommands
+ Code
+ Text
Run all
Copy to Drive
Connect
Install hydroserverpy
For this workshop, we will use Google Colab to run the exercises. Before starting, run the code cell below to install the required version of the hydroserverpy package. The current version of hydroserverpy used for this training is 1.11.3.
!pip install hydroserverpy==1.11.3
Show hidden output
Import Required Packages
The following packages and module
hydroserverpy – Connects to
pandas – Reads, organizes, ar
datetime – Works with dates a
getpass – Allows you to enter your HydroServer password securely without displaying it on the screen.
Warning: This notebook was not authored by Google
This notebook is being loaded from GitHub. It may request access to your data stored with Google, or read data and credentials from other sessions. Please review the source code before executing this notebook.
Cancel
Run anyway
# Import HydroServer to connect to and manage HydroServer resources
from hydroserverpy import HydroServer
# Import pandas to read, organize, and process the sensor data
import pandas as pd
# Import datetime to work with dates and times
from datetime import datetime
If you see this message when opening the Google Colab notebook, click **Run anyway.**
Remember this

HydroServer_Exercise2_Kenya.ipynb   Save in GitHub to keep changes

File   Edit   View   Insert   Runtime   Tools   Help

Commands   + Code   + Text   Run all   Copy to Drive   Reconnect

Share

IMPORTANT

If you see this message while running the code, click **Reconnect**. Do not rerun the cells you already completed, as this may create duplicate resources.

Exercise 2 - Kenya: Loading Real-time Sensor D

Overview

In this exercise, we will use the **Kanzenze Hydrological Station in Rwanda** as an example to demonstrate how real-time telemetry data can be loaded into HydroServer. Kanzenze is an exi... modern telemetry system through the **Nile Basin Initiative**

The station provides several hydrological time serie... observations. For this exercise, we will use the **Stag... ptember 2022 to the present.** The data can be accessed di...

Following this example, you will create a monitoring... ll use this datastream to load observations using the HydroSe...

Runtime disconnected

Your runtime has been disconnected due to inactivity or reaching its maximum duration. Learn more

If you are interested in longer runtimes with more lenient timeouts, you may want to check out Colab Pro.

Close   Reconnect

1. Getting Started

Install hydroserverpy

For this workshop, we will use Google Colab to run the exercises. Before starting, run the code cell below to install the required version of the hydroserverpy package. For this exercise, we will connect to the HydroServer Playground instance at playg... current version of hydroserverpy used for this training is 1.11.3.

```
!pip install hydroserverpy==1.11.3
```

DON'T FORGET!

Variables   Terminal   3:35 AM

Getting Started with HydroServer Programmatically

Python code

Install **hydroserverpy** package

```
!pip install hydroserverpy==1.11.3
```

REMINDER

Import the **Python packages** needed for the exercise

```
# Import HydroServer to connect to and manage HydroServer resources
from hydroserverpy import HydroServer

# Import pandas to read, organize, and process the sensor data
import pandas as pd

# Import datetime to work with dates and times
from datetime import datetime

# Import getpass to securely enter your HydroServer password
from getpass import getpass
```

These steps are the same as in the previous exercises.

Getting Started with HydroServer Programmatically

```
Collecting dnspython>=2.0.0 (from email-validator>=2.0.0->pydantic[email]>=2.6->hydroserverpy==1.11.3)
  Downloading dnspython-2.8.0-py3-none-any.whl.metadata (5.7 kB)
Downloading hydroserverpy-1.11.3-py3-none-any.whl (113 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 113.4/113.4 kB 5.6 MB/s eta 0:00:00
Downloading croniter-6.2.4-py3-none-any.whl (46 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 46.7/46.7 kB 3.3 MB/s eta 0:00:00
Downloading jmespath-1.1.0-py3-none-any.whl (20 kB)
Downloading python_crontab-3.4.0-py3-none-any.whl (27 kB)
Downloading email_validator-2.3.0-py3-none-any.whl (35 kB)
Downloading dnspython-2.8.0-py3-none-any.whl (331 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 331.1/331.1 kB 18.2 MB/s eta 0:00:00
Installing collected packages: python-crontab, jmespath, dnspython, email-validator, croniter, hydroserverpy
Successfully installed croniter-6.2.4 dnspython-2.8.0 email-validator-2.3 ... 0
```

When the packages have been successfully installed and imported, you will see these messages!

[4] ✓ 0s

```python
# Import HydroServer to connect to and manage HydroServer resources
from hydroserverpy import HydroServer

# Import pandas to read, organize, and process the sensor data
import pandas as pd

# Import datetime to work with dates and times
from datetime import datetime

# Import getpass to securely enter your HydroServer password
from getpass import getpass
```

Programmatically – Set the Initial Parameters for Login

Set the initial parameters to connect:

- Host
- E-mail
- Password

HydroServer Playground URL

```python
# Set initial parameters to connect to HydroServer
hydroserver_host = 'https://playground.hydroserver.org'

# Change the email and password below to your HydroServer username and password
hydroserver_email = 'svicario@lincolninst.edu'
hydroserver_password = getpass('Enter your HydroServer password: ')
```

IMPORTANT!

When you run the code, you will be prompted to enter your password.

```python
# Set initial parameters to connect to HydroServer
hydroserver_host = 'https://playground.hydroserver.org'

# Change the email and password below to your HydroServer username and password
hydroserver_email = 'svicario@lincolninst.edu' #'user@youremail.com'
hydroserver_password = getpass('Enter your HydroServer password: ') #getpass('Enter your HydroServer password: ')
```

```
Enter your HydroServer password: ···········
```

REMINDER

These steps are the same as in the previous exercises.

Programmatically – Log in into HydroServer

The next step is to **establish a connection with HydroServer** using the parameters you just set up.

```python
# Initialize HydroServer connection with credentials.
hs = HydroServer(
    host='https://playground.hydroserver.org',
    email='your@email.com',
    password='yourpassword'
)
print('\nSuccessfully connected to HydroServer!')
```

REMINDER

```python
# Initialize HydroServer connection with credentials.
hs = HydroServer(
    host=hydroserver_host,
    email=hydroserver_email,
    password=hydroserver_password
)

print('\nSuccessfully connected to HydroServer!')
```

```
...
Successfully connected to HydroServer!
```

If the connection is successful, you will see this message in your console.

These steps are the same as in the previous exercises.

Now, you need to retrieve the Universally Unique Identifiers (UUIDs) of some of the resources you created in Exercise 1.

You will use these IDs to specify where the new datastream will be created.

Get your workspace UUID

Next, retrieve your **Workspace ID**.

You will use it to specify the workspace where the datastream and its associated metadata will be created.

```
# Enter the name of the workspace you created in Exercise 1
workspace_name = "Rwanda Training 2026 - your name"

workspaces = hs.workspaces.list(
    is_associated=True
).fetch_all()

workspace_id = next(
    workspace.uid
    for workspace in workspaces.items
    if workspace.name == workspace_name
)

print("Selected workspace:", workspace_name)
print("Workspace ID:", workspace_id)
```

```
... Selected workspace: Rwanda Training 2026 - Sara
    Workspace ID: 01a07c59-74f1-7d05-b3ed-35526bd6e89e
```

Get your monitoring site UUID

For **Rwanda and Ethiopia**, you also need to retrieve the **Monitoring Site ID** because you created the Kanzenze Hydrological Station monitoring site in Exercise 1.

You will use this ID to specify the monitoring site where the new datastream will be created.

```
# Select the Kanzenze monitoring site
monitoring_site_name = "Kanzenze Hydrological Station"

monitoring_sites = hs.things.list(
  workspace=workspace_id
).fetch_all()

monitoring_site_id = next(
  s.uid for s in monitoring_sites.items
  if s.name == monitoring_site_name
)

print("Monitoring Site ID:", monitoring_site_id)
```

```
Monitoring Site ID: 01a07c59-7b78-7b49-8bb7-ef222c45d5f7
```

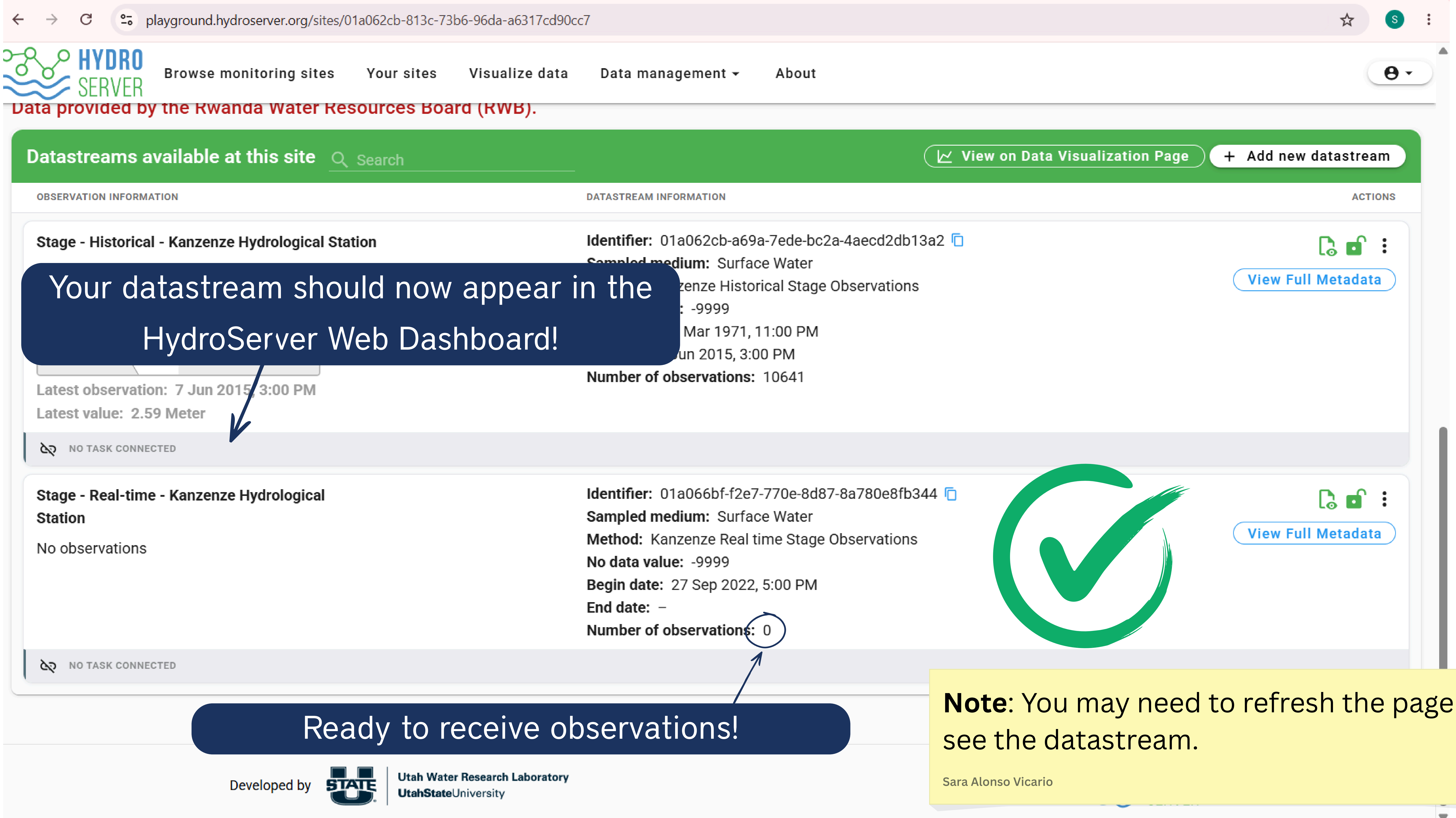
The next steps are the same as in Exercise 1:

- Create a **monitoring site** (Kenya and Uganda).
- Create a **datastream** (Kenya, Uganda, Rwanda, and Ethiopia).

We will run the Python code together during the training, so there is no need to describe each step in detail here.

For detailed instructions on a similar process, please refer to the Session 5a presentation.

After running the code, go to the HydroServer Web Dashboard, where you will see the datastream you just created.

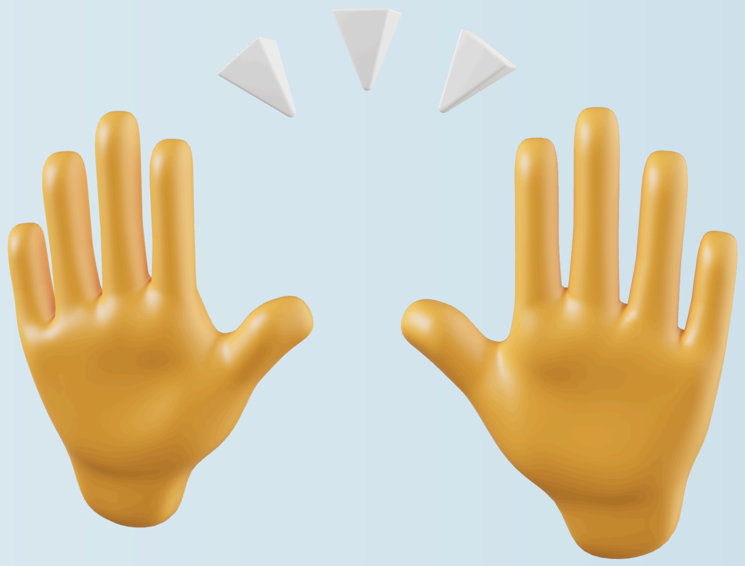


Your datastream should now appear in the HydroServer Web Dashboard!

Ready to receive observations!

Note: You may need to refresh the page to see the datastream.

Sara Alonso Vicario



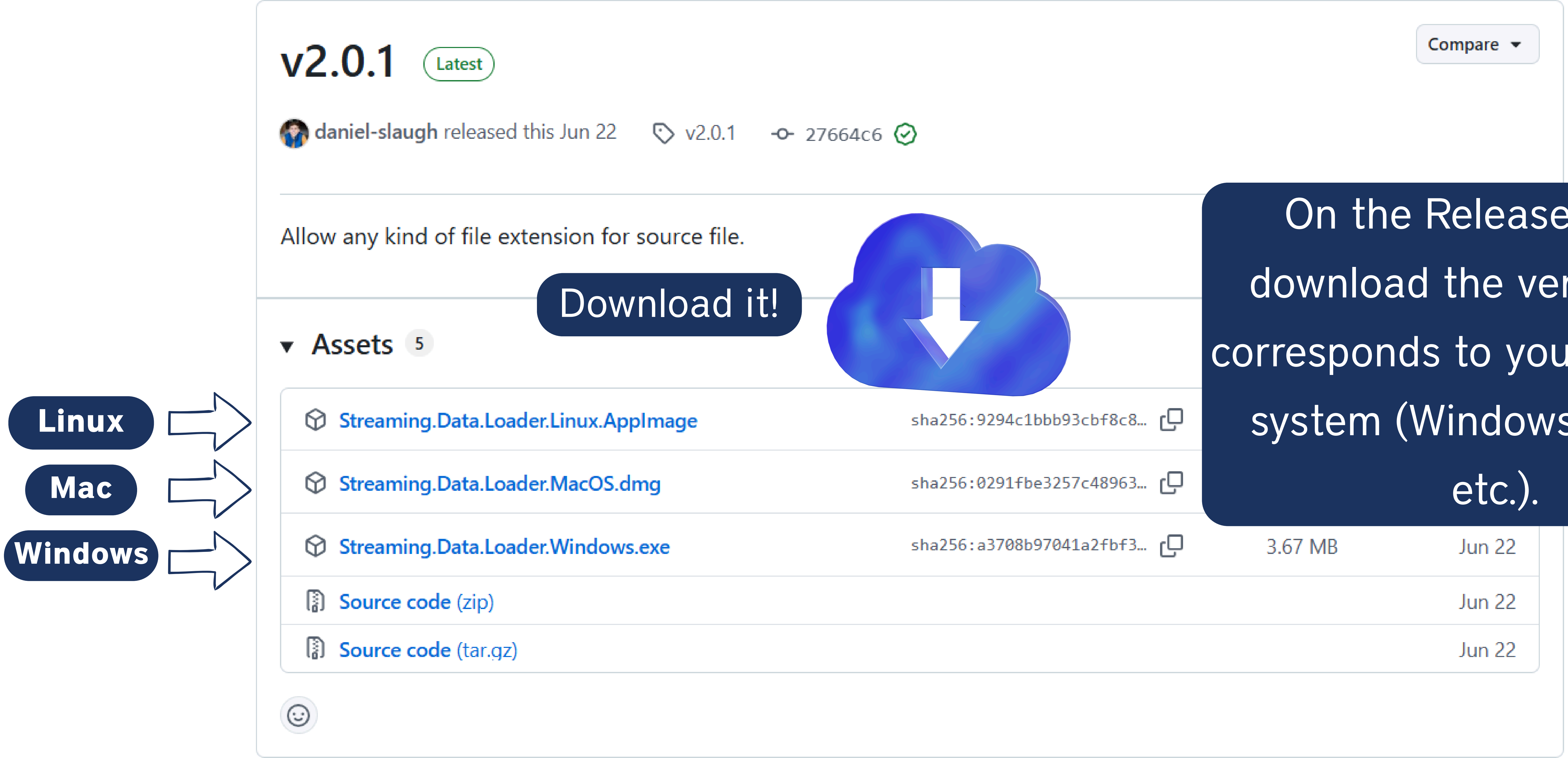
Datastream Created!

Now, we're ready to use the Streaming Data Loader!

Install the Streaming Data Loader

Download the Streaming Data Loader

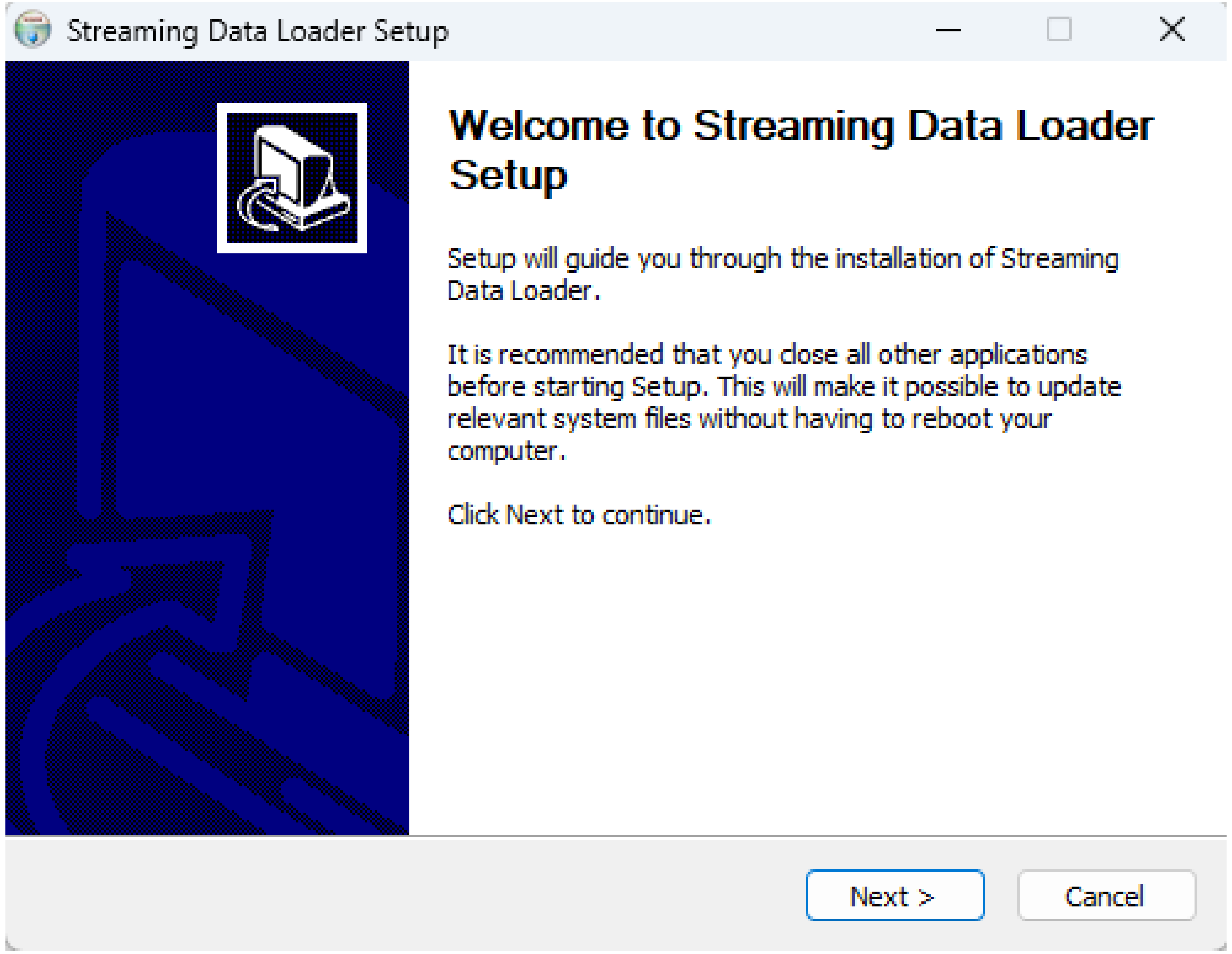
Go to Streaming Data Loader Releases

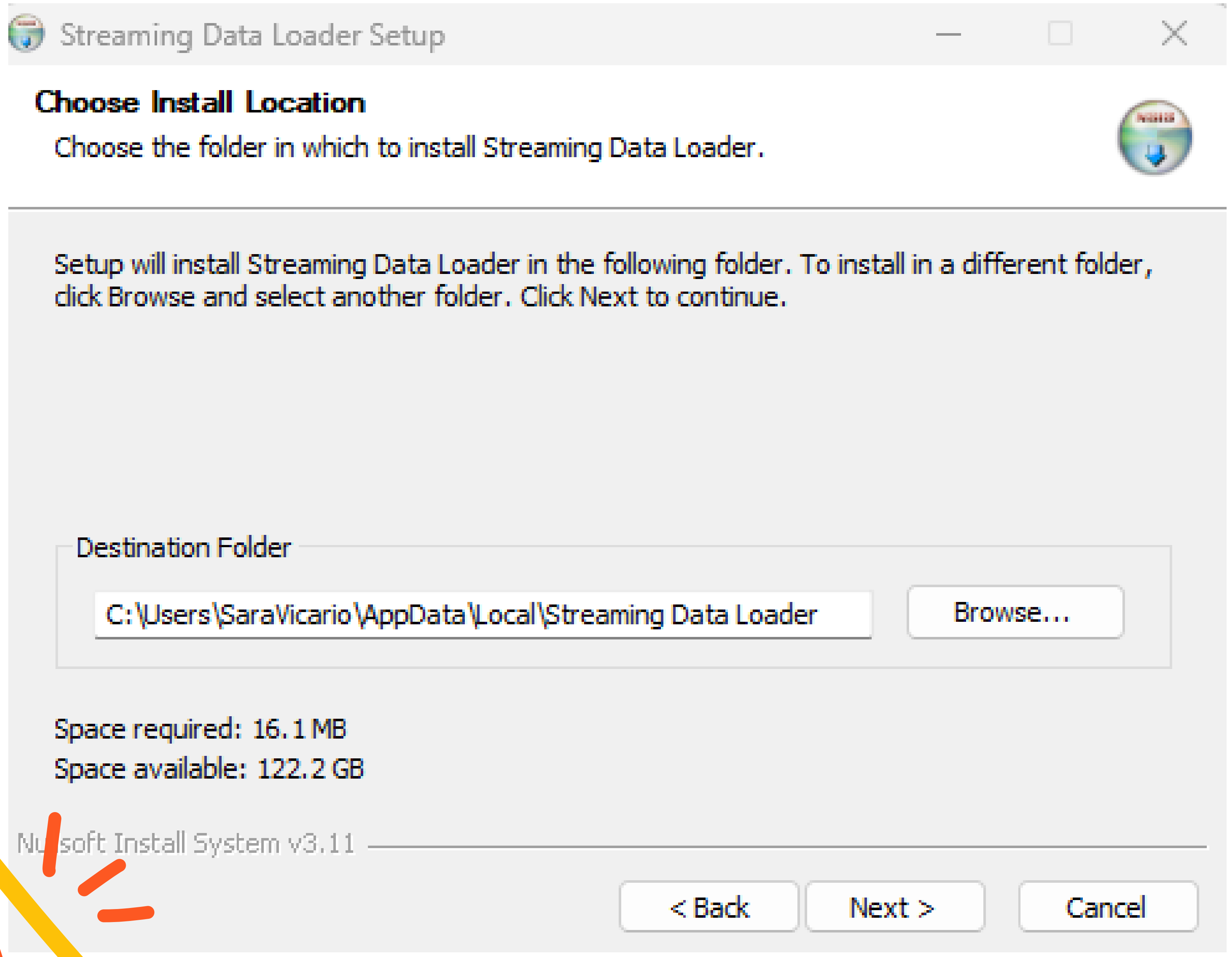


On the Releases page, download the version that corresponds to your operating system (Windows, macOS, etc.).

INSTALLING THE STREAMING DATA LOADER

Once the Streaming Data Loader has been downloaded, open the downloaded file. The installation window shown below will appear.



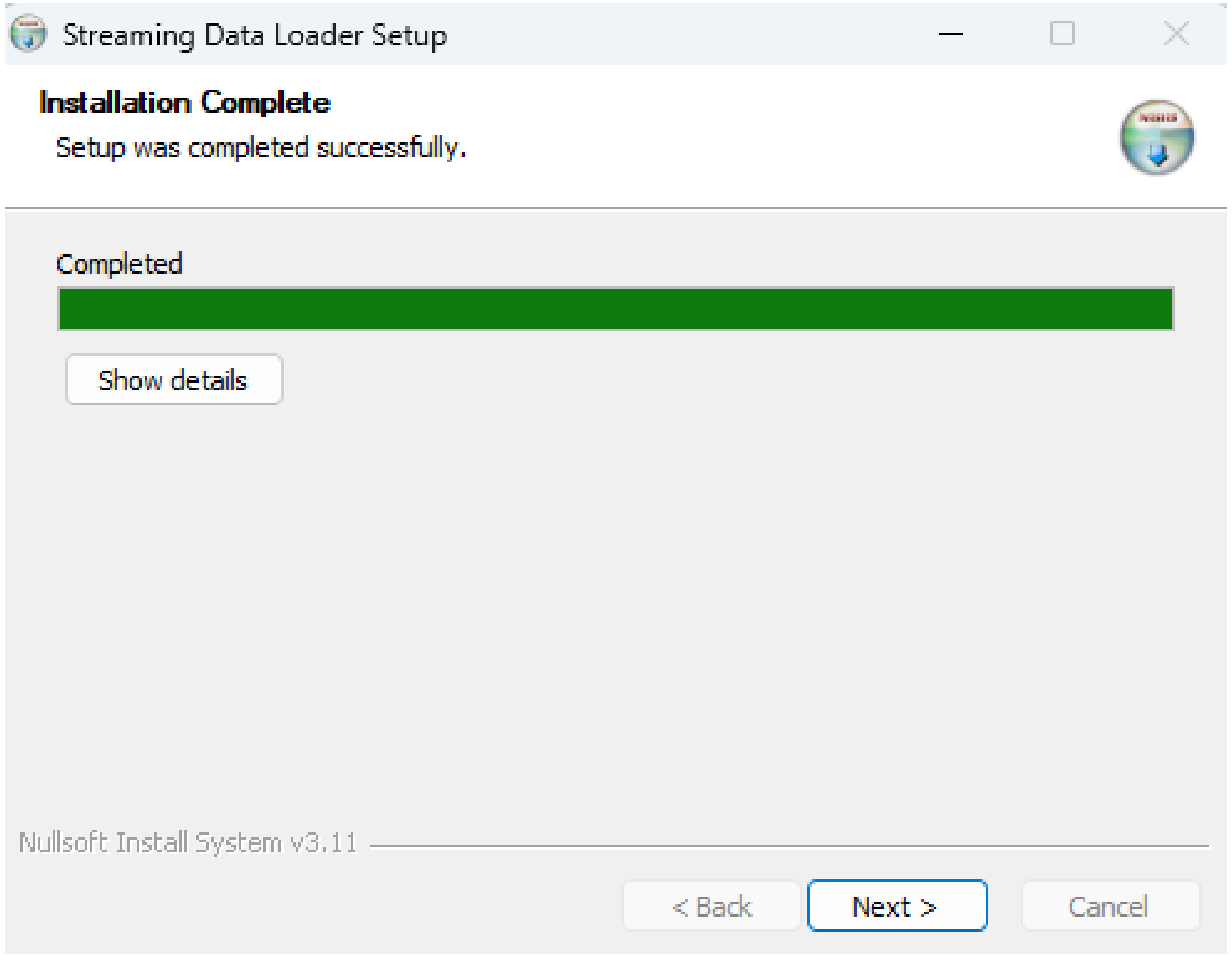


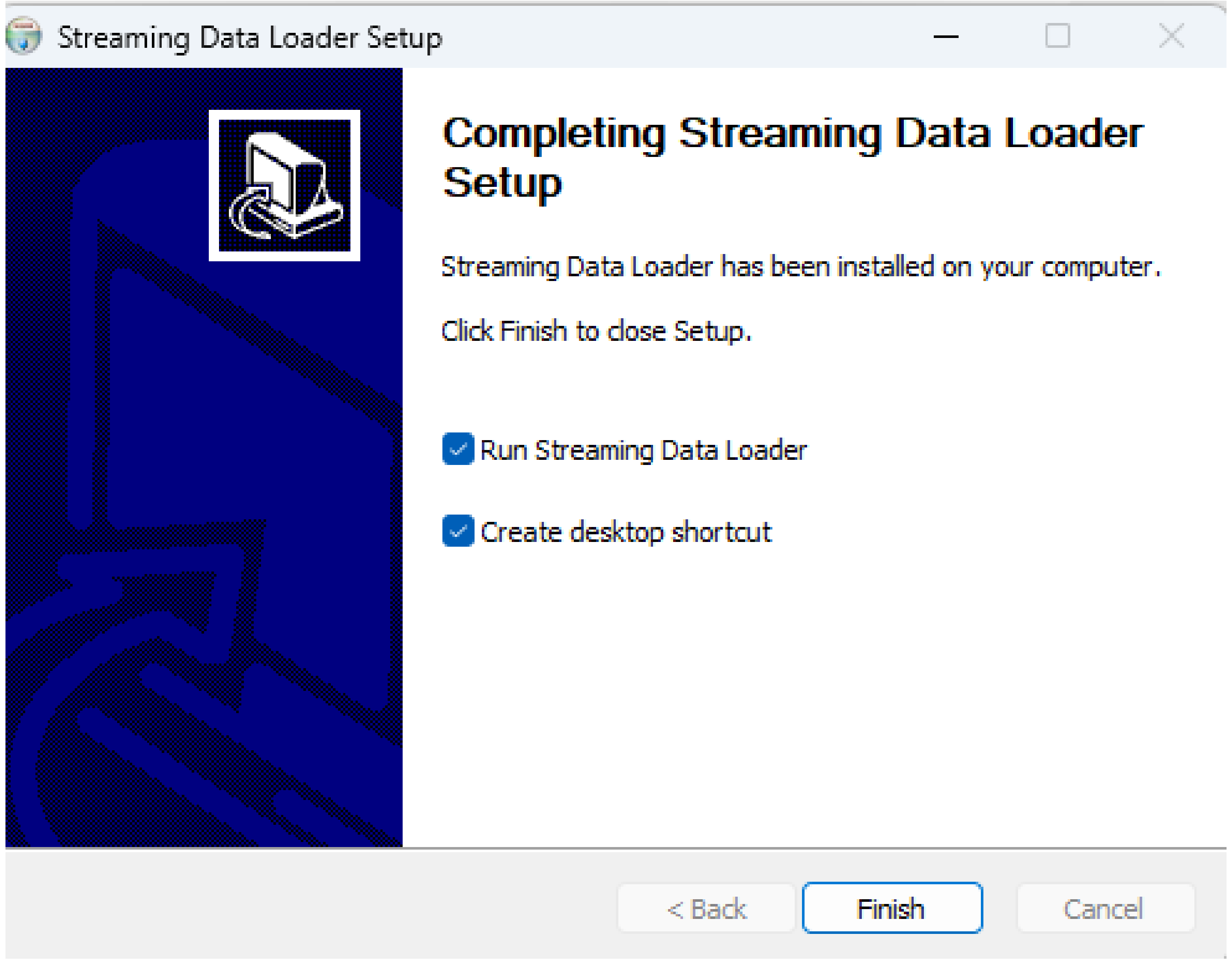
- **Windows: Save it to the C: drive.**
- **Mac: Save it to /Users/Shared/.**

INSTALLING THE STREAMING DATA LOADER

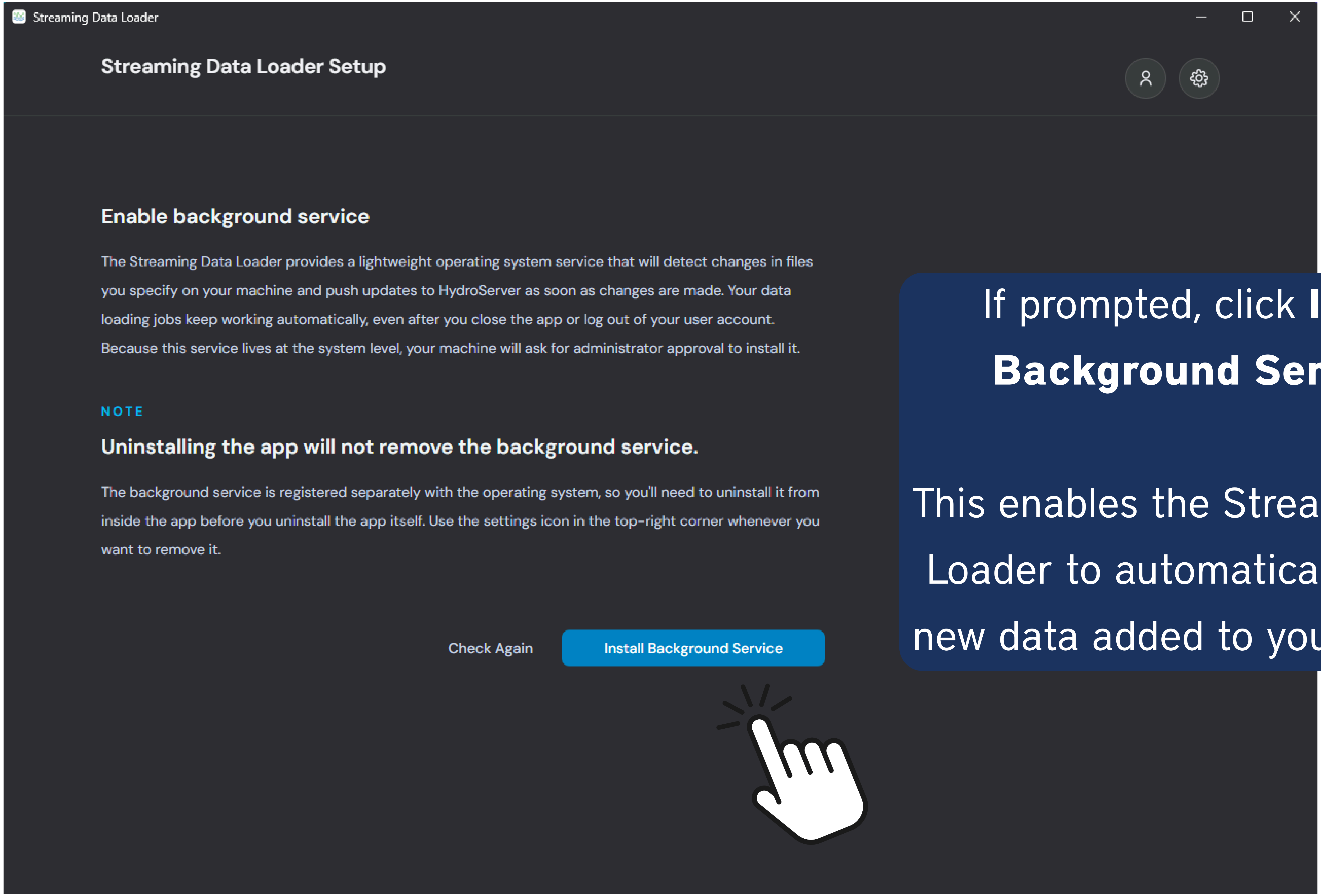
Once the installation is complete, click Next.

Make sure Run Streaming Data Loader is selected, then click Finish to launch the application.





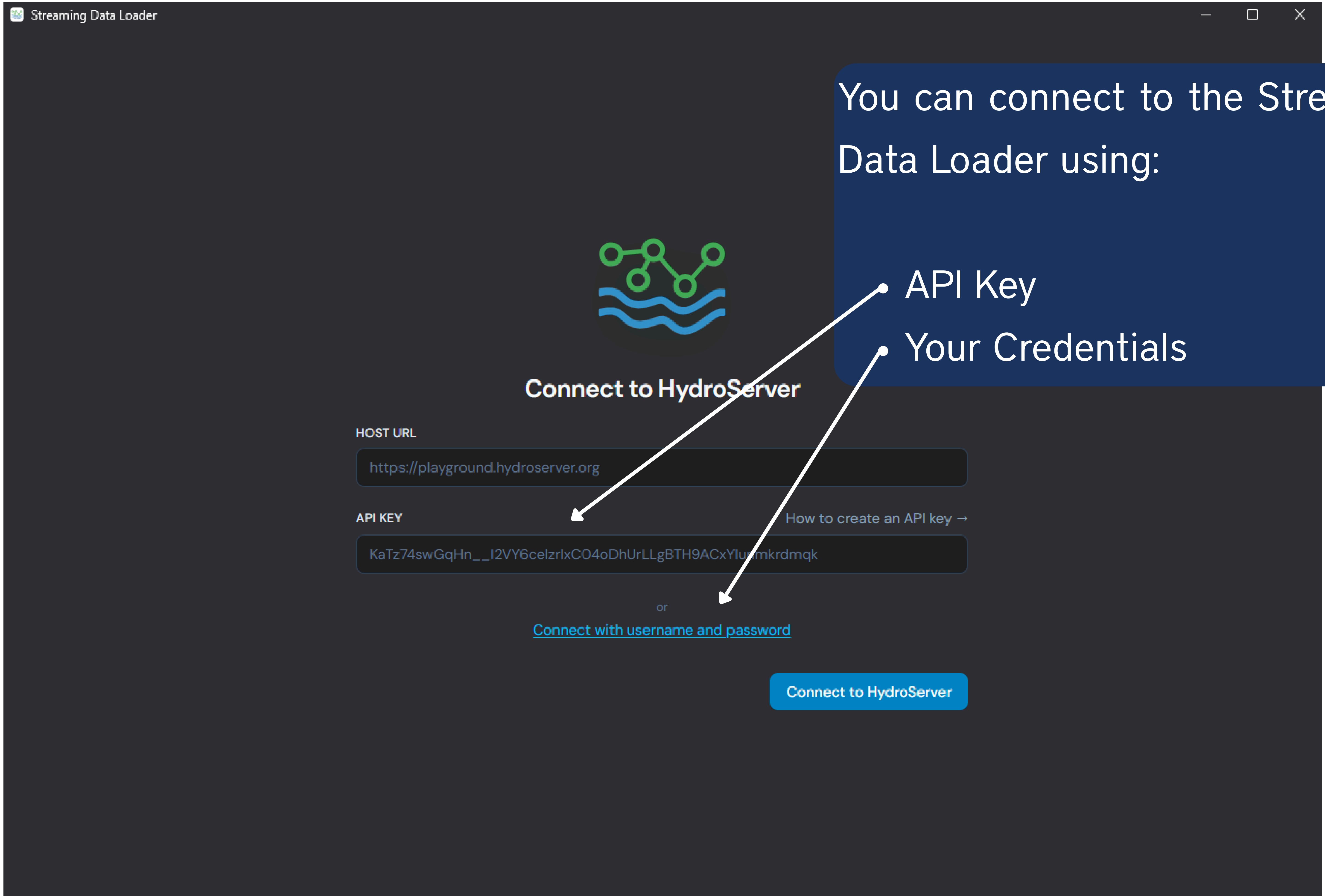
INSTALLING THE STREAMING DATA LOADER



If prompted, click **Install Background Service**.

This enables the Streaming Data Loader to automatically upload new data added to your CSV file.

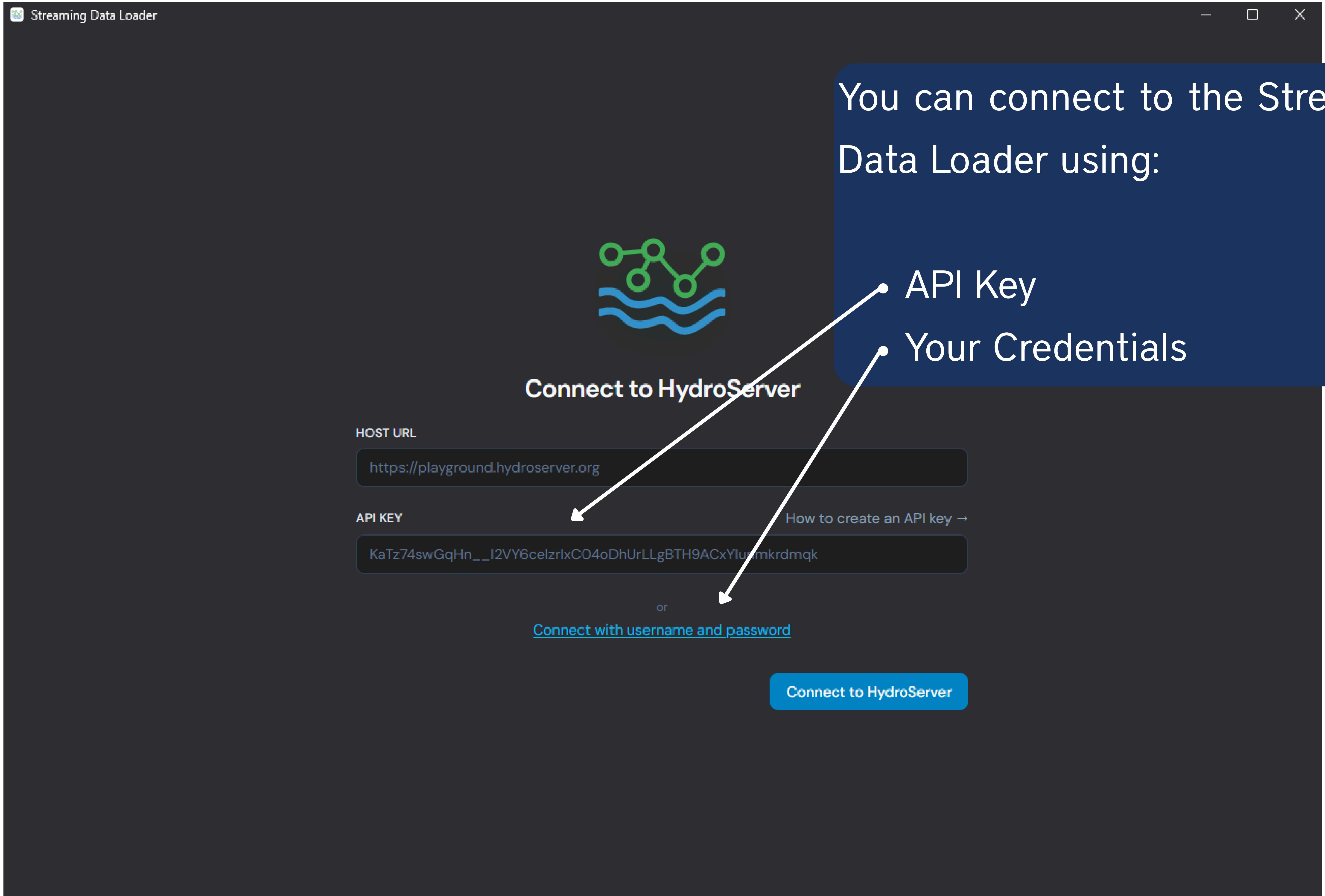
STREAMING DATA LOADER



You can connect to the Streaming Data Loader using:

- API Key
- Your Credentials

STREAMING DATA LOADER



You can connect to the Streaming Data Loader using:

- API Key
- Your Credentials

Connecting the Streaming Data Loader to HydroServer

You can connect using either your HydroServer credentials or an API key.

For **automated and long-running workflows,** an **API key is recommended** because it avoids storing or repeatedly using your personal password.

Why Use an API Key with the Streaming Data Loader?

An API key allows the Streaming Data Loader to securely connect to HydroServer

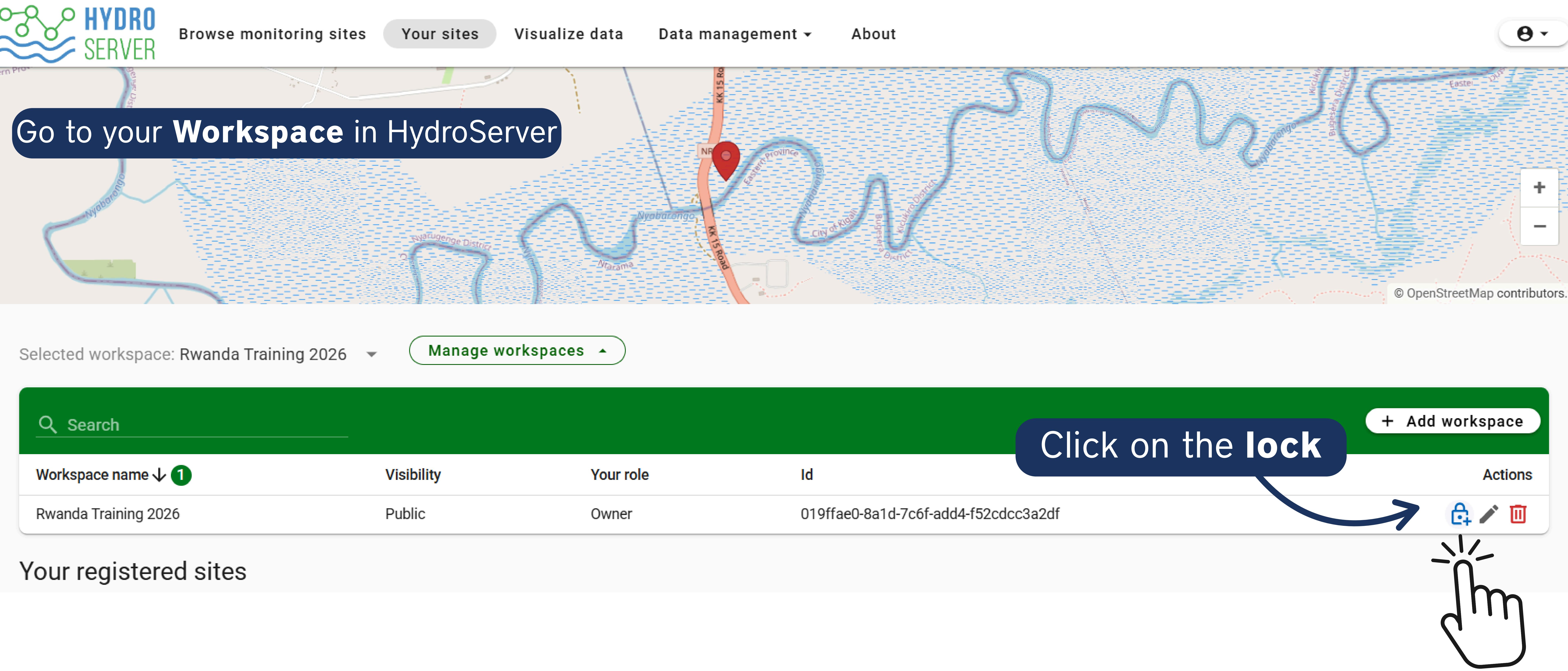
- Let applications authenticate automatically.
- Avoid storing your username and password in scripts.
- Can usually be revoked or regenerated if compromised.
- Allow automated workflows to run unattended (e.g., scheduled data uploads).

GET AN API KEY

Web Dashboard

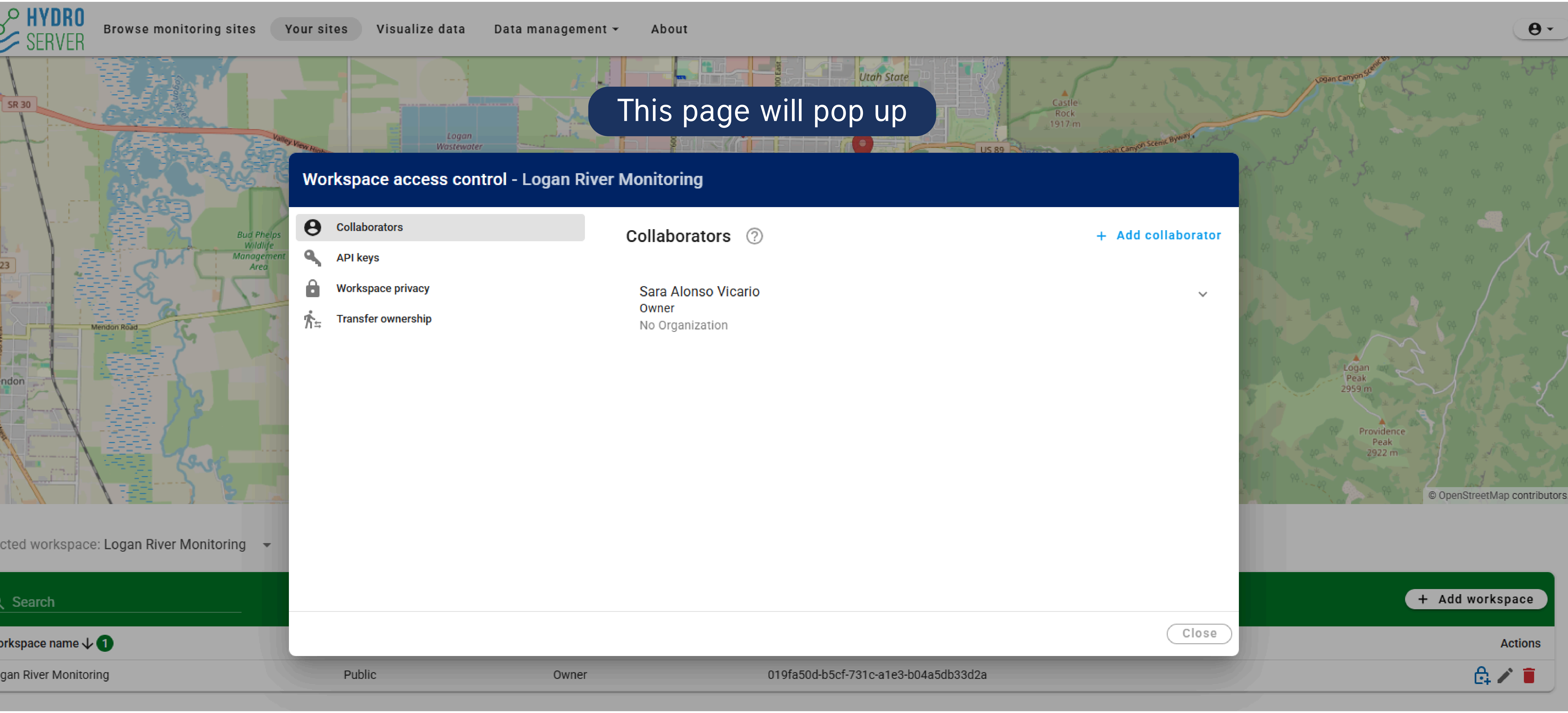
Go to your **Workspace** in HydroServer

Click on the **lock**



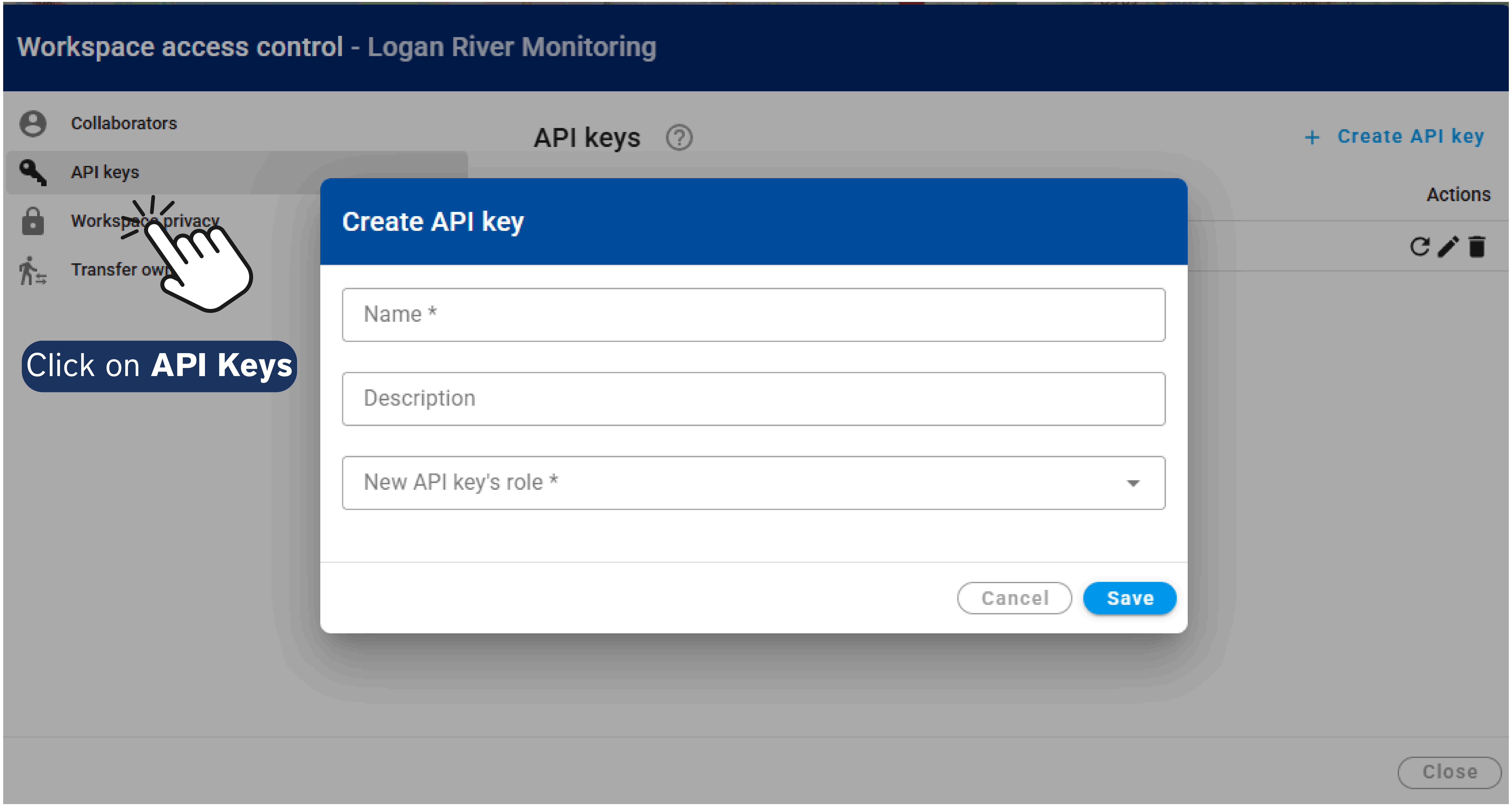
GET AN API KEY

Web Dashboard



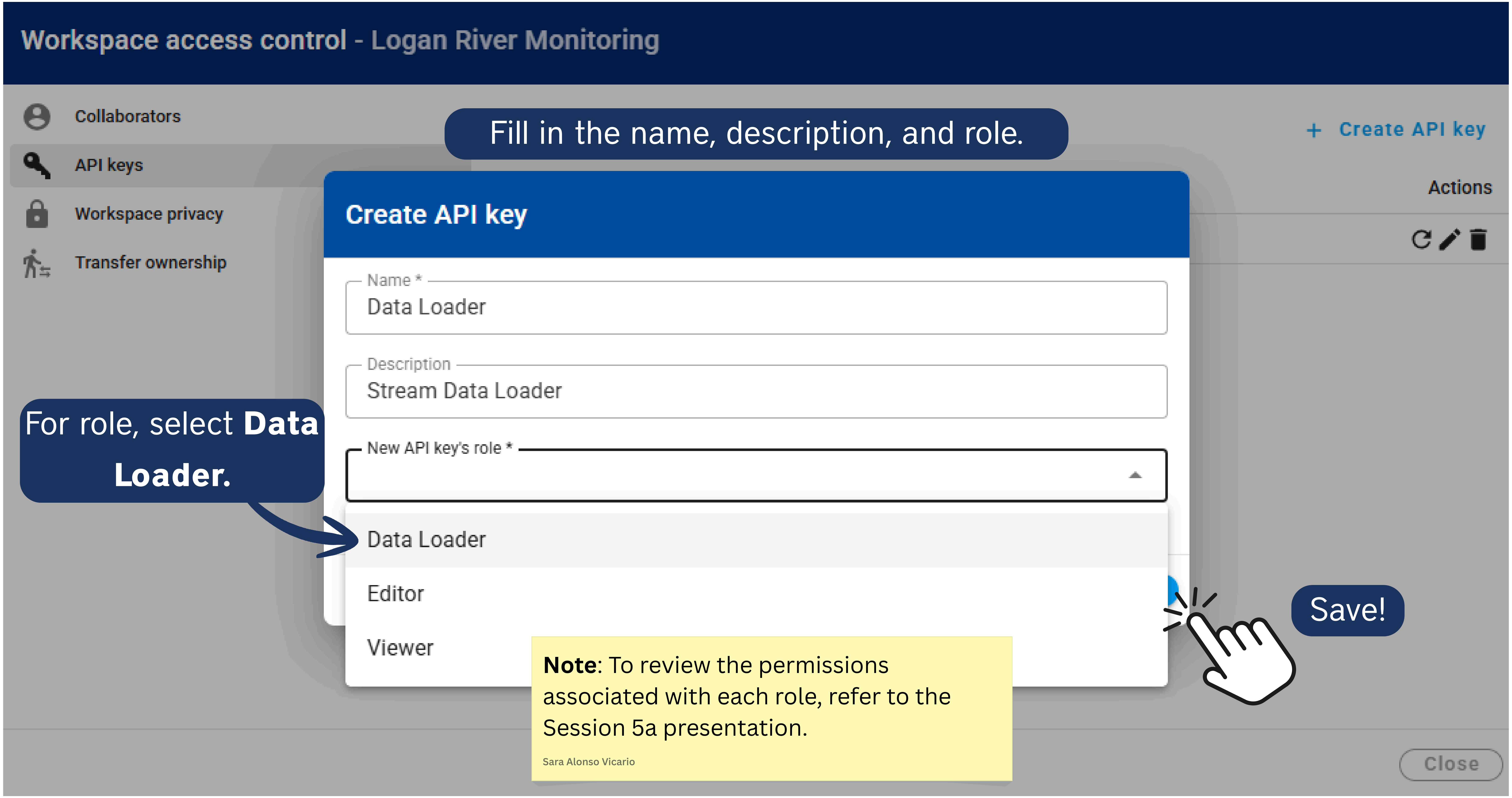
GET AN API KEY

Web Dashboard

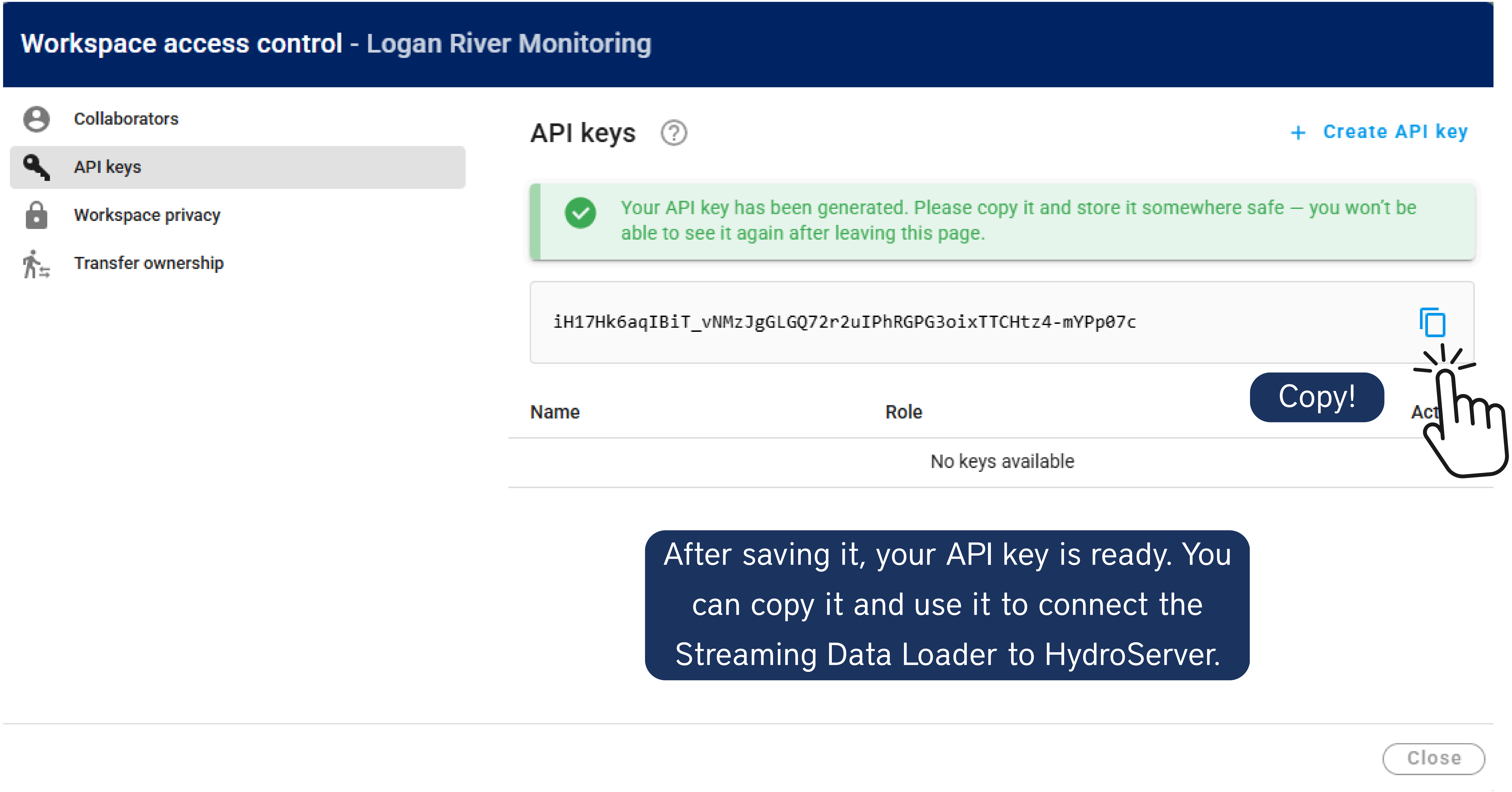


GET AN API KEY

Web Dashboard



Your API key is now ready to use!



After saving it, your API key is ready. You can copy it and use it to connect the Streaming Data Loader to HydroServer.

You can also regenerate the API Key

Workspace access control - Logan River Monitoring

- Collaborators
- API keys
- Workspace privacy
- Transfer ownership

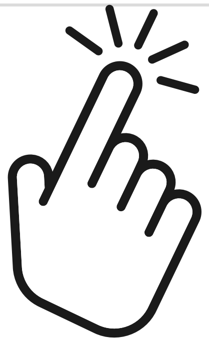
API keys

+ Create API key

Your API key has been generated. Please copy it and store it somewhere safe — you won't be able to see it again after leaving this page.

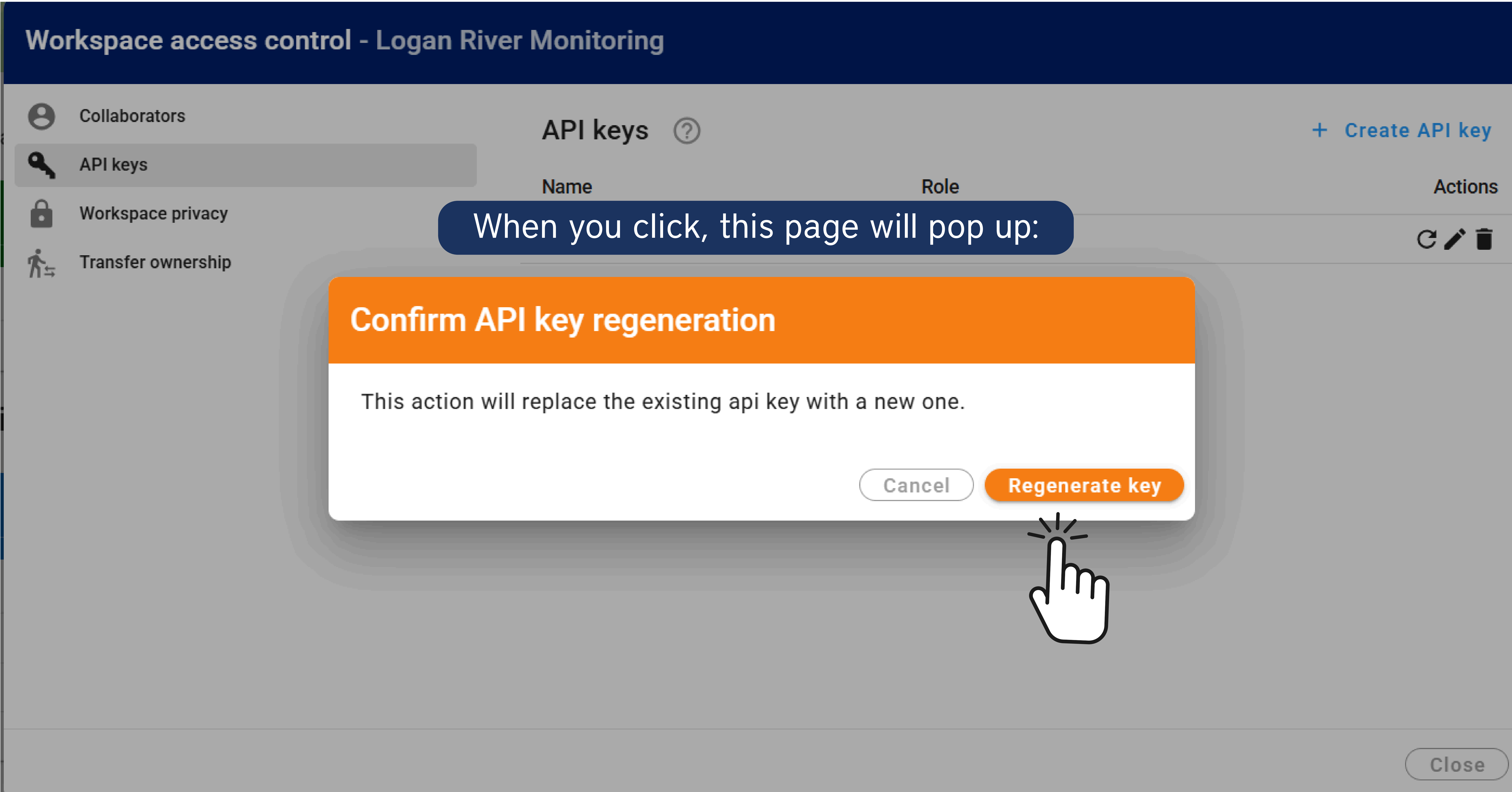
```
MyOiXranci3qzI8lfiCwFY8GtIU3YHy-LMi3JdAMY_pkQf_LZX-UTkQ
```

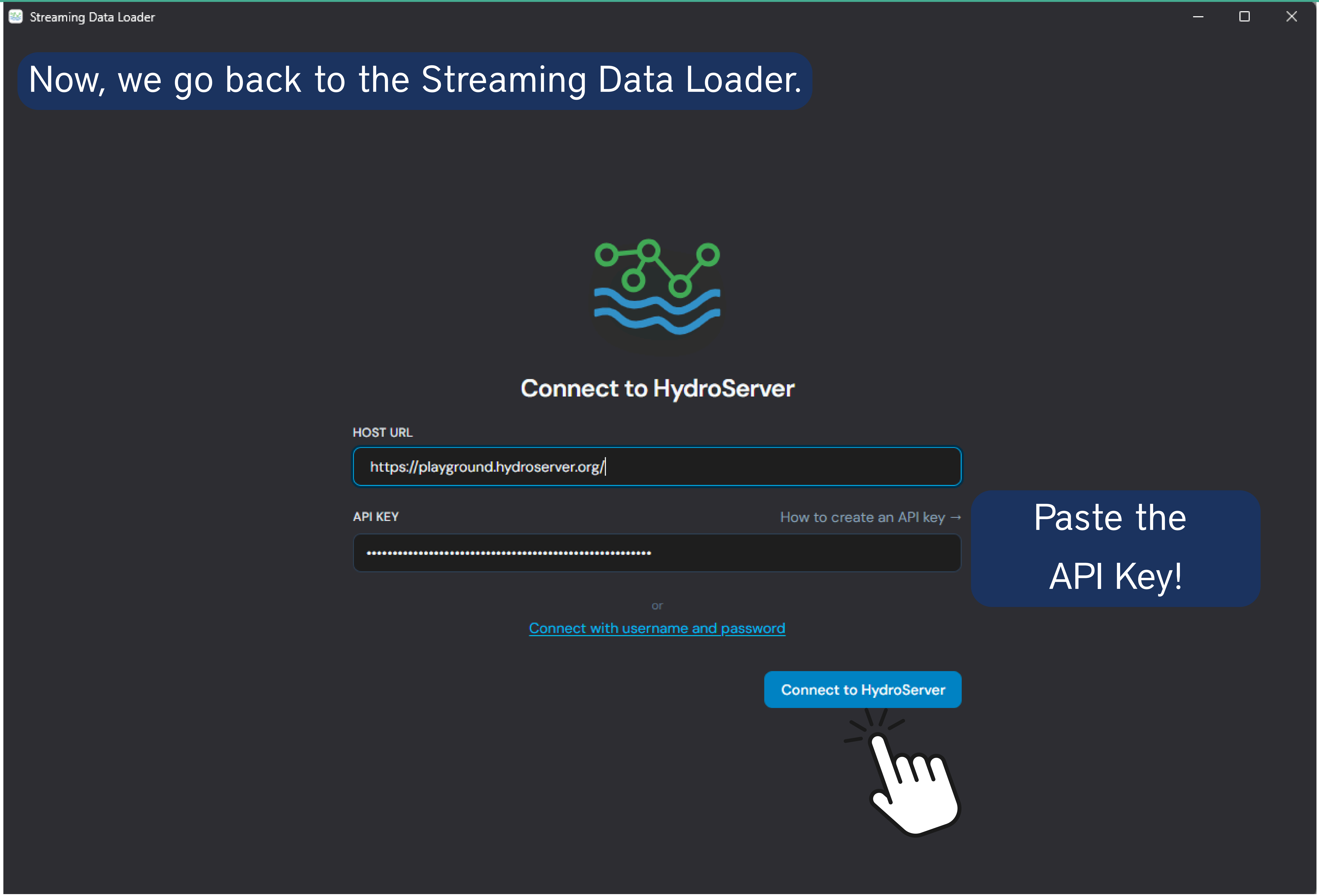
Name	Role	Actions
Data Loader	Data Loader	



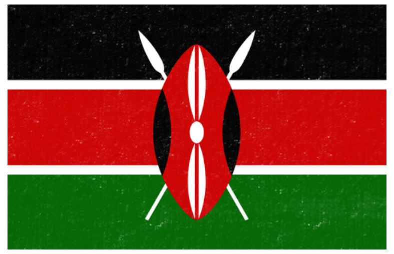
If you forget the API Key, you can regenerate it clicking

Close


Workspace access control - Logan River Monitoring
Collaborators
API keys
Workspace privacy
Transfer ownership
API keys
+ Create API key
Name
Role
Actions
When you click, this page will pop up:
Confirm API key regeneration
This action will replace the existing api key with a new one.
Cancel
Regenerate key
Close

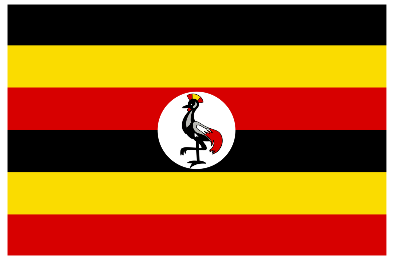

Streaming Data Loader
Now, we go back to the Streaming Data Loader.
Connect to HydroServer
HOST URL
https://playground.hydroserver.org/
API KEY
How to create an API key →
or
Connect with username and password
Connect to HydroServer
Paste the
API Key!

You can access the real-time stage data from the Kanzenze Hydrological Station, which we will upload using the Streaming Data Loader, through the following link for your country:



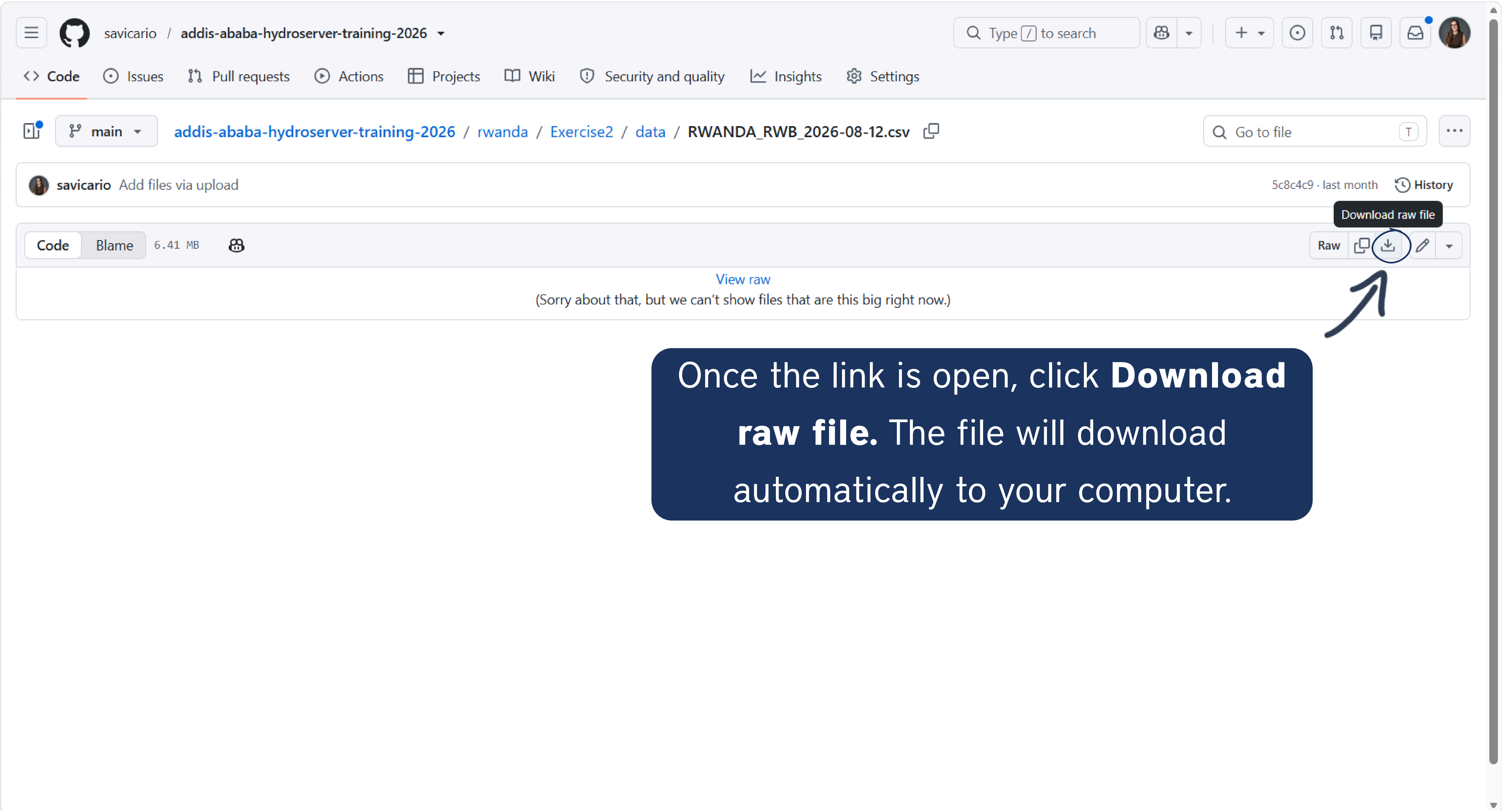
Kenya: [Access Data Exercise 2]

Rwanda: [Access Data Exercise 2]



Uganda: [Access Data Exercise 2]

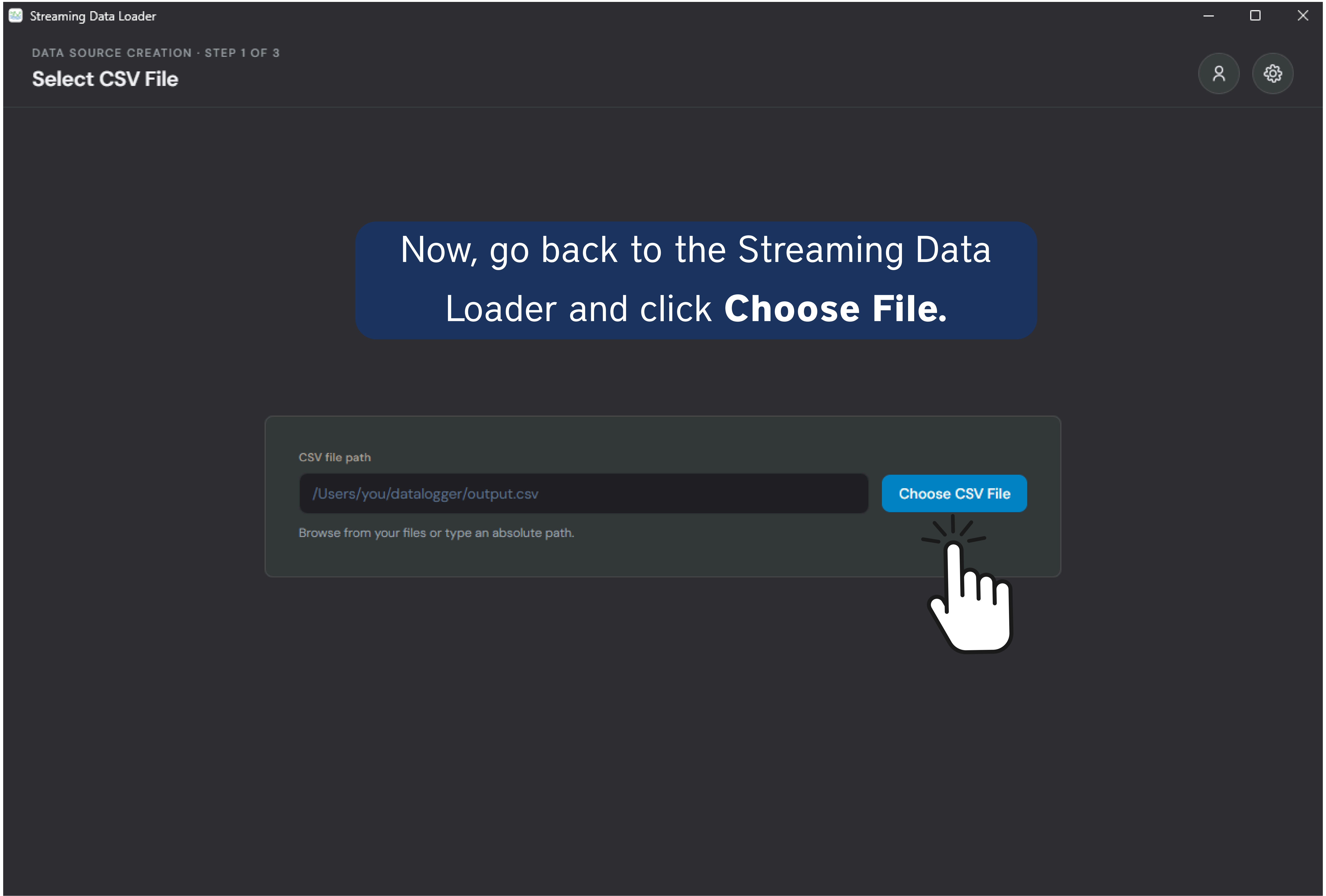
Ethiopia: [Access Data Exercise 2]

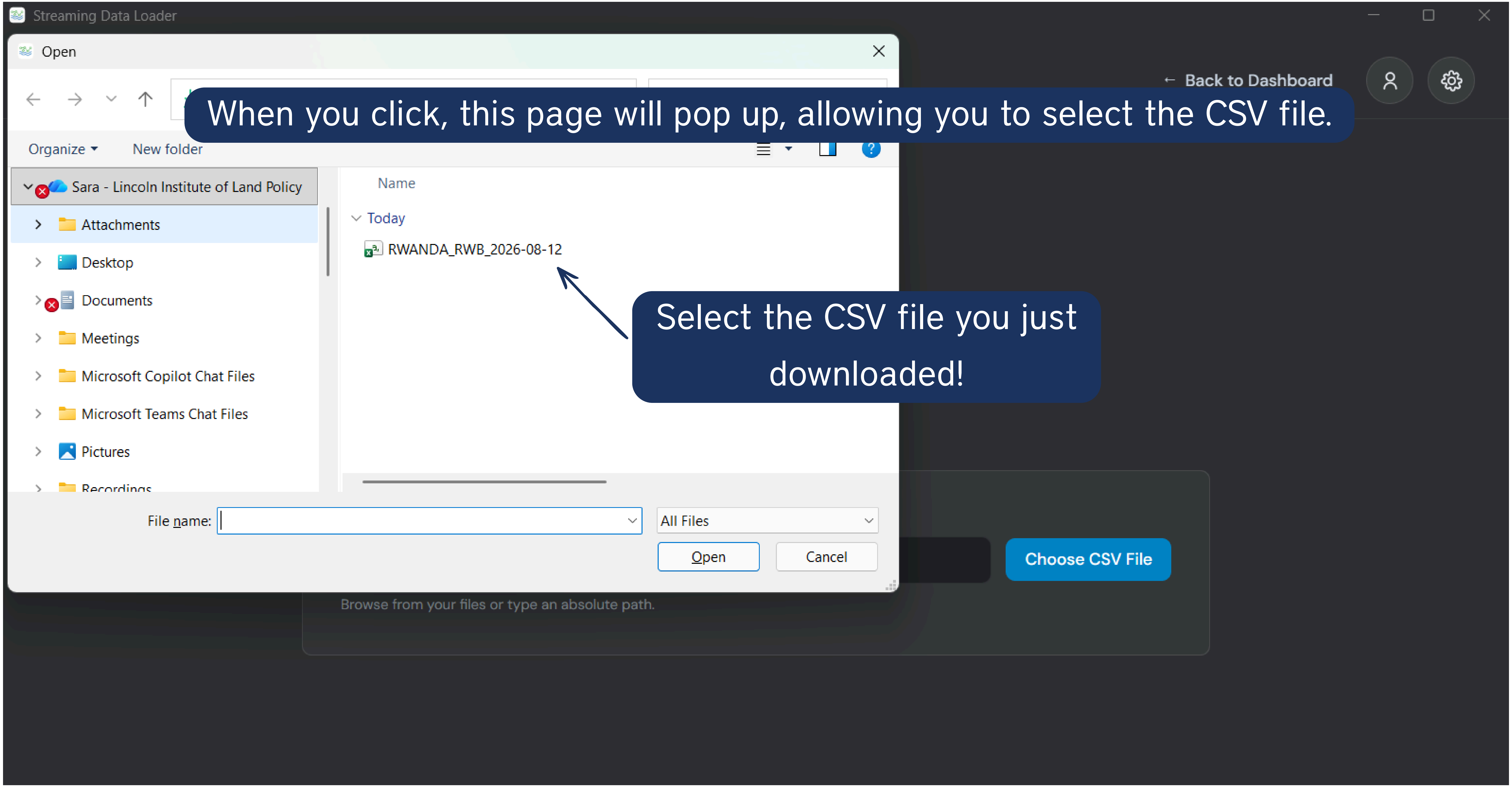


Once the link is open, click **Download raw file.** The file will download automatically to your computer.

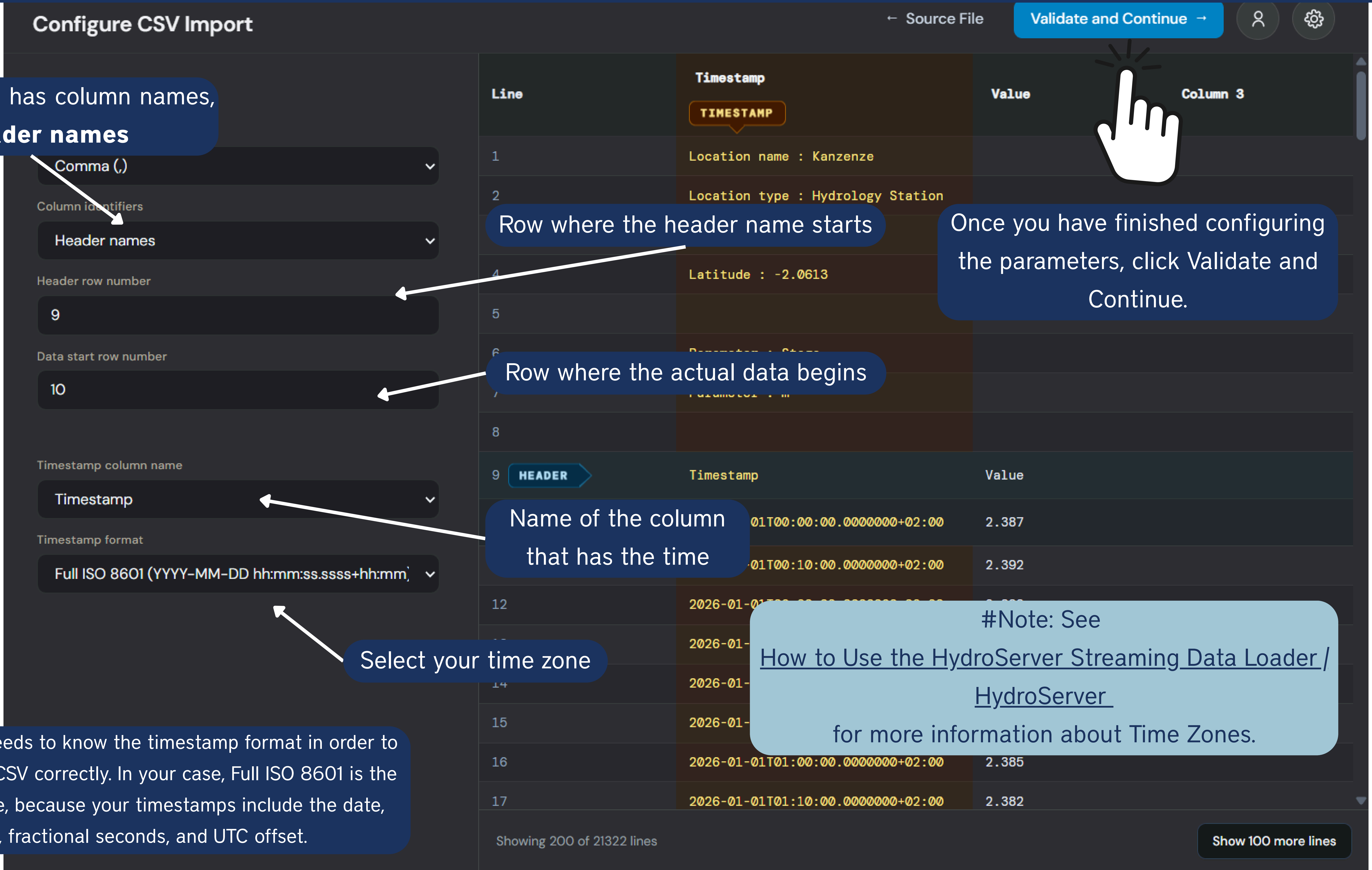
If you open the CSV file you just downloaded, this is what the data will look like:

Location name : Kanzenze	
Location type : Hydrology Station	
Identifier : 259501	
Latitude : -2.0613	Longitude : SRID : 4326
Parameter : Stage	
Parameter : m	
Timestamp	Value
2026-01-01T00:00:00.0000000+02:00	2.387
2026-01-01T00:10:00.0000000+02:00	2.392
2026-01-01T00:20:00.0000000+02:00	2.392
2026-01-01T00:30:00.0000000+02:00	2.395
2026-01-01T00:40:00.0000000+02:00	2.389
2026-01-01T00:50:00.0000000+02:00	2.386
2026-01-01T01:00:00.0000000+02:00	2.385
2026-01-01T01:10:00.0000000+02:00	2.382
2026-01-01T01:20:00.0000000+02:00	2.384
2026-01-01T01:30:00.0000000+02:00	2.385
2026-01-01T01:40:00.0000000+02:00	2.377
2026-01-01T01:50:00.0000000+02:00	2.377
2026-01-01T02:00:00.0000000+02:00	2.374
2026-01-01T02:10:00.0000000+02:00	2.374
2026-01-01T02:20:00.0000000+02:00	2.37
2026-01-01T02:30:00.0000000+02:00	2.366
2026-01-01T02:40:00.0000000+02:00	2.368
2026-01-01T02:50:00.0000000+02:00	2.365
2026-01-01T03:00:00.0000000+02:00	2.359
2026-01-01T03:10:00.0000000+02:00	2.36
2026-01-01T03:20:00.0000000+02:00	2.359
2026-01-01T03:30:00.0000000+02:00	2.364
2026-01-01T03:40:00.0000000+02:00	2.362


Streaming Data Loader
DATA SOURCE CREATION · STEP 1 OF 3
Select CSV File
Now, go back to the Streaming Data Loader and click **Choose File.**
CSV file path
/Users/you/datalogger/output.csv
Choose CSV File
Browse from your files or type an absolute path.


Streaming Data Loader
Open
← Back to Dashboard
When you click, this page will pop up, allowing you to select the CSV file.
Organize
New folder
Sara - Lincoln Institute of Land Policy
Attachments
Desktop
Documents
Meetings
Microsoft Copilot Chat Files
Microsoft Teams Chat Files
Pictures
Recordings
Name
Today
RWANDA_RWB_2026-08-12
Select the CSV file you just downloaded!
File name:
All Files
Open
Cancel
Choose CSV File
Browse from your files or type an absolute path.

After selecting the file, configure the parameters that tell the Streaming Data Loader how to read your CSV file.



If you data has column names, select **Header names**

Row where the header name starts

Row where the actual data begins

Name of the column that has the time

Select your time zone

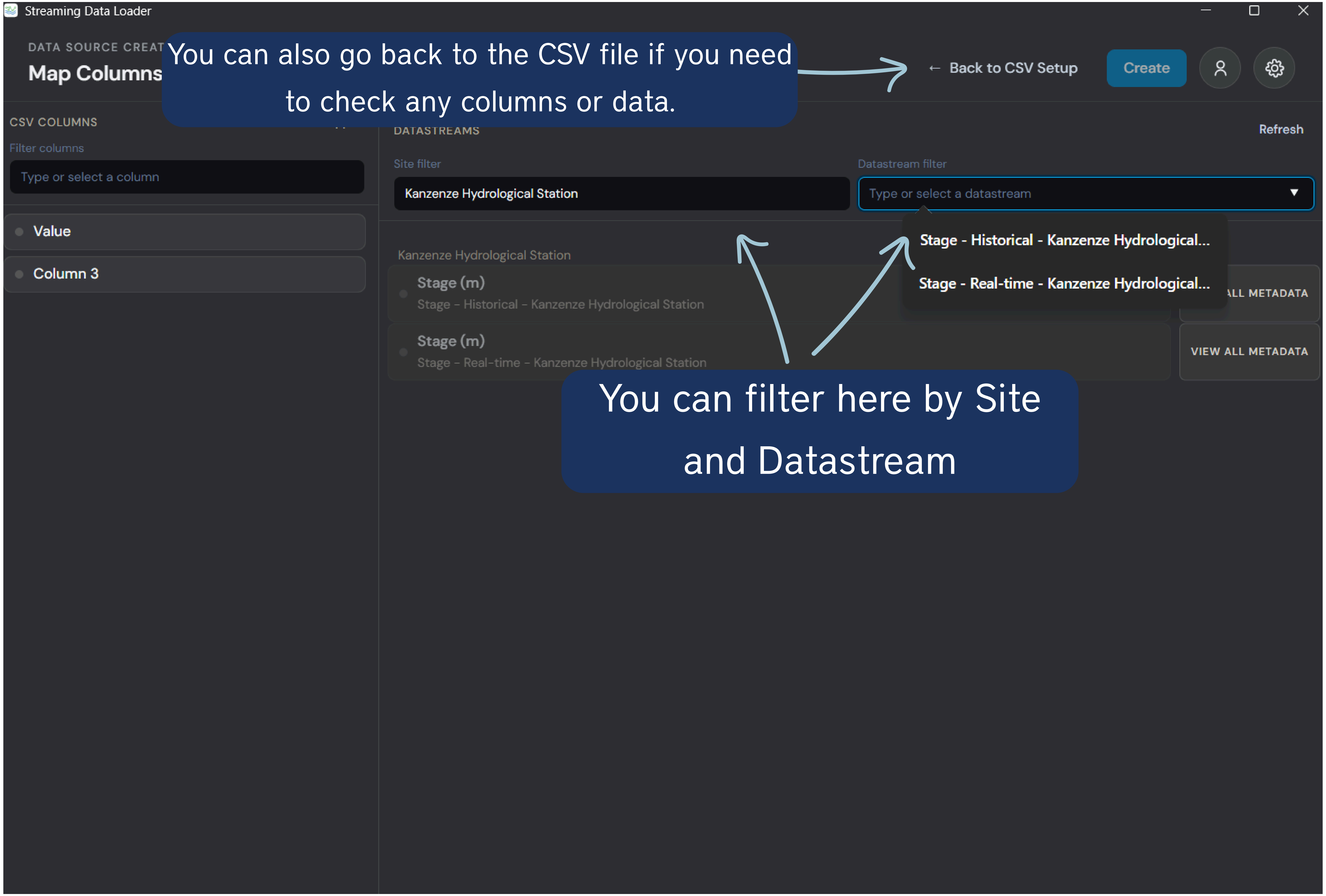
Once you have finished configuring the parameters, click Validate and Continue.

#Note: See [How to Use the HydroServer Streaming Data Loader | HydroServer](#) for more information about Time Zones.

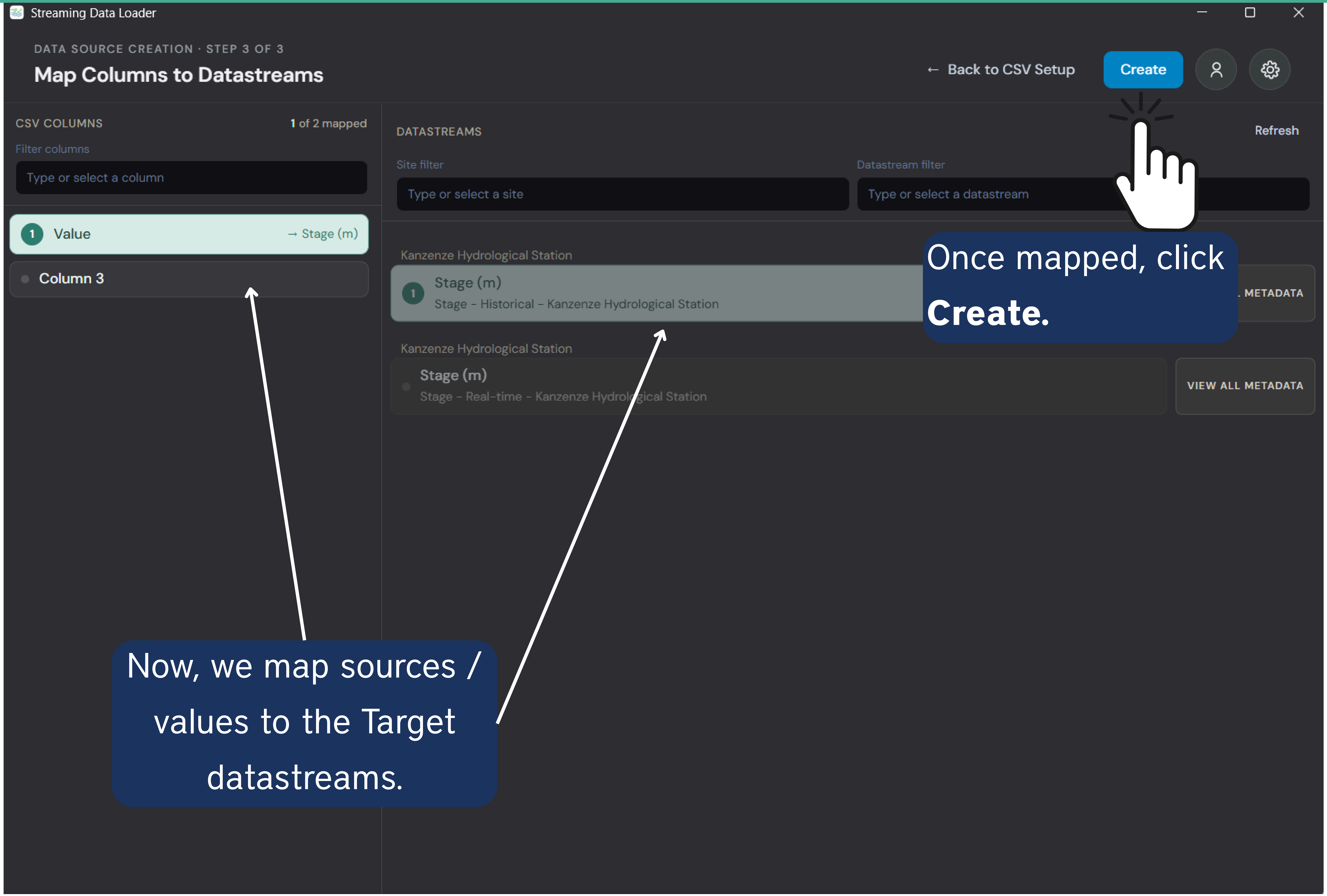
HydroServer needs to know the timestamp format in order to interpret your CSV correctly. In your case, Full ISO 8601 is the correct choice, because your timestamps include the date, time, fractional seconds, and UTC offset.

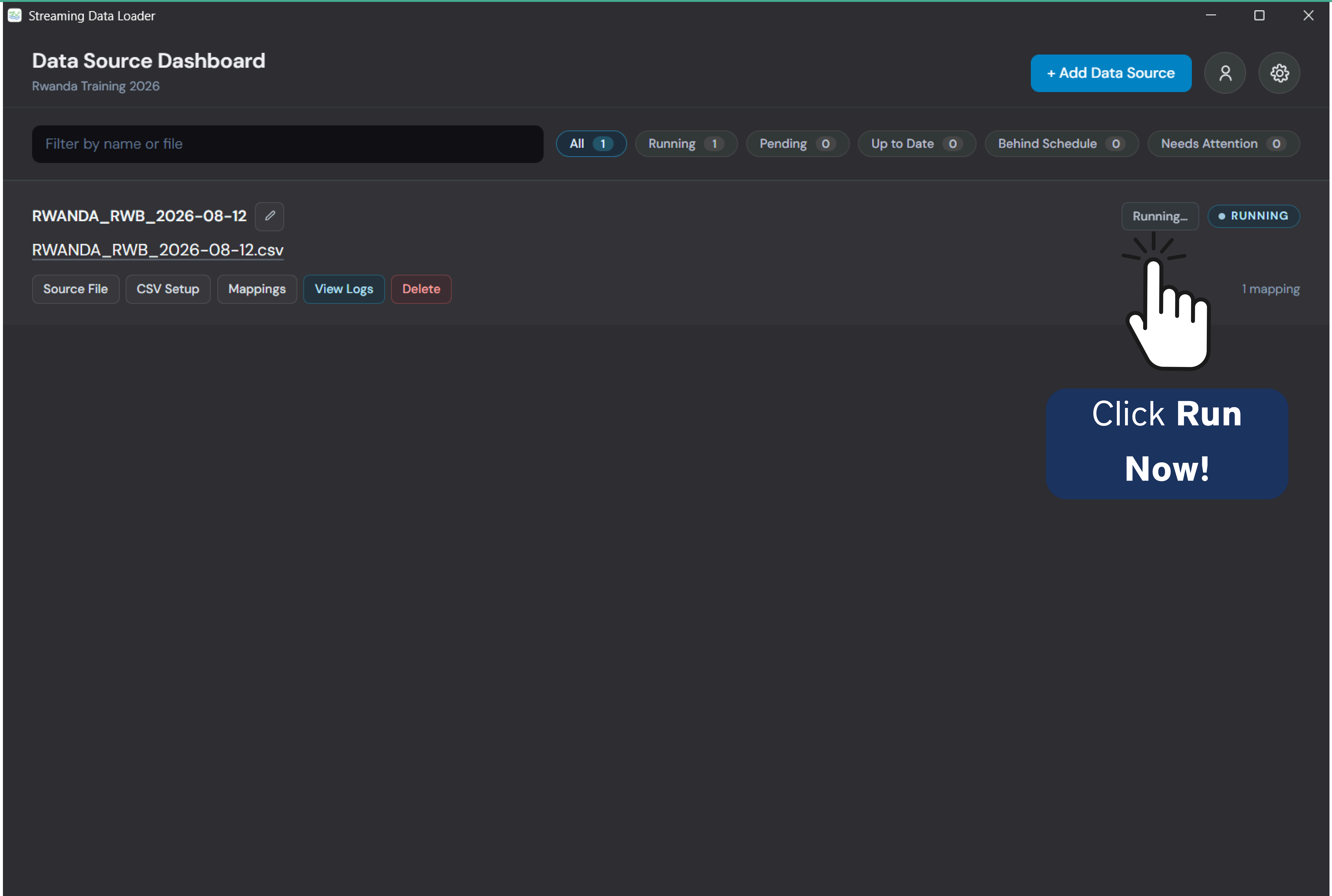
Parameters you need to configure in the Streaming Data Loader after selecting your CSV file:

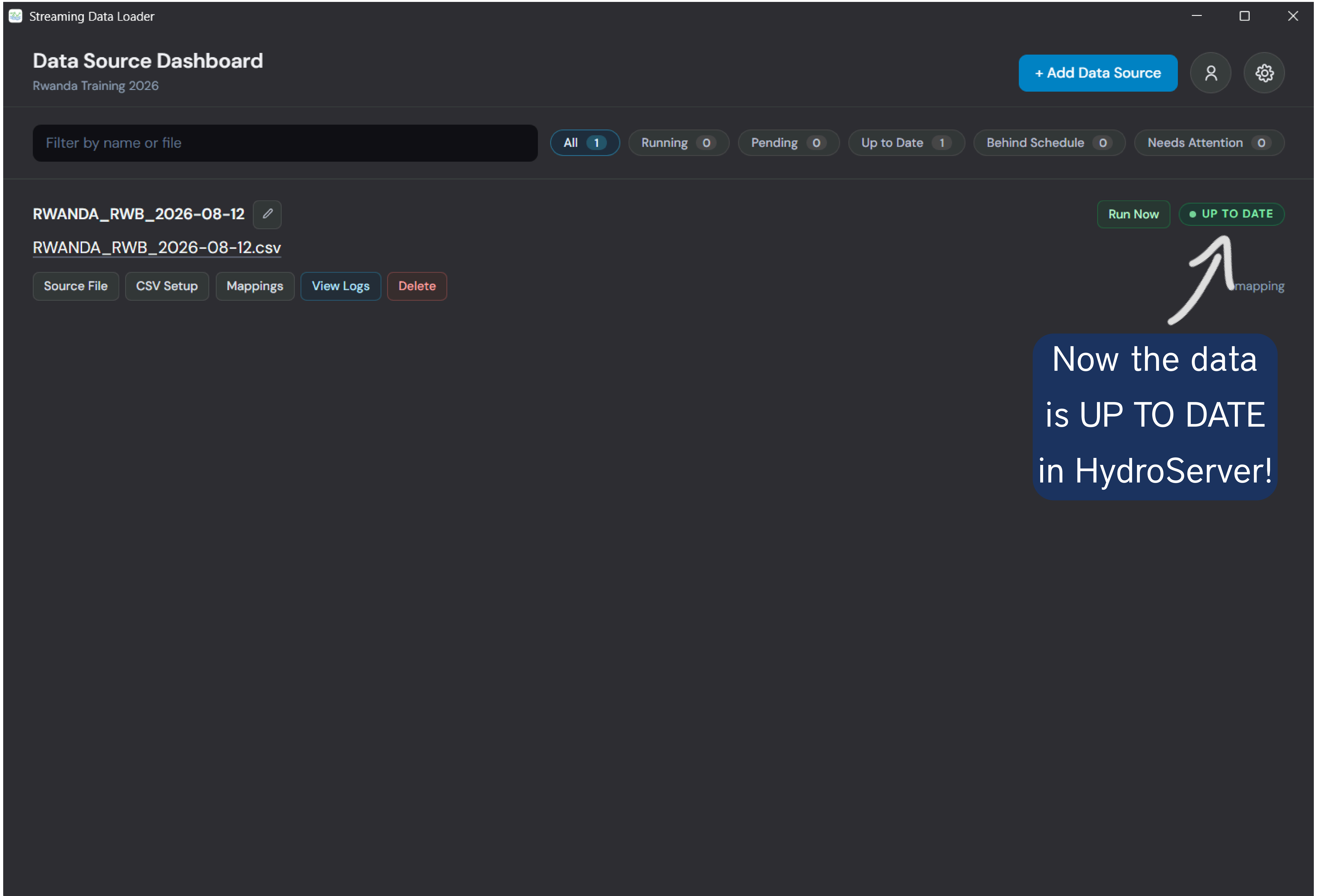
- **Header row number**: Row containing the column headers.
- **Data start row number**: Row where the observations begin.
- **Timestamp column name**: Column containing the timestamps.
- **Timestamp format**: Format used for the timestamps in your CSV file.

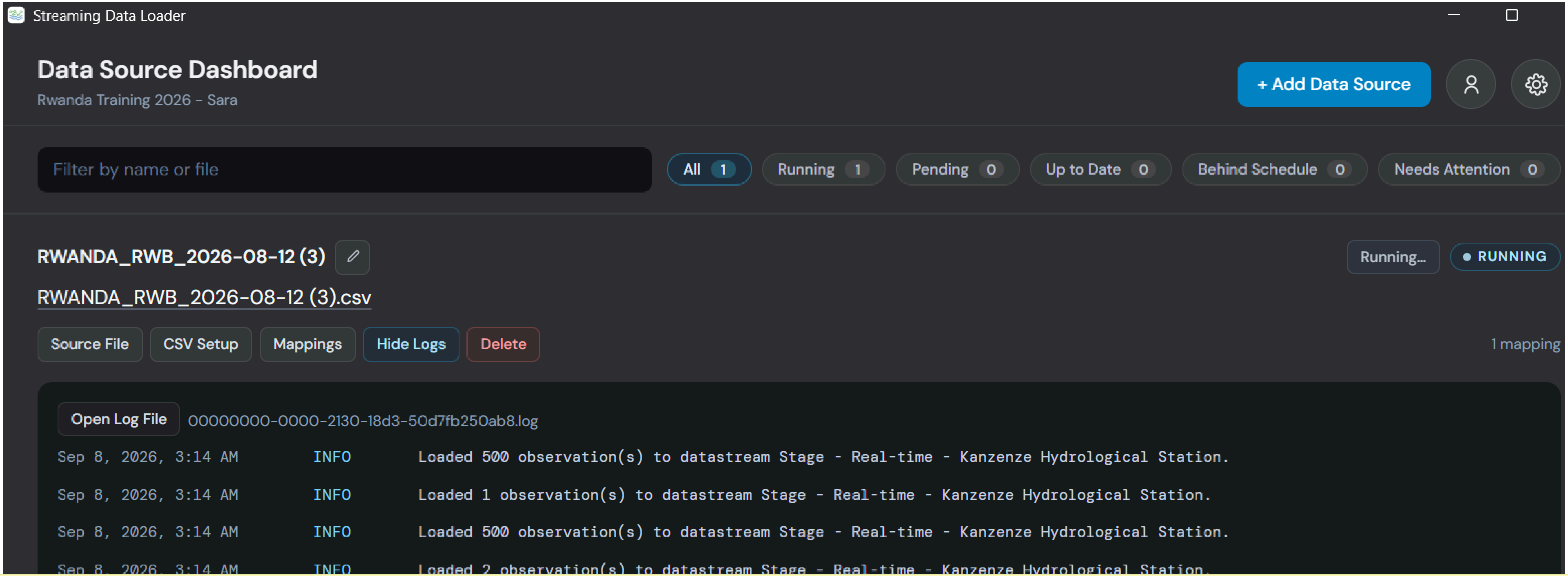


Streaming Data Loader
DATA SOURCE CREAT
Map Columns
You can also go back to the CSV file if you need to check any columns or data.
← Back to CSV Setup
Create
CSV COLUMNS
Filter columns
Type or select a column
Value
Column 3
DATASTREAMS
Refresh
Site filter
Kanzenze Hydrological Station
Datastream filter
Type or select a datastream
Stage - Historical - Kanzenze Hydrological...
Stage - Real-time - Kanzenze Hydrological...
Kanzenze Hydrological Station
Stage (m)
Stage - Historical - Kanzenze Hydrological Station
Stage (m)
Stage - Real-time - Kanzenze Hydrological Station
VIEW ALL METADATA
You can filter here by Site and Datastream


Streaming Data Loader
DATA SOURCE CREATION · STEP 3 OF 3
Map Columns to Datastreams
← Back to CSV Setup
Create
CSV COLUMNS
1 of 2 mapped
Filter columns
Type or select a column
1 Value
→ Stage (m)
Column 3
DATASTREAMS
Refresh
Site filter
Type or select a site
Datastream filter
Type or select a datastream
Kanzenze Hydrological Station
Stage (m)
Stage - Historical - Kanzenze Hydrological Station
METADATA
Kanzenze Hydrological Station
Stage (m)
Stage - Real-time - Kanzenze Hydrological Station
VIEW ALL METADATA
Once mapped, click **Create.**
Now, we map sources / values to the Target datastreams.


Streaming Data Loader
Data Source Dashboard
Rwanda Training 2026
+ Add Data Source
Filter by name or file
All 1
Running 1
Pending 0
Up to Date 0
Behind Schedule 0
Needs Attention 0
RWANDA_RWB_2026-08-12
RWANDA_RWB_2026-08-12.csv
Source File
CSV Setup
Mappings
View Logs
Delete
Running...
RUNNING
1 mapping
Click Run Now!


Streaming Data Loader
Data Source Dashboard
Rwanda Training 2026
+ Add Data Source
Filter by name or file
All 1
Running 0
Pending 0
Up to Date 1
Behind Schedule 0
Needs Attention 0
RWANDA_RWB_2026-08-12
RWANDA_RWB_2026-08-12.csv
Source File
CSV Setup
Mappings
View Logs
Delete
Run Now
UP TO DATE
mapping
Now the data is UP TO DATE in HydroServer!

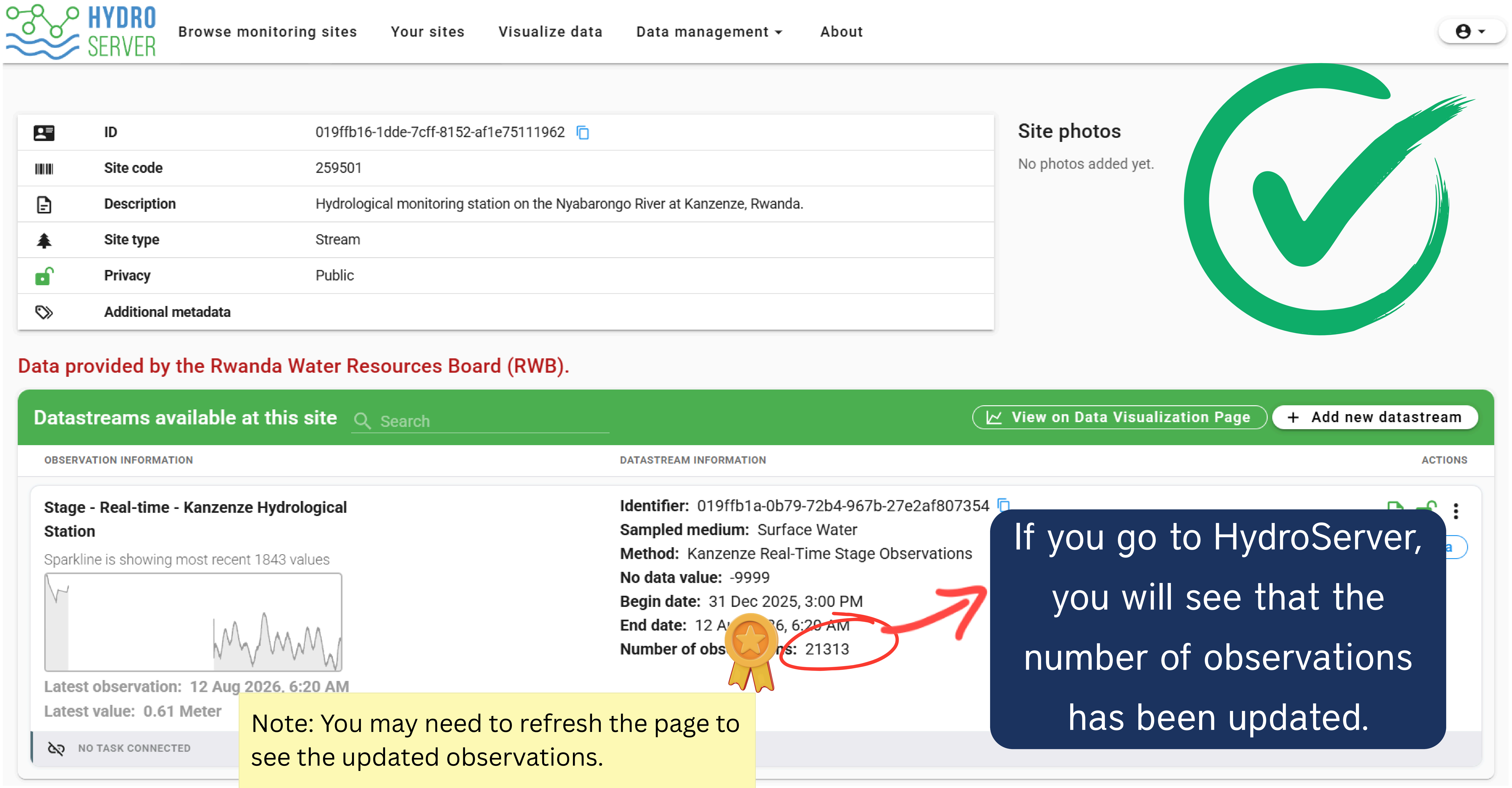


Note: If your file is large, the job may remain in Pending or Running status for some time while the data are being processed. For this reason, in this exercise we are only working with data from 2026. The complete dataset from September 28, 2022 up to August 12, 2026, contains more than 150,000 observations.

This is more likely to happen during the initial upload, when a large batch of observations is loaded.

Subsequent runs may be faster because the Streaming Data Loader only needs to upload new observations.

After completing the steps in the Streaming Data Loader, go to the HydroServer Web Management App. You will see that the number of observations in your datastream has been updated.



If you go to HydroServer, you will see that the number of observations has been updated.

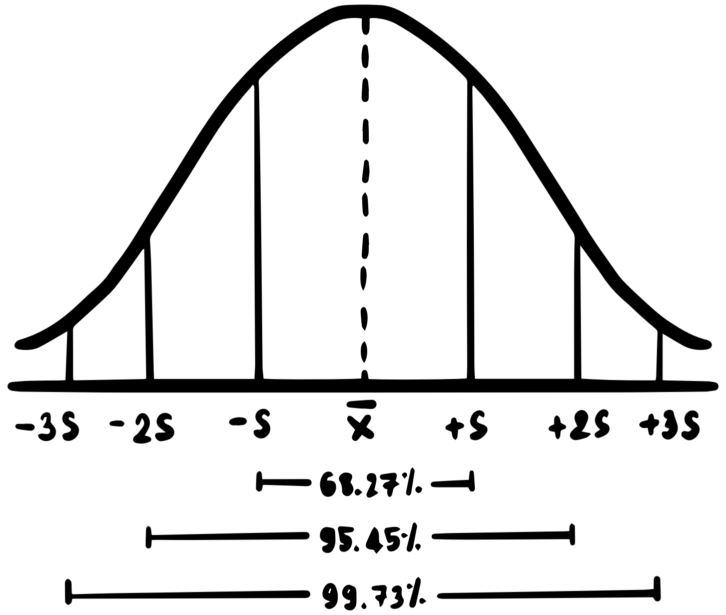
Note: You may need to refresh the page to see the updated observations.

Sara Alonso Vicario

This was an example of how to use the **Streaming Data Loader.**

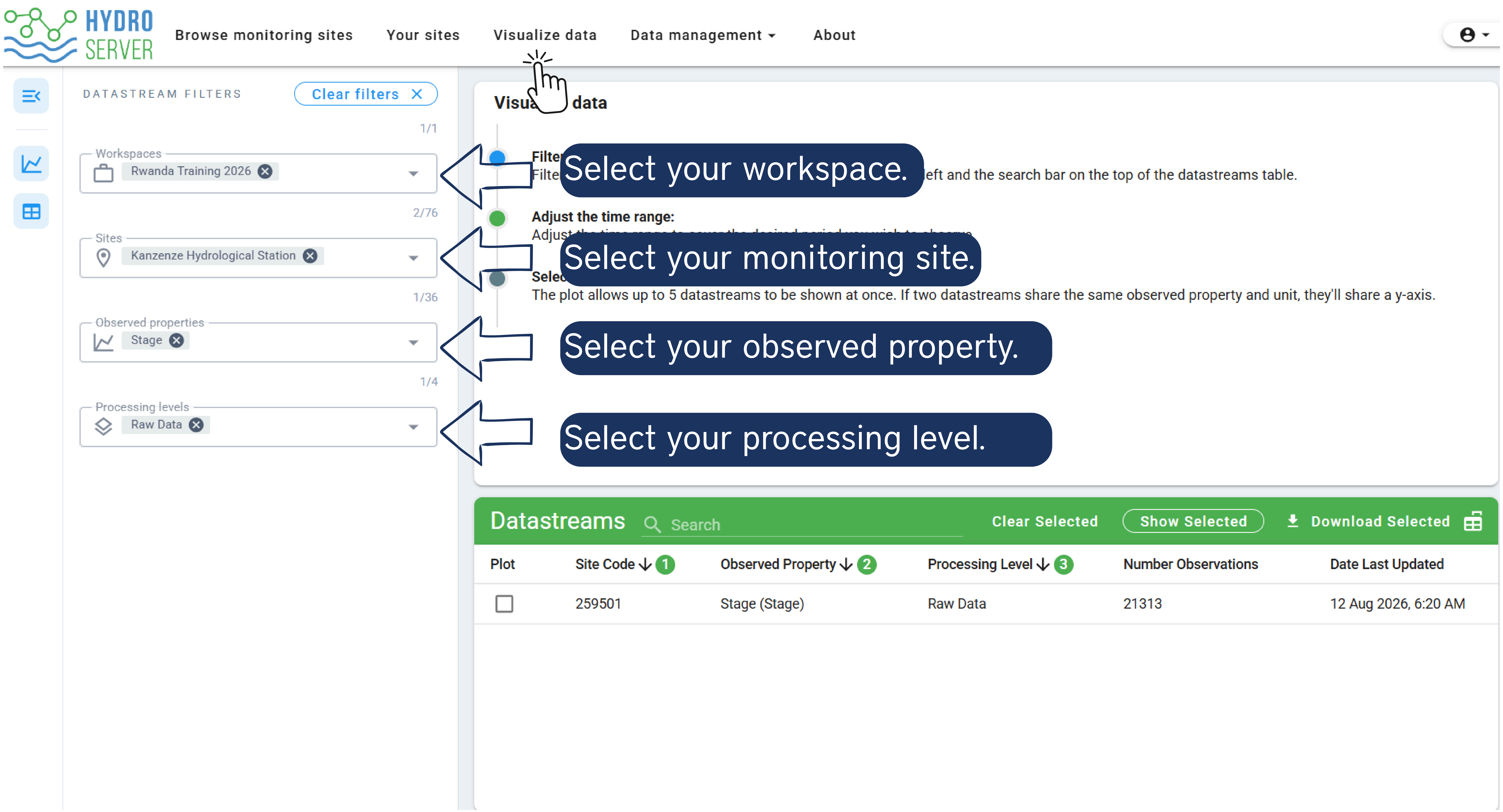
To automate the process, you need to select a CSV file on your computer or server that is regularly updated with new data received from your sensors/dataloggers.

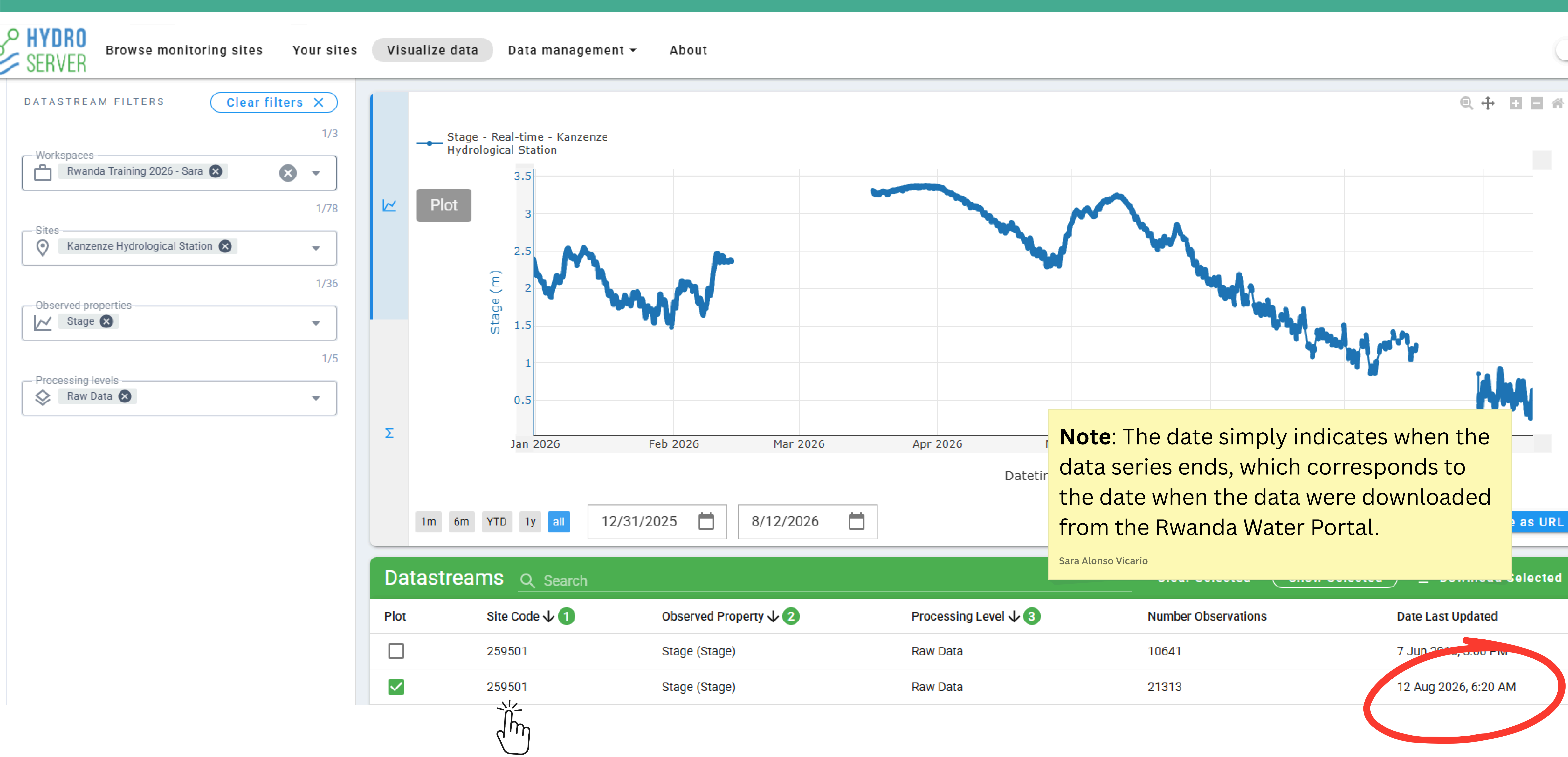
Now, let's go back to the HydroServer Web Dashboard to visualize the data we just uploaded using the Streaming Data Loader and explore some statistics for the observations!



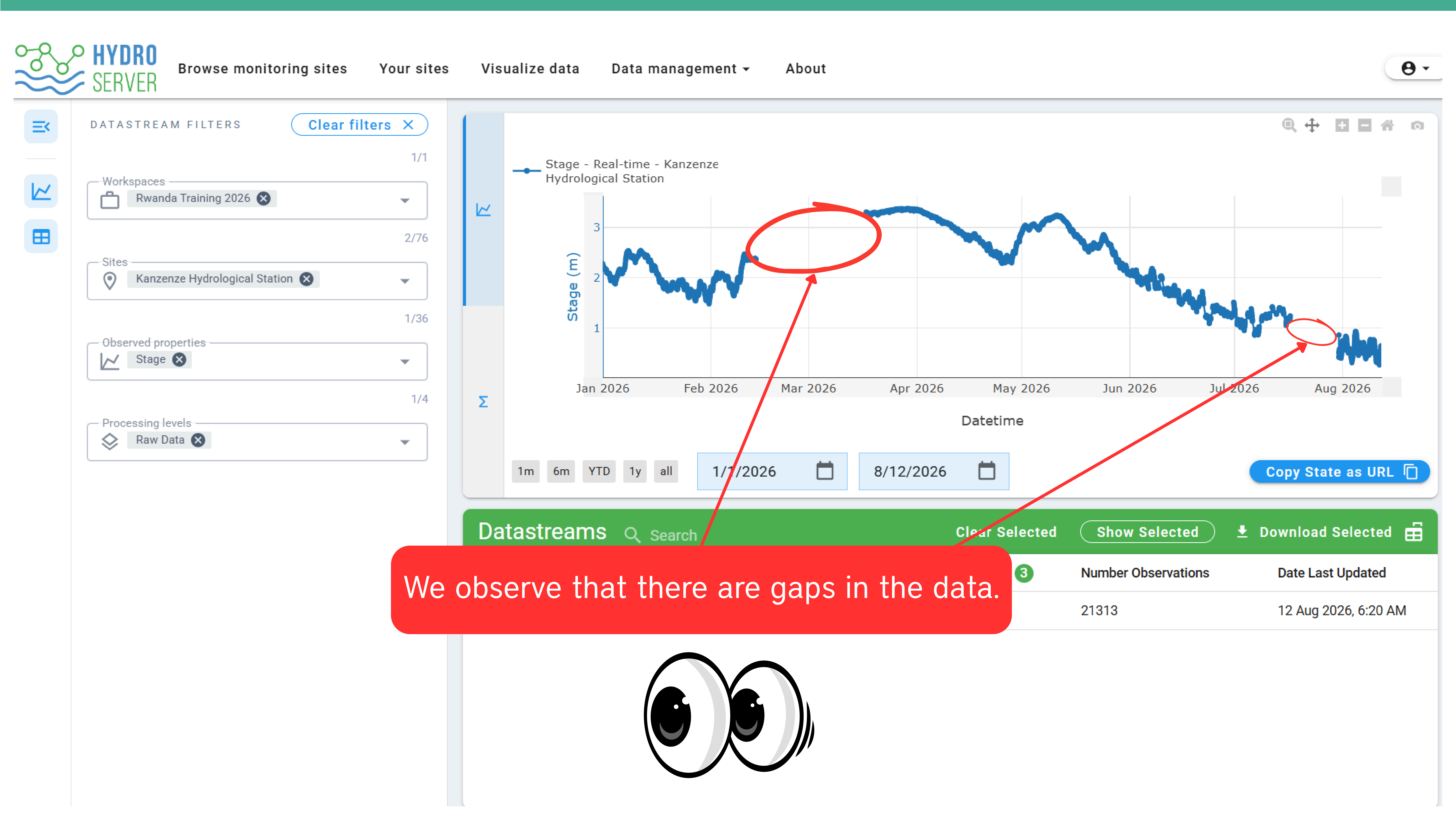
Data Visualization in HydroServer

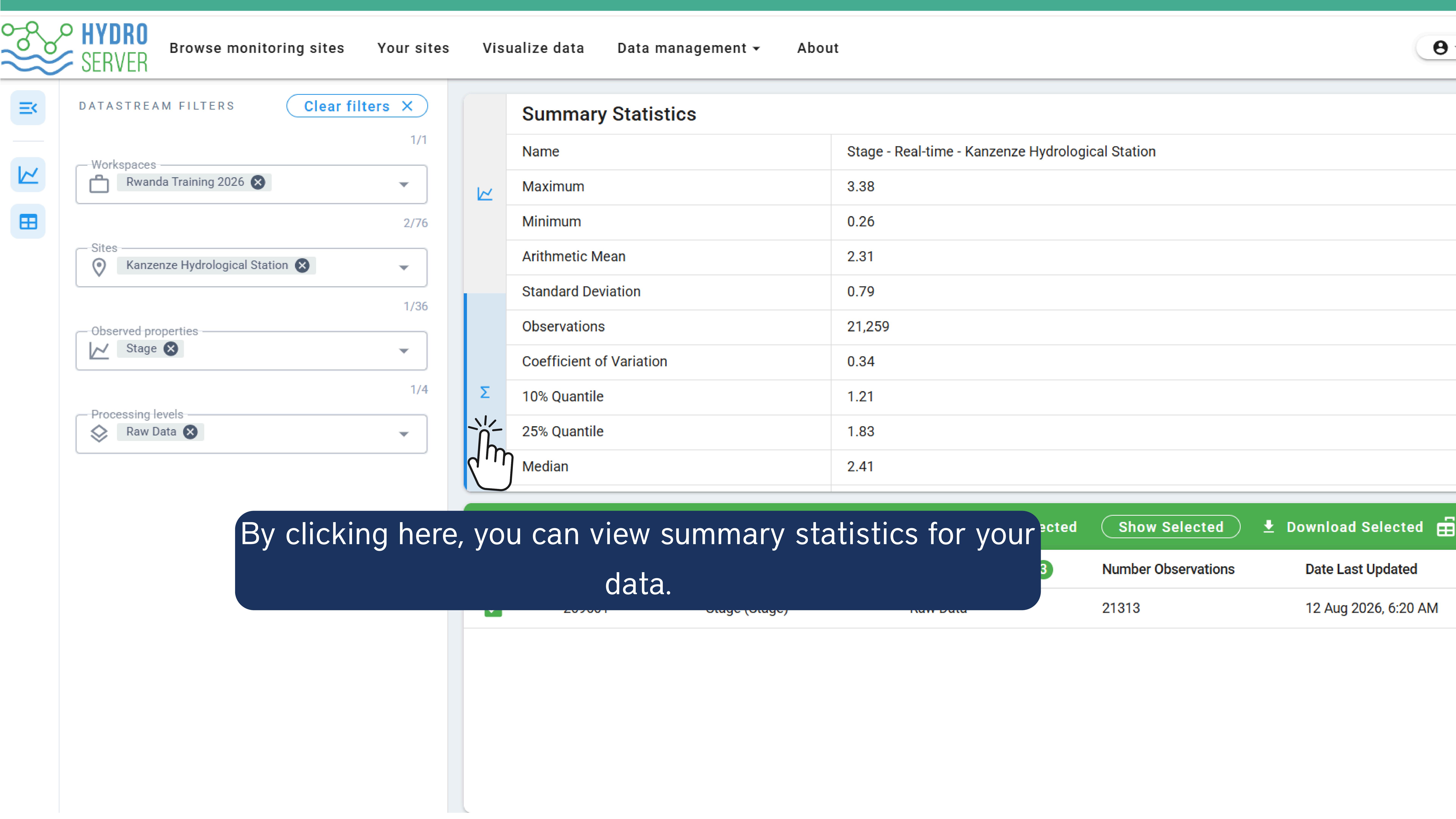
Go to the HydroServer Web Dashboard and click Visualize Data.





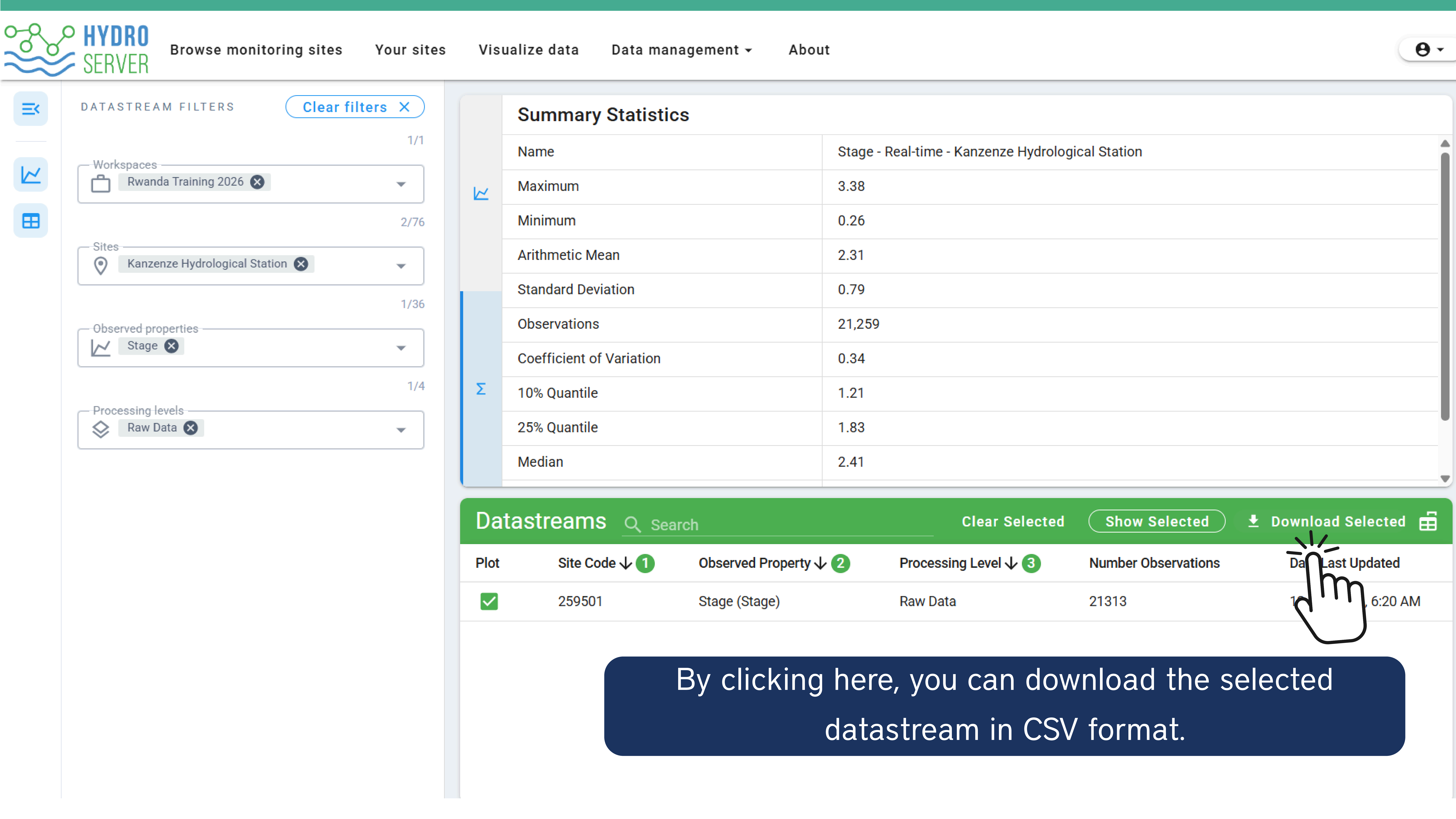
Now, select the datastream you are interested in. You should select the most recently updated datastream from 2026.


HYDRO SERVER
Browse monitoring sites
Your sites
Visualize data
Data management
About
DATASTREAM FILTERS
Clear filters
1/1
Workspaces
Rwanda Training 2026
2/76
Sites
Kanzenze Hydrological Station
1/36
Observed properties
Stage
1/4
Processing levels
Raw Data
Stage - Real-time - Kanzenze Hydrological Station
Stage (m)
3
2
1
Jan 2026
Feb 2026
Mar 2026
Apr 2026
May 2026
Jun 2026
Jul 2026
Aug 2026
Datetime
1m
6m
YTD
1y
all
1/1/2026
8/12/2026
Copy State as URL
Datastreams
Search
Clear Selected
Show Selected
Download Selected
3
Number Observations
Date Last Updated
21313
12 Aug 2026, 6:20 AM
We observe that there are gaps in the data.


HYDRO SERVER
Browse monitoring sites
Your sites
Visualize data
Data management
About
DATASTREAM FILTERS
Clear filters
1/1
Workspaces
Rwanda Training 2026
2/76
Sites
Kanzenze Hydrological Station
1/36
Observed properties
Stage
1/4
Processing levels
Raw Data
Summary Statistics
Name
Stage - Real-time - Kanzenze Hydrological Station
Maximum
3.38
Minimum
0.26
Arithmetic Mean
2.31
Standard Deviation
0.79
Observations
21,259
Coefficient of Variation
0.34
10% Quantile
1.21
25% Quantile
1.83
Median
2.41
Show Selected
Download Selected
Number Observations
Date Last Updated
21313
12 Aug 2026, 6:20 AM
By clicking here, you can view summary statistics for your data.

HydroServer automatically calculates summary statistics for each datastream, including:

- **Maximum** - Highest observed value.
- **Minimum** - Lowest observed value.
- **Arithmetic Mean** - Average of all observations.
- **Standard Deviation** - How much the observations vary around the mean.
- **Number of Observations** -Total number of observations.
- **Coefficient of Variation** - Variability relative to the mean (Standard Deviation/mean).
- **10% Quantile** - Value below which 10% of observations fall.
- **25% Quantile** - Value below which 25% of observations fall.
- **Median (50% Quantile)** - Middle value of the observations.
- **75% Quantile** - Value below which 75% of observations fall.
- **90% Quantile** - Value below which 90% of observations fall.


HYDRO SERVER
Browse monitoring sites
Your sites
Visualize data
Data management
About
DATASTREAM FILTERS
Clear filters
1/1
Workspaces
Rwanda Training 2026
2/76
Sites
Kanzenze Hydrological Station
1/36
Observed properties
Stage
1/4
Processing levels
Raw Data
Summary Statistics
Name
Stage - Real-time - Kanzenze Hydrological Station
Maximum
3.38
Minimum
0.26
Arithmetic Mean
2.31
Standard Deviation
0.79
Observations
21,259
Coefficient of Variation
0.34
10% Quantile
1.21
25% Quantile
1.83
Median
2.41
Datastreams
Search
Clear Selected
Show Selected
Download Selected
Plot
Site Code 1
Observed Property 2
Processing Level 3
Number Observations
Last Updated
259501
Stage (Stage)
Raw Data
21313
6:20 AM
By clicking here, you can download the selected datastream in CSV format.

We identified some data gaps in our observations.

Let’s address them by performing a data quality control analysis!

Data Quality Control in HydroServer

Data Quality Control (QC) is the process of **checking, identifying, and correcting potential problems in observational data** to ensure that the data are reliable and suitable for analysis.

What can you do with the HydroServer QC App?

- **Filter suspicious observations** using thresholds, time windows, rate-of-change limits, gap detection, or persistence checks.

Examples of Available Filter & Operations

- **Edit selected observations**: change values, interpolate, correct drift, shift timestamps, delete or add observations, and assign quality qualifiers.

- **Save your QC workflow as a JSON file** that can be reused later.

- **Submit quality-controlled observations back to HydroServer**.

Quality Control Workflow in HydroServer

- **Select a datastream** to review. 
- **Inspect and filter observations** for potential issues. 
- **Apply QC operations** to correct, delete, interpolate, or add data. 
- **Assign data-quality qualifiers** to selected observations. 
- **Save the changes** to HydroServer. 

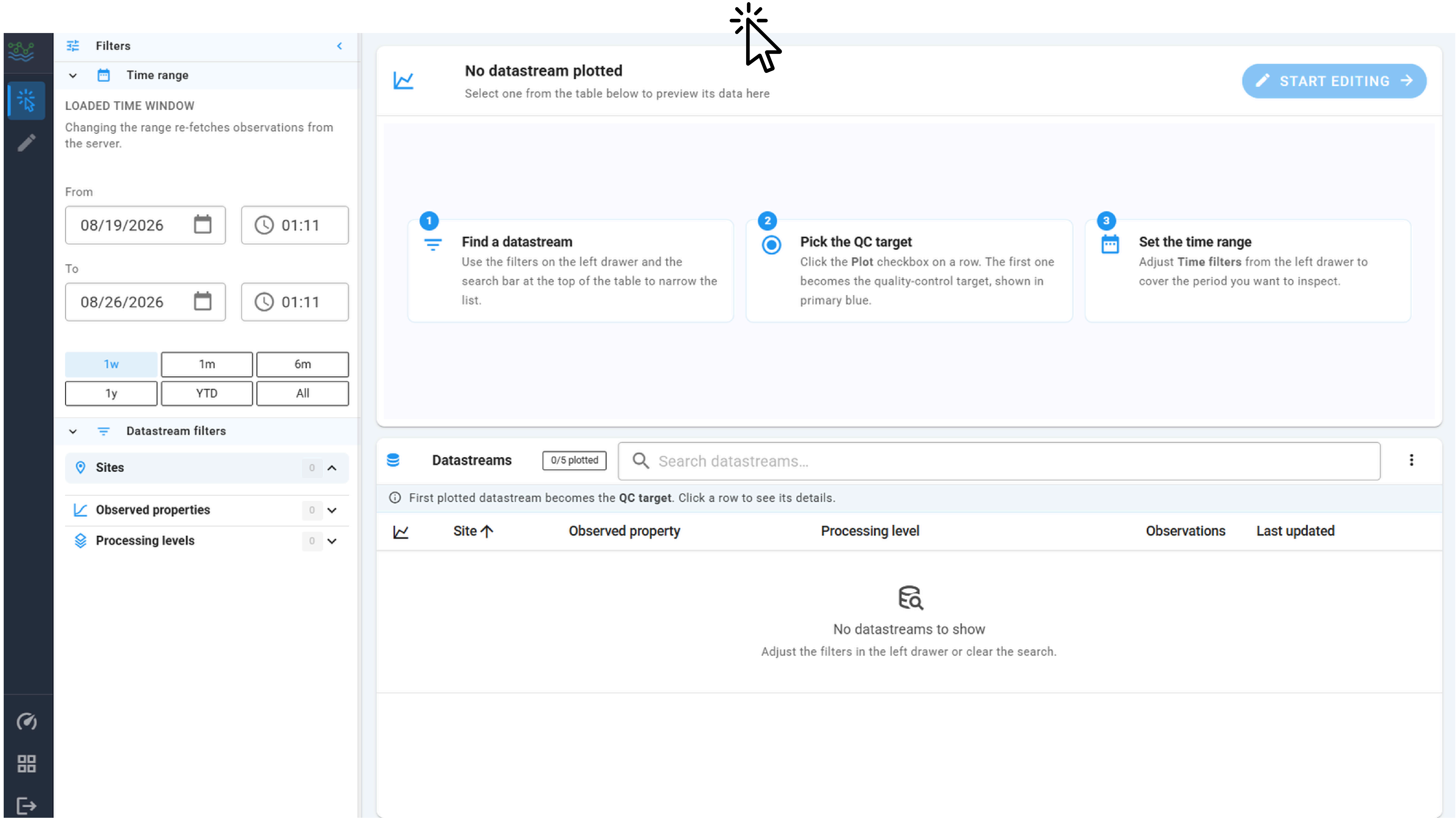
QC App Availability Timeline

The HydroServer Quality Control (QC) App is currently **under active development.**

- It is **currently available only in the HydroServer Playground.**
- It is expected to be officially released in 2026.

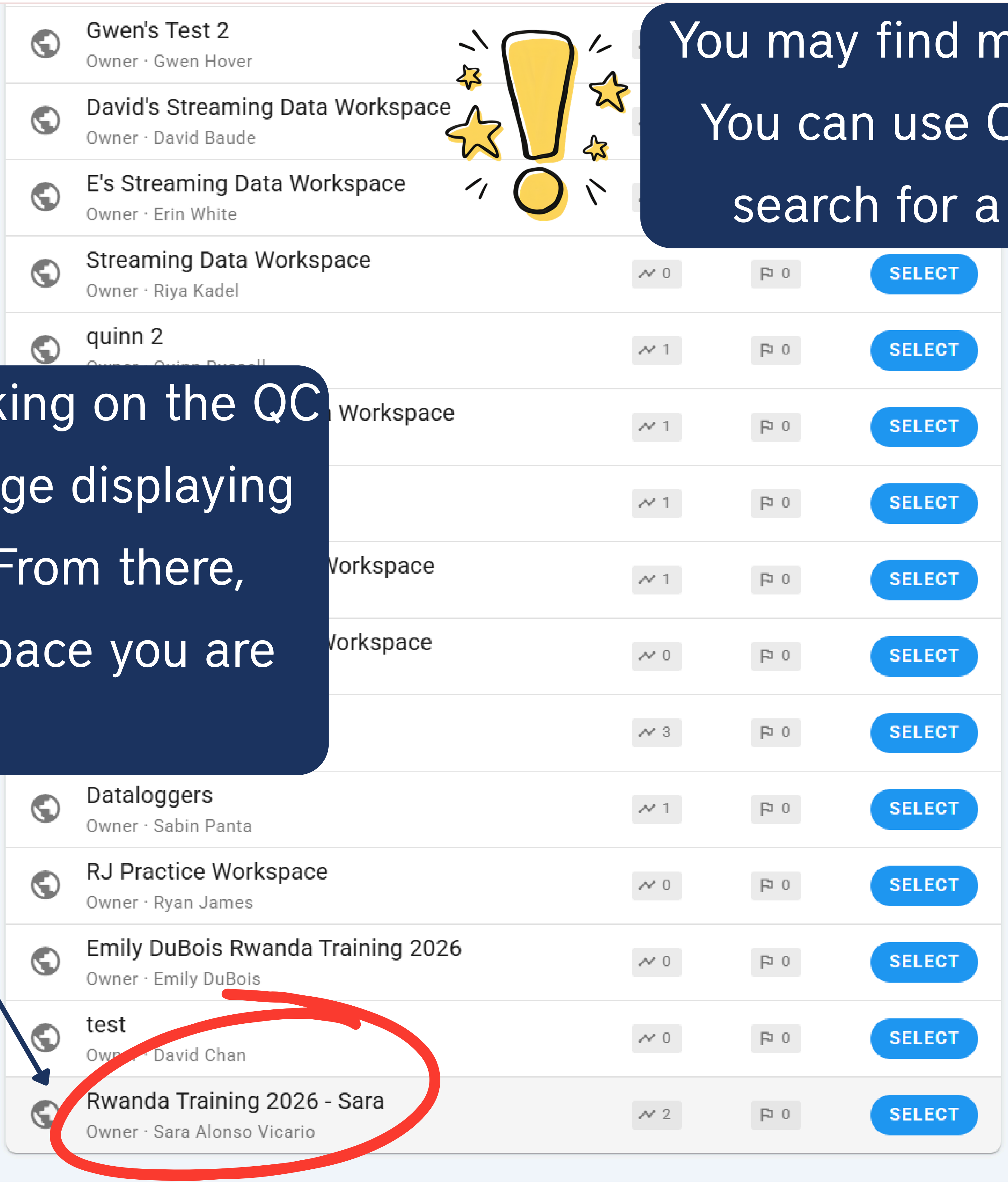
Try the HydroServer QC App here:

HydroServer QC App

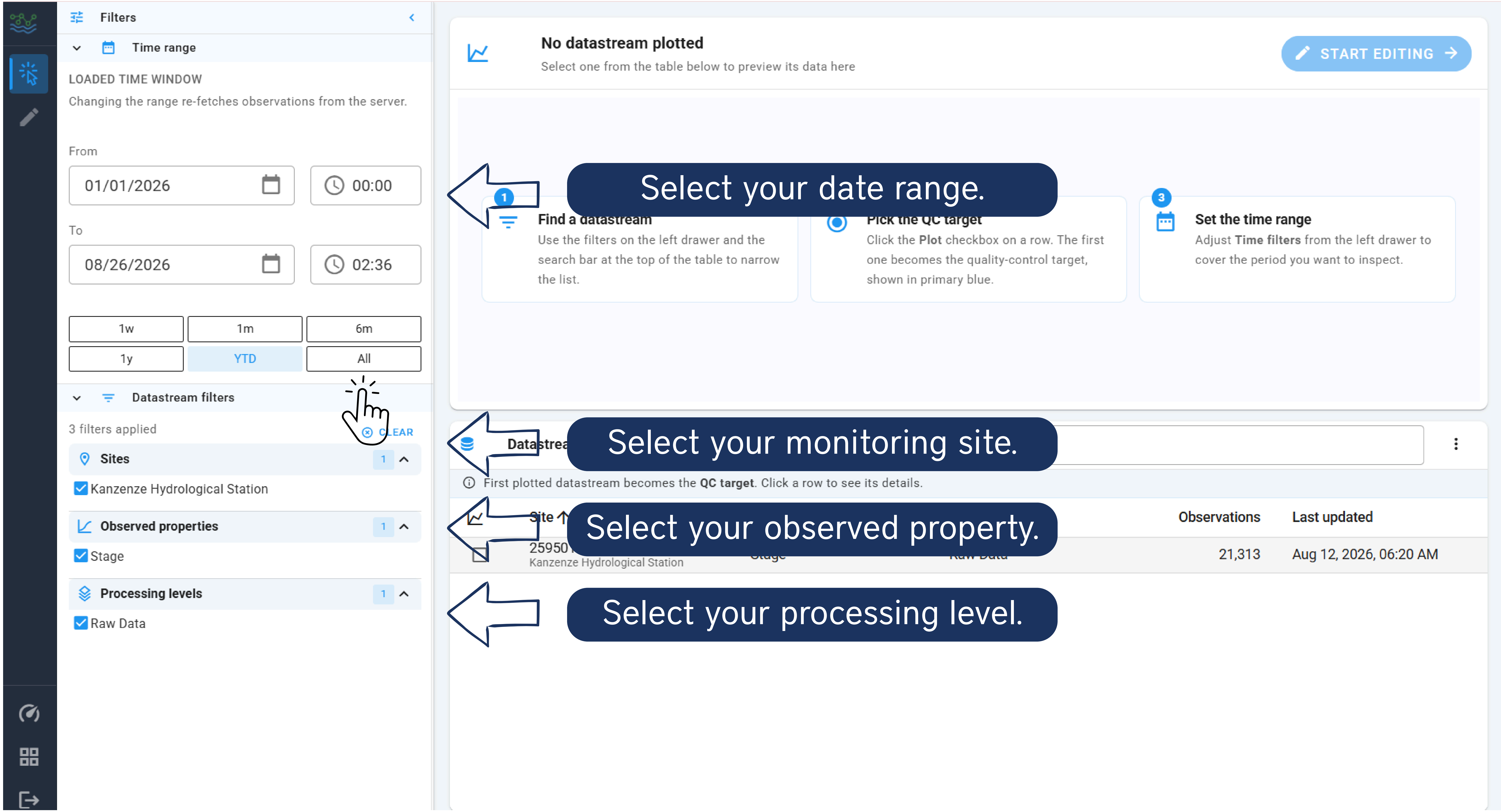


You may find many public workspaces. You can use Ctrl + F on Windows to search for a workspace by name.

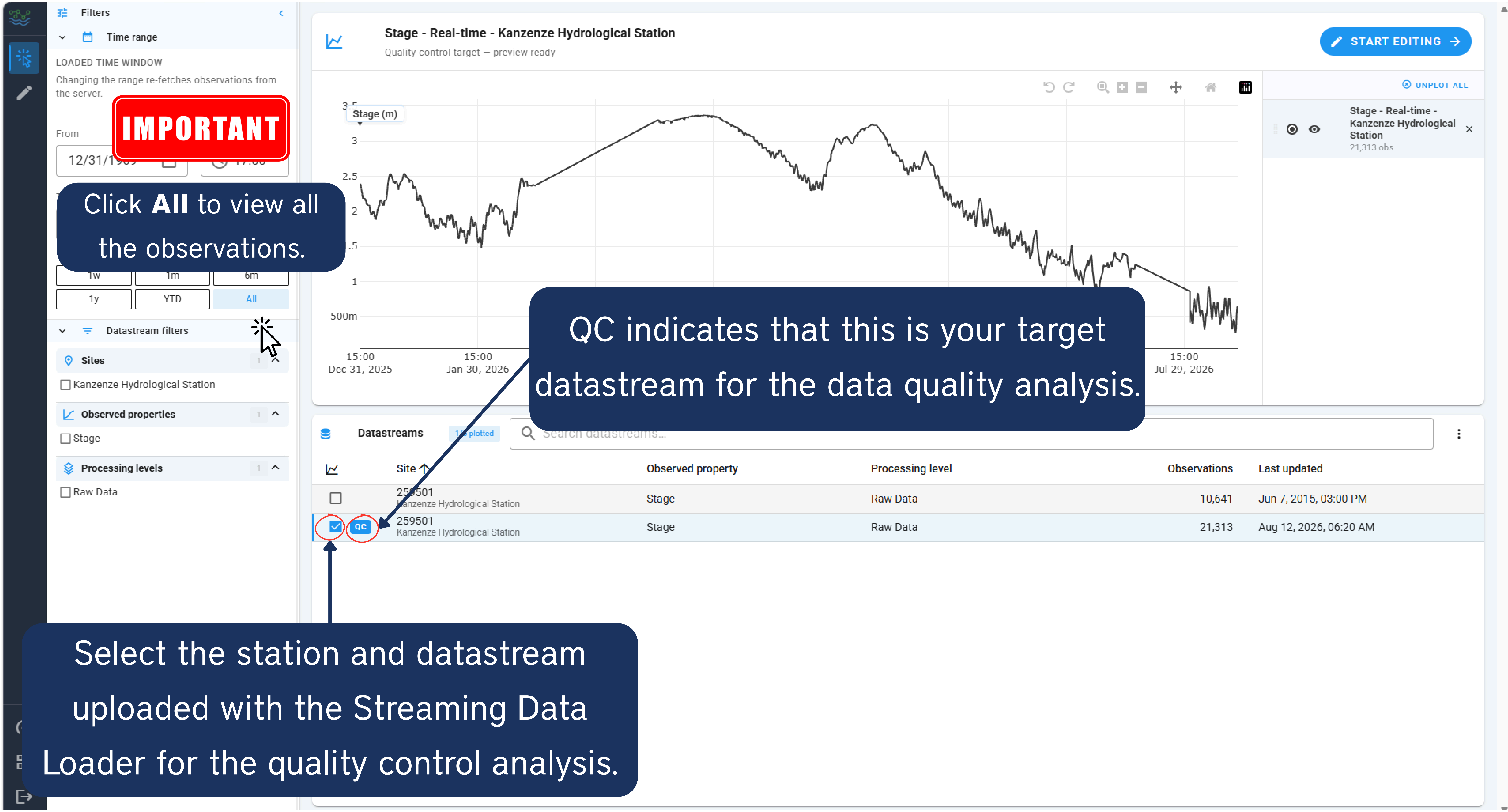
If you are logged out, clicking on the QC App will take you to a page displaying all public workspaces. From there, simply select the workspace you are working on.

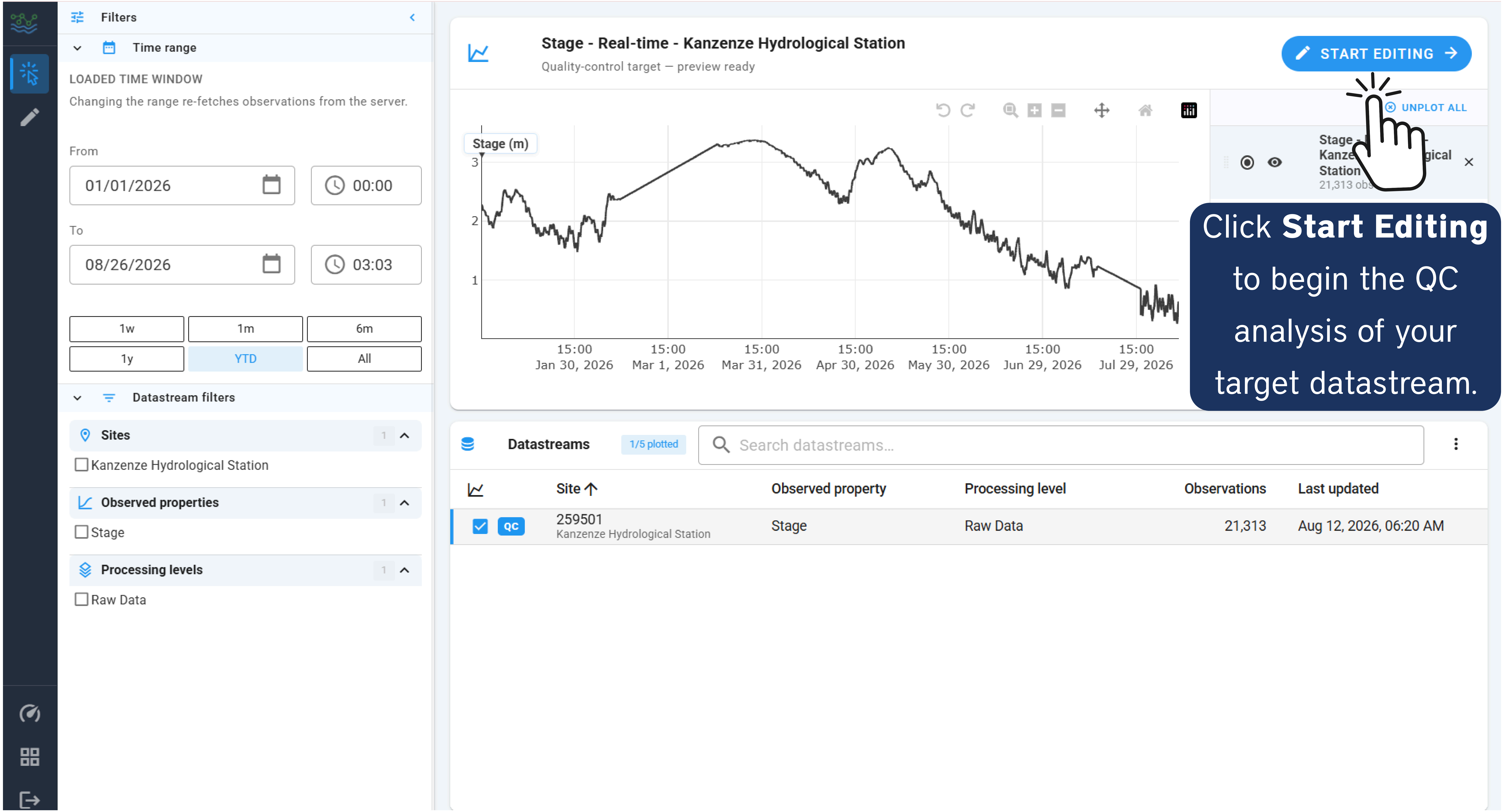


Once you have selected the workspace, the following page will appear:

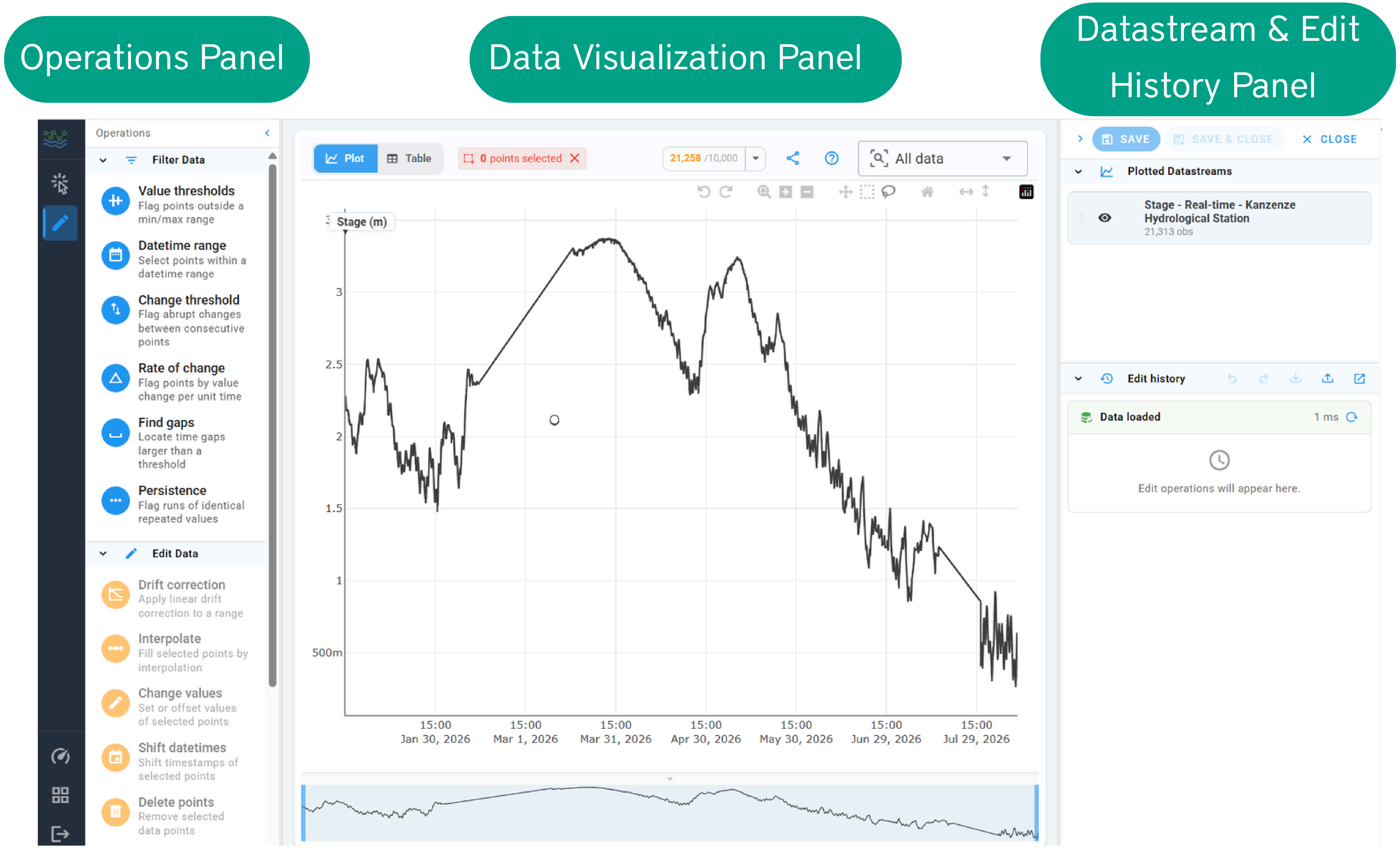


Once the selection is complete, the datastreams associated with your monitoring site will appear.

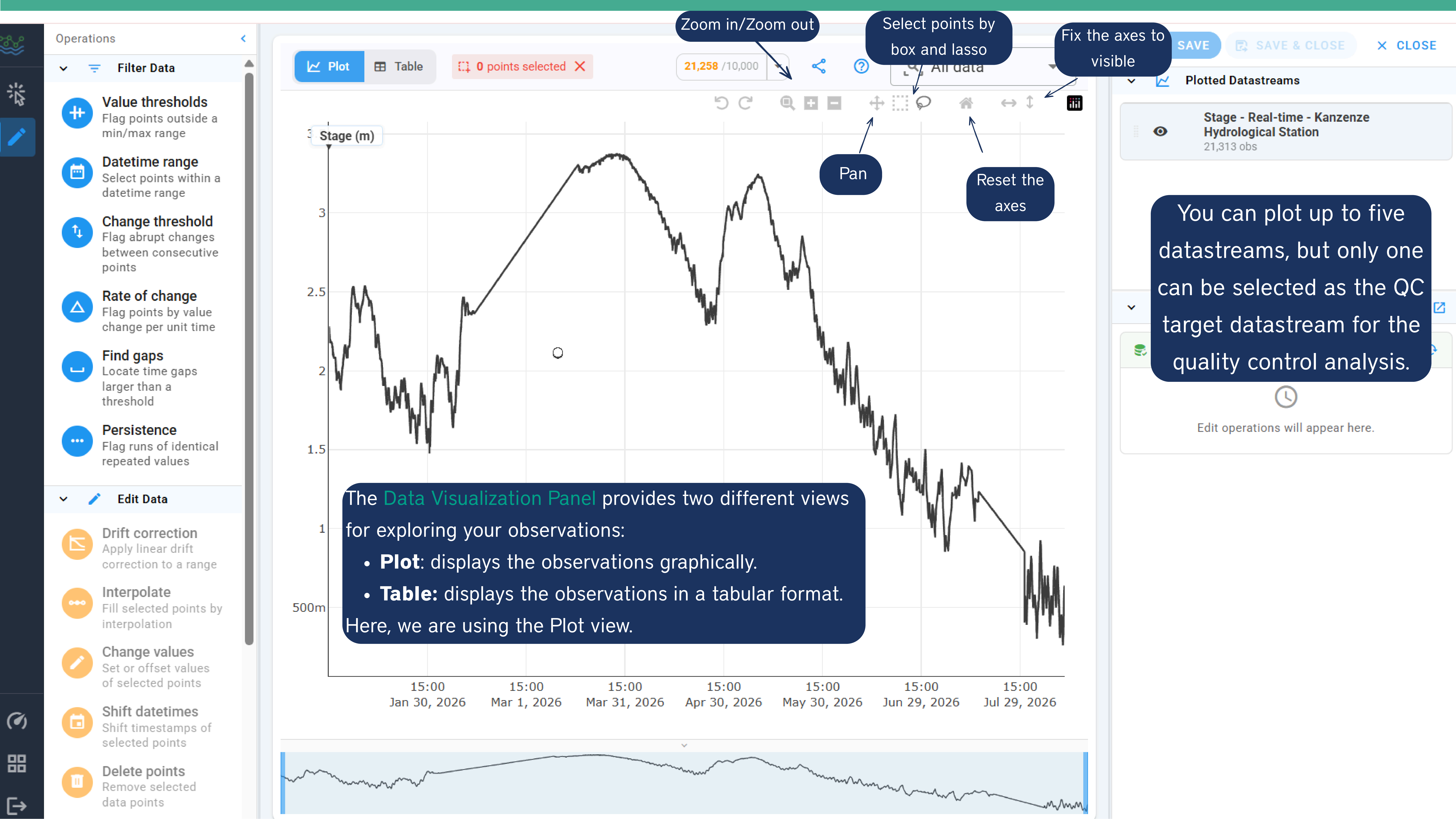


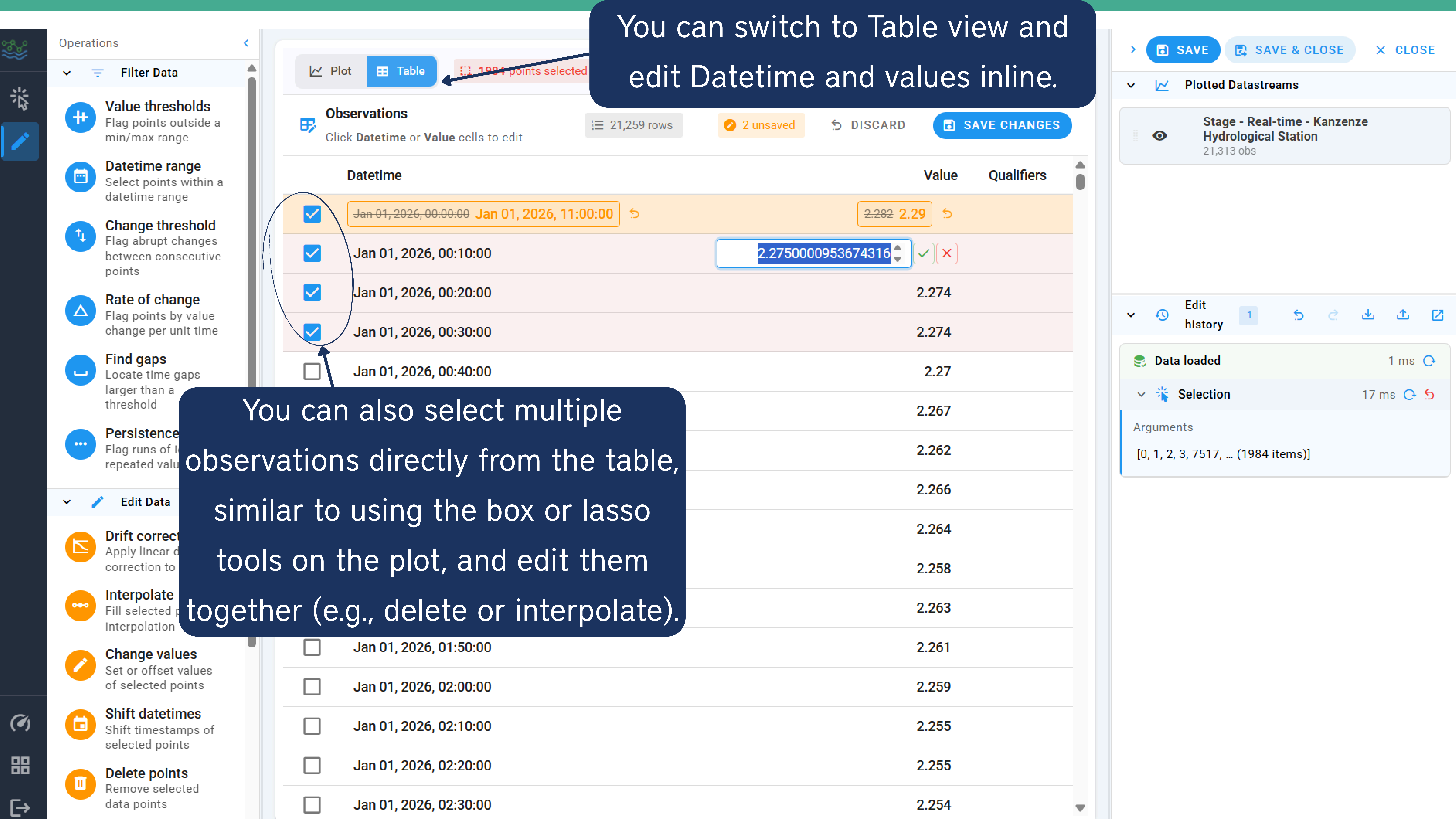

Filters
Time range
LOADED TIME WINDOW
Changing the range re-fetches observations from the server.
From
01/01/2026
00:00
To
08/26/2026
03:03
1w
1m
6m
1y
YTD
All
Datastream filters
Sites
Kanzenze Hydrological Station
Observed properties
Stage
Processing levels
Raw Data
Stage - Real-time - Kanzenze Hydrological Station
Quality-control target — preview ready
START EDITING
UNPLOT ALL
Stage (m)
15:00 Jan 30, 2026
15:00 Mar 1, 2026
15:00 Mar 31, 2026
15:00 Apr 30, 2026
15:00 May 30, 2026
15:00 Jun 29, 2026
15:00 Jul 29, 2026
Click Start Editing to begin the QC analysis of your target datastream.
Datastreams
1/5 plotted
Search datastreams...
Site
Observed property
Processing level
Observations
Last updated
QC
259501
Kanzenze Hydrological Station
Stage
Raw Data
21,313
Aug 12, 2026, 06:20 AM

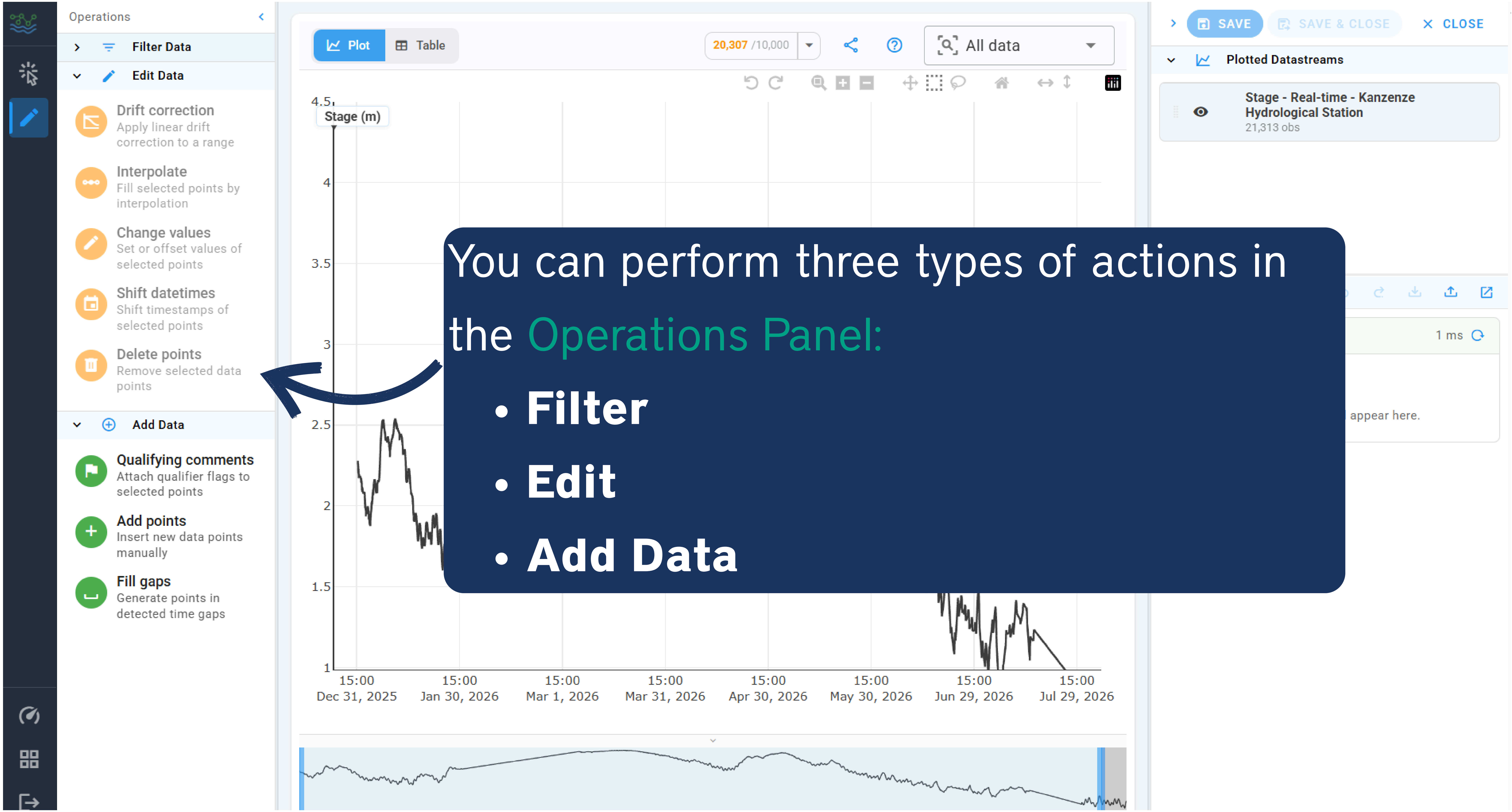
The QC App editing interface is divided into three main areas:



1. Operations Panel: Select QC and data-editing operations.
2. Data Visualization Panel: View, explore, and select observations.
3. Datastream & Edit History Panel: View datastreams and track applied edits.


Zoom in/Zoom out
Select points by box and lasso
Fix the axes to visible
SAVE
SAVE & CLOSE
CLOSE
Operations
Filter Data
Value thresholds
Flag points outside a min/max range
Datetime range
Select points within a datetime range
Change threshold
Flag abrupt changes between consecutive points
Rate of change
Flag points by value change per unit time
Find gaps
Locate time gaps larger than a threshold
Persistence
Flag runs of identical repeated values
Edit Data
Drift correction
Apply linear drift correction to a range
Interpolate
Fill selected points by interpolation
Change values
Set or offset values of selected points
Shift datetimes
Shift timestamps of selected points
Delete points
Remove selected data points
Plot
Table
0 points selected
21,258 /10,000
All data
Stage (m)
Pan
Reset the axes
Plotted Datastreams
Stage - Real-time - Kanzenze Hydrological Station
21,313 obs
You can plot up to five datastreams, but only one can be selected as the QC target datastream for the quality control analysis.
Edit operations will appear here.
The Data Visualization Panel provides two different views for exploring your observations:
• **Plot**: displays the observations graphically.
• **Table:** displays the observations in a tabular format.
Here, we are using the Plot view.
3
2.5
2
1.5
1
500m
15:00 Jan 30, 2026
15:00 Mar 1, 2026
15:00 Mar 31, 2026
15:00 Apr 30, 2026
15:00 May 30, 2026
15:00 Jun 29, 2026
15:00 Jul 29, 2026


You can switch to Table view and edit Datetime and values inline.
You can also select multiple observations directly from the table, similar to using the box or lasso tools on the plot, and edit them together (e.g., delete or interpolate).
Operations
Filter Data
Value thresholds
Flag points outside a min/max range
Datetime range
Select points within a datetime range
Change threshold
Flag abrupt changes between consecutive points
Rate of change
Flag points by value change per unit time
Find gaps
Locate time gaps larger than a threshold
Persistence
Edit Data
Interpolate
Change values
Set or offset values of selected points
Shift datetimes
Shift timestamps of selected points
Delete points
Remove selected data points
Plot
Table
Observations
Click Datetime or Value cells to edit
21,259 rows
2 unsaved
DISCARD
SAVE CHANGES
Datetime
Value
Qualifiers
Jan 01, 2026, 00:00:00 Jan 01, 2026, 11:00:00
2.282 2.29
Jan 01, 2026, 00:10:00
2.2750000953674316
Jan 01, 2026, 00:20:00 2.274
Jan 01, 2026, 00:30:00 2.274
Jan 01, 2026, 00:40:00 2.27
2.267
2.262
2.266
2.264
2.258
2.263
Jan 01, 2026, 01:50:00 2.261
Jan 01, 2026, 02:00:00 2.259
Jan 01, 2026, 02:10:00 2.255
Jan 01, 2026, 02:20:00 2.255
Jan 01, 2026, 02:30:00 2.254
SAVE
SAVE & CLOSE
CLOSE
Plotted Datastreams
Stage - Real-time - Kanzenze Hydrological Station
21,313 obs
Edit history
1
Data loaded 1 ms
Selection 17 ms
Arguments
[0, 1, 2, 3, 7517, ... (1984 items)]


You can perform three types of actions in the Operations Panel:
• Filter
• Edit
• Add Data
Operations
Filter Data
Edit Data
Drift correction
Apply linear drift correction to a range
Interpolate
Fill selected points by interpolation
Change values
Set or offset values of selected points
Shift datetimes
Shift timestamps of selected points
Delete points
Remove selected data points
Add Data
Qualifying comments
Attach qualifier flags to selected points
Add points
Insert new data points manually
Fill gaps
Generate points in detected time gaps
Plot
Table
20,307 /10,000
All data
Stage (m)
SAVE
SAVE & CLOSE
CLOSE
Plotted Datastreams
Stage - Real-time - Kanzenze Hydrological Station
21,313 obs
1 ms
appear here.
15:00 Dec 31, 2025
15:00 Jan 30, 2026
15:00 Mar 1, 2026
15:00 Mar 31, 2026
15:00 Apr 30, 2026
15:00 May 30, 2026
15:00 Jun 29, 2026
15:00 Jul 29, 2026

These are the types of operations you can perform, organized into three categories: Filter Data, Edit Data, and Add Data.

Filter

Filter Data

Value thresholds
Flag points outside a min/max range

Datetime range
Select points within a datetime range

Change threshold
Flag abrupt changes between consecutive points

Rate of change
Flag points by value change per unit time

Find gaps
Locate time gaps larger than a threshold

Persistence
Flag runs of identical repeated values

Edit

Edit Data

Drift correction
Apply linear drift correction to a range

Interpolate
Fill selected points by interpolation

Change values
Set or offset values of selected points

Shift datetimes
Shift timestamps of selected points

Delete points
Remove selected data points

Add Data

Add Data

Qualifying comments
Attach qualifier flags to selected points

Add points
Insert new data points manually

Fill gaps
Generate points in detected time gaps

Here, you can repeat the operation or undo it.
All the operations you perform are recorded in the Datastream & Edit History Panel.
Operations
Filter Data
Value thresholds
Flag points outside a min/max range
Datetime range
Select points within a datetime range
Change threshold
Flag abrupt changes between consecutive points
Rate of change
Flag points by value change per unit time
Find gaps
Locate time gaps larger than a threshold
Persistence
Flag runs of identical repeated values
Edit Data
Drift correction
Apply linear drift correction to a range
Interpolate
Fill selected points by interpolation
Change values
Set or offset values of selected points
Shift datetimes
Shift timestamps of selected points
Delete points
Remove selected data points
Plot
Table
4503 points selected
21,258 /10,000
All data
Stage (m)
Jan 2026
Feb 2026
Mar 2026
Apr 2026
May 2026
Jun 2026
Jul 2026
Aug 2026
SAVE
SAVE & CLOSE
CLOSE
Plotted Datastreams
Stage - Real-time - Kanzenze Hydrological Station
21,313 obs
Edit history
Data loaded
Datetime Range
4 ms
Arguments
1767250800000
1769952000000
Drift correction
Apply linear drift correction to a range
Drift correction
1 consecutive group
Groups
4503 points
starts Jan 01, 2026, 00:00:00
Drift amount
1

We observed gaps in the data.

Now, we will find the gaps and fill them using interpolation.

Filter

Filter Data

- **Value thresholds** — Flag points outside a min/max range
- **Datetime range** — Select points within a datetime range
- **Change threshold** — Flag abrupt changes between consecutive points
- **Rate of change** — Flag points by value change per unit time
- **Find gaps** — Locate time gaps larger than a threshold
- **Persistence** — Flag runs of identical repeated values

Edit

Edit Data

- **Drift correction** — Apply linear drift correction to a range
- **Interpolate** — Fill selected points by interpolation
- **Change values** — Set or offset values of selected points
- **Shift datetimes** — Shift timestamps of selected points
- **Delete points** — Remove selected data points

Add Data

Add Data

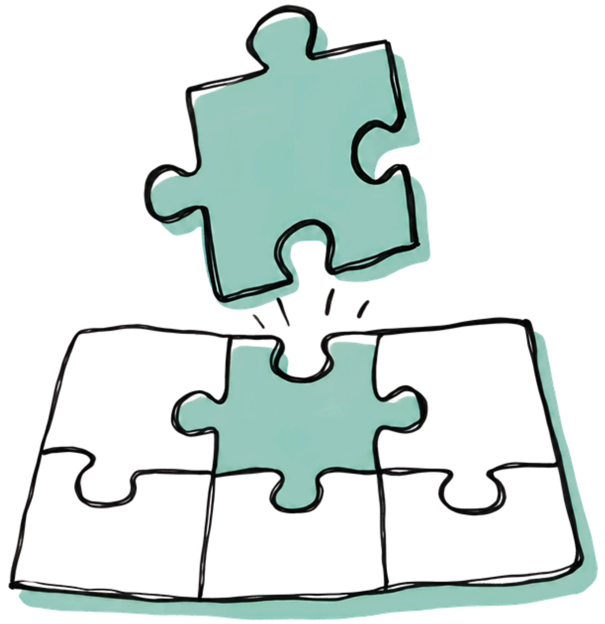
- **Qualifying comments** — Attach qualifier flags to selected points
- **Add points** — Insert data points manually
- **Fill gaps** — Generate points in detected time gaps

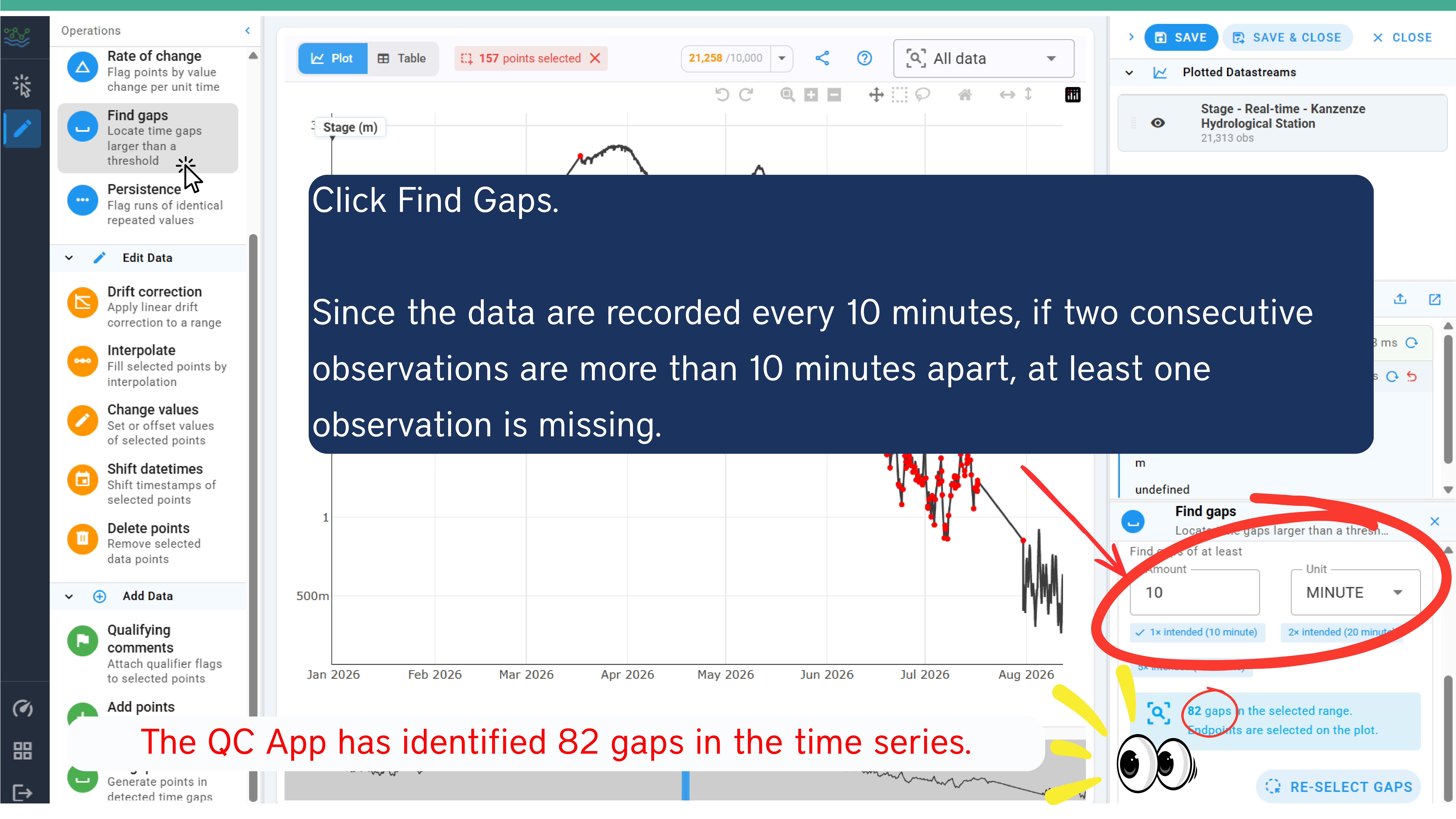
Note: Additionally, you can add qualifying comments to indicate that the filled values are interpolated rather than observed data.

Sara Alonso Vicario

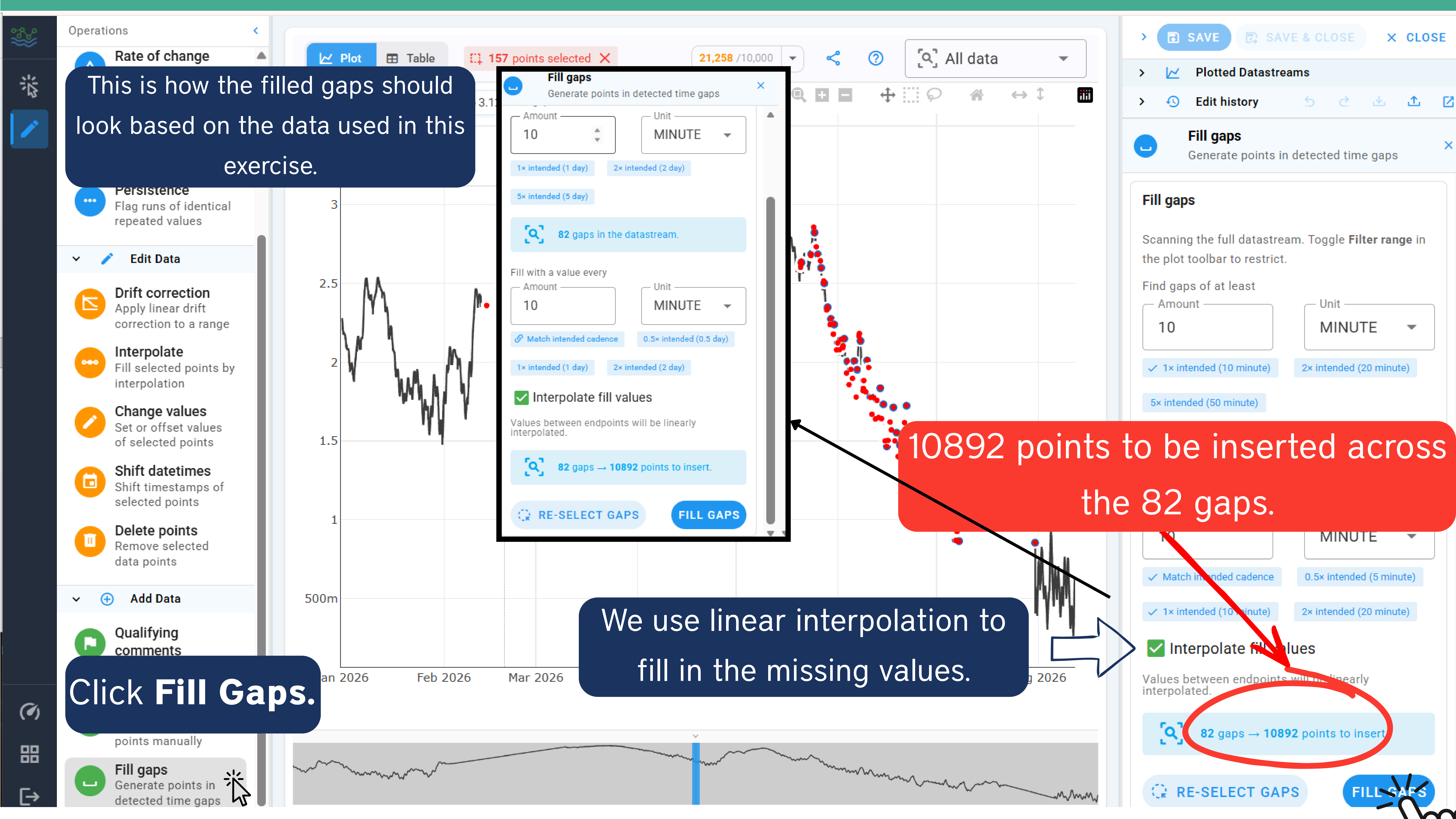
Then, you need to perform three operations in order:

1. **Filter Data - Find Gaps**: Identify the gaps in the time series.
2. **Add Data - Fill Gaps**: Fill the identified gaps using linear interpolation.
3. **Add Data - Qualifying Comments**: Flag the filled values as interpolated rather than observed.





Operations
Rate of change
Flag points by value change per unit time
Find gaps
Locate time gaps larger than a threshold
Persistence
Flag runs of identical repeated values
Edit Data
Drift correction
Apply linear drift correction to a range
Interpolate
Fill selected points by interpolation
Change values
Set or offset values of selected points
Shift datetimes
Shift timestamps of selected points
Delete points
Remove selected data points
Add Data
Qualifying comments
Attach qualifier flags to selected points
Add points
Generate points in detected time gaps
Plot
Table
157 points selected
21,258 /10,000
All data
Stage (m)
1
500m
Jan 2026
Feb 2026
Mar 2026
Apr 2026
May 2026
Jun 2026
Jul 2026
Aug 2026
SAVE
SAVE & CLOSE
CLOSE
Plotted Datastreams
Stage - Real-time - Kanzenze Hydrological Station
21,313 obs
m
undefined
Find gaps
Find gaps of at least
Amount
10
Unit
MINUTE
1× intended (10 minute)
82 gaps in the selected range.
Endpoints are selected on the plot.
RE-SELECT GAPS
Click Find Gaps.
Since the data are recorded every 10 minutes, if two consecutive observations are more than 10 minutes apart, at least one observation is missing.
The QC App has identified 82 gaps in the time series.



This is how the filled gaps should look based on the data used in this exercise.
10892 points to be inserted across the 82 gaps.
We use linear interpolation to fill in the missing values.
Click **Fill Gaps.**
Operations
Rate of change
Persistence
Flag runs of identical repeated values
Edit Data
Drift correction
Apply linear drift correction to a range
Interpolate
Fill selected points by interpolation
Change values
Set or offset values of selected points
Shift datetimes
Shift timestamps of selected points
Delete points
Remove selected data points
Add Data
Qualifying comments
points manually
Fill gaps
Generate points in detected time gaps
Plot
Table
157 points selected
21,258 /10,000
All data
Fill gaps
Generate points in detected time gaps
Amount
10
Unit
MINUTE
1× intended (1 day)
2× intended (2 day)
5× intended (5 day)
82 gaps in the datastream.
Fill with a value every
Match intended cadence
0.5× intended (0.5 day)
Interpolate fill values
Values between endpoints will be linearly interpolated.
82 gaps → 10892 points to insert.
RE-SELECT GAPS
FILL GAPS
SAVE
SAVE & CLOSE
CLOSE
Plotted Datastreams
Edit history
Scanning the full datastream. Toggle Filter range in the plot toolbar to restrict.
Find gaps of at least
1× intended (10 minute)
2× intended (20 minute)
5× intended (50 minute)
0.5× intended (5 minute)
Jan 2026
Feb 2026
Mar 2026
Aug 2026
3
2.5
2
1.5
1
500m


Operations
Rate of change
Flag points by value change per unit time
Find gaps
Locate time gaps larger than a threshold
Persistence
Flag runs of identical repeated values
Edit Data
Drift correction
Apply linear drift correction to a range
Interpolate
Fill selected points by interpolation
Change values
Set or offset values of selected points
Shift datetimes
Shift timestamps of selected points
Delete points
Qualifying comments
Attach qualifier flags to selected points
Add points
Insert new data points manually
Fill gaps
Generate points in detected time gaps
Plot
Table
10892 points selected
32,150 /10,000
All data
x Mar 24, 2026, 13:12:48 Stage 0.9424 m
Stage (m)
3
2.5
2
1.5
1
Jan 2026
Feb 2026
Mar 2026
Apr 2026
May 2026
Jun 2026
Jul 2026
Aug 2026
SAVE
SAVE & CLOSE
CLOSE
Plotted Datastreams
Edit history
2
Data loaded
44 ms
Fill Gaps
117 ms
Arguments
[10, "m"]
[10, "m"]
true
-9999
[1767250800000, 1786540800000]
Selection
18 ms
All 10,892 inserted points are shown in red.

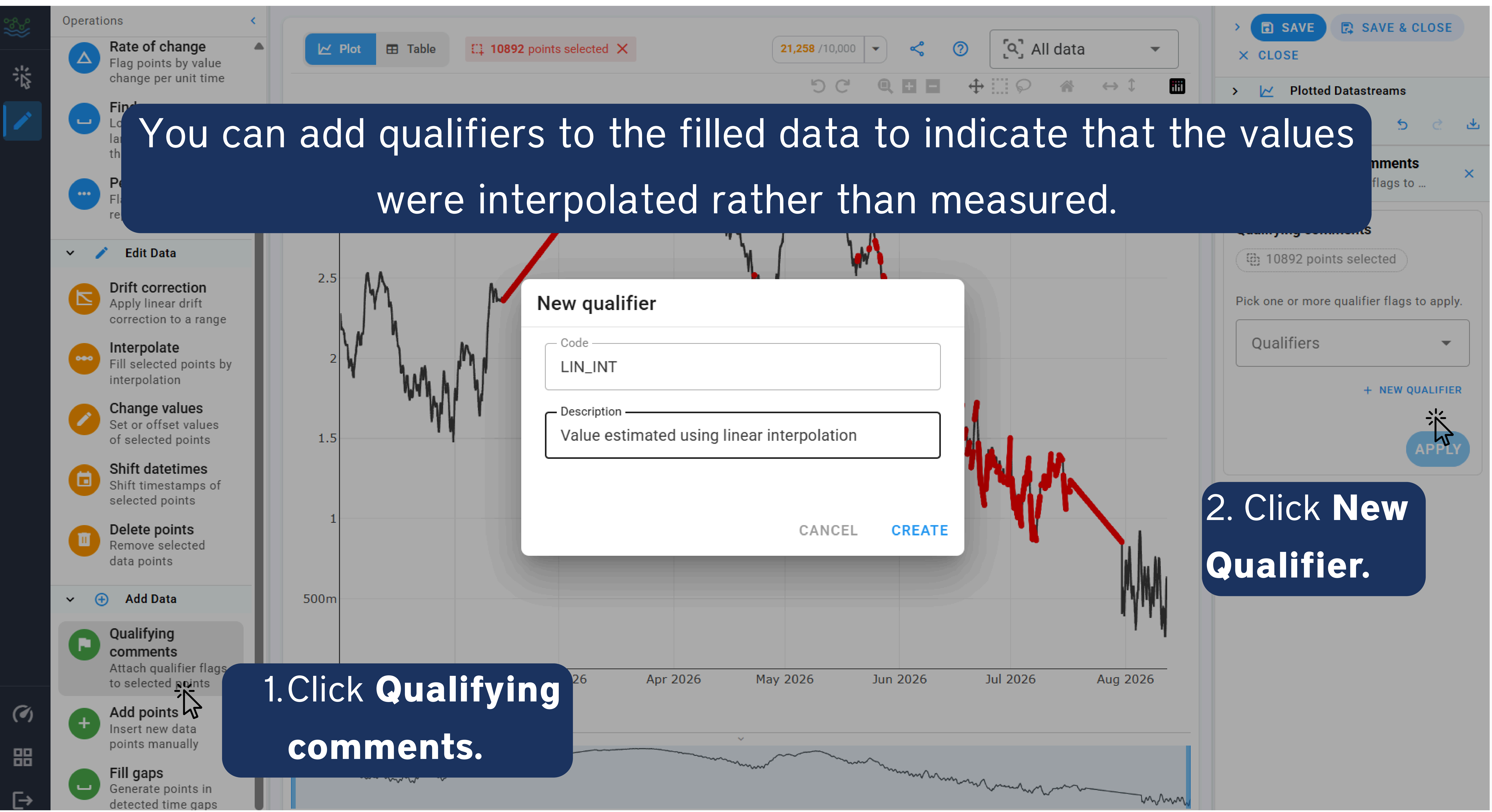
If you switch to the Table View in the Data Visualization Panel, you will see the values that were added to fill the gaps. The image below shows an example of one of the filled data gaps.

Data gap:

☑	Apr 22, 2026, 13:20:00	2.473
☑	Apr 22, 2026, 14:30:00	2.481

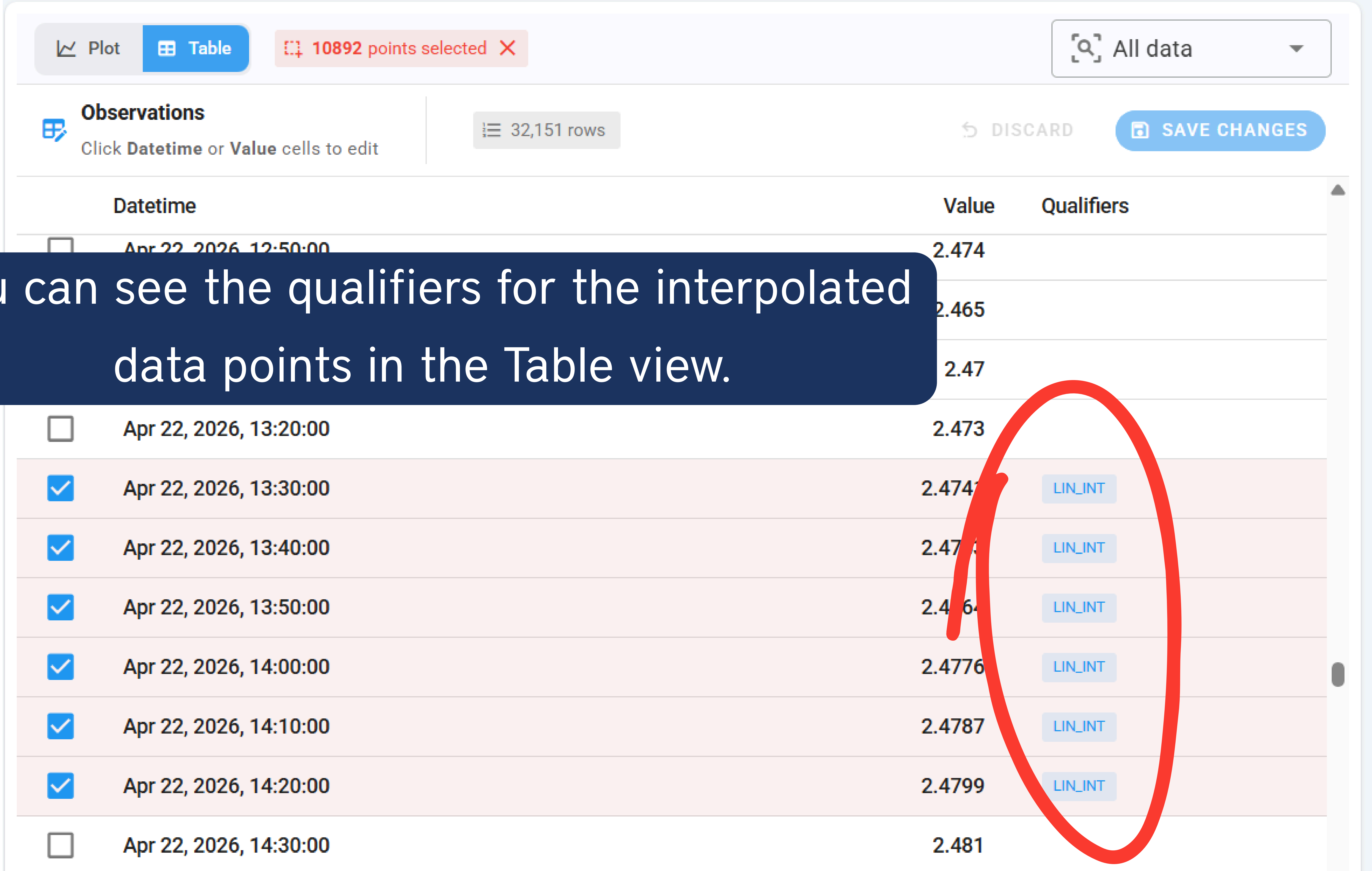
The data gap identified above have been filled using linear interpolation.

☐	Apr 22, 2026, 13:20:00	2.473
☑	Apr 22, 2026, 13:30:00	2.4741
☑	Apr 22, 2026, 13:40:00	2.4753
☑	Apr 22, 2026, 13:50:00	2.4764
☑	Apr 22, 2026, 14:00:00	2.4776
☑	Apr 22, 2026, 14:10:00	2.4787
☑	Apr 22, 2026, 14:20:00	2.4799
☐	Apr 22, 2026, 14:30:00	2.481



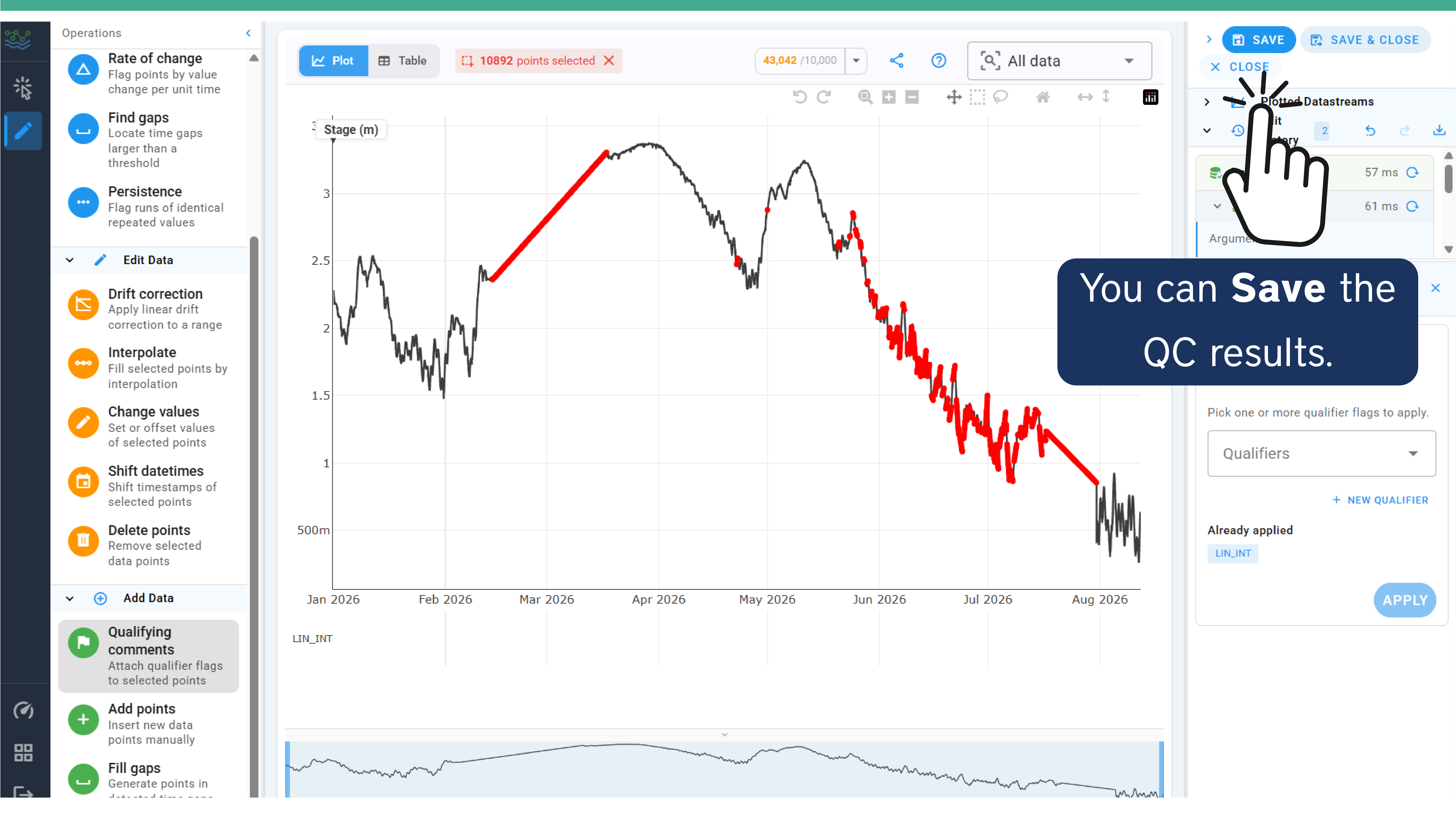
You can add qualifiers to the filled data to indicate that the values were interpolated rather than measured.
New qualifier
Code
LIN_INT
Description
Value estimated using linear interpolation
CANCEL
CREATE
1. Click Qualifying comments.
2. Click New Qualifier.
Operations
Rate of change
Flag points by value change per unit time
Edit Data
Drift correction
Apply linear drift correction to a range
Interpolate
Fill selected points by interpolation
Change values
Set or offset values of selected points
Shift datetimes
Shift timestamps of selected points
Delete points
Remove selected data points
Add Data
Qualifying comments
Attach qualifier flags to selected points
Add points
Insert new data points manually
Fill gaps
Generate points in detected time gaps
Plot
Table
10892 points selected
21,258 /10,000
All data
SAVE
SAVE & CLOSE
CLOSE
Plotted Datastreams
10892 points selected
Pick one or more qualifier flags to apply.
Qualifiers
NEW QUALIFIER
APPLY
2.5
2
1.5
1
500m
Apr 2026
May 2026
Jun 2026
Jul 2026
Aug 2026

You can see the qualifiers for the interpolated data points in the Table view.

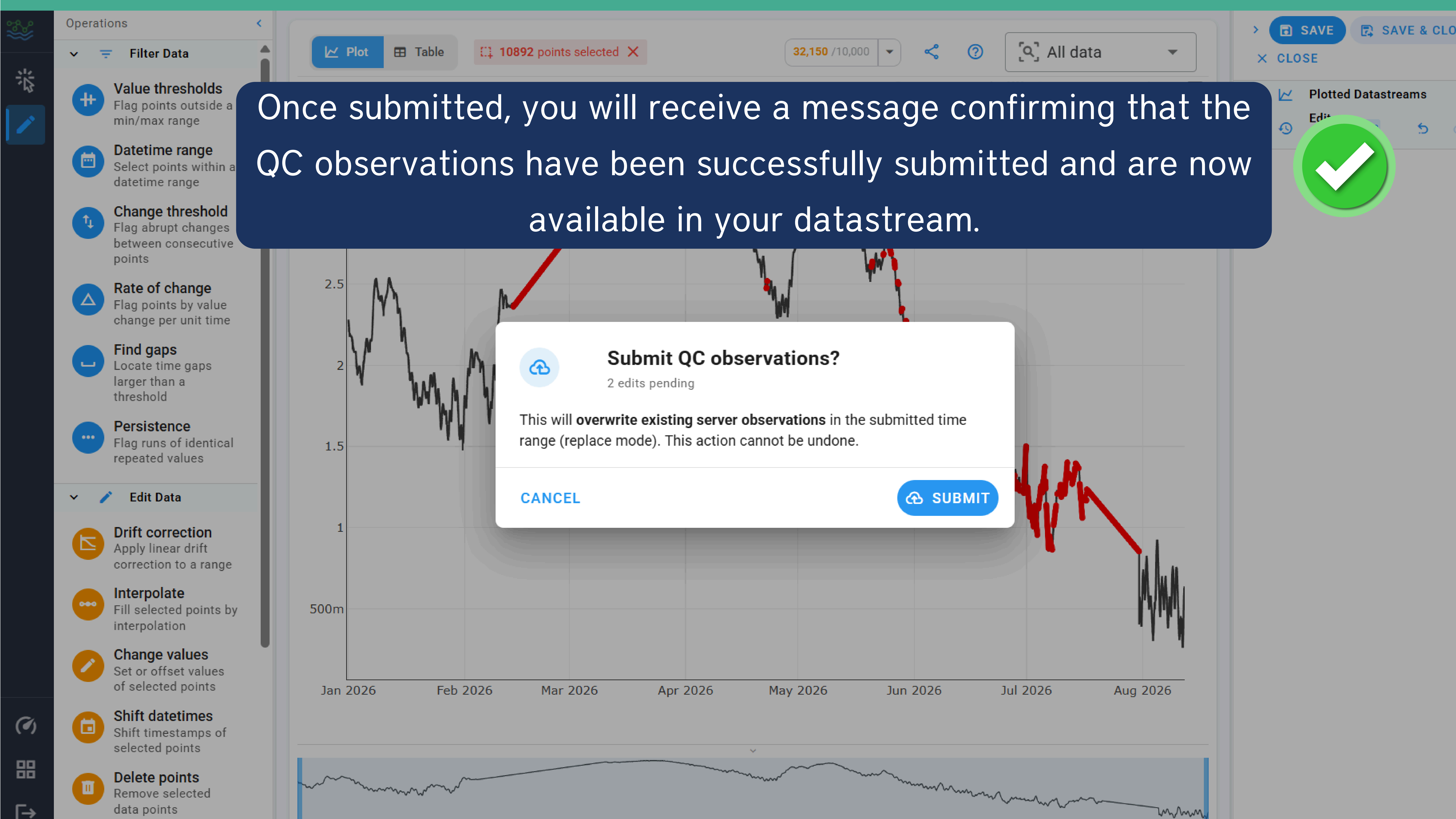


You can download your edit history by clicking **the downward arrow.**

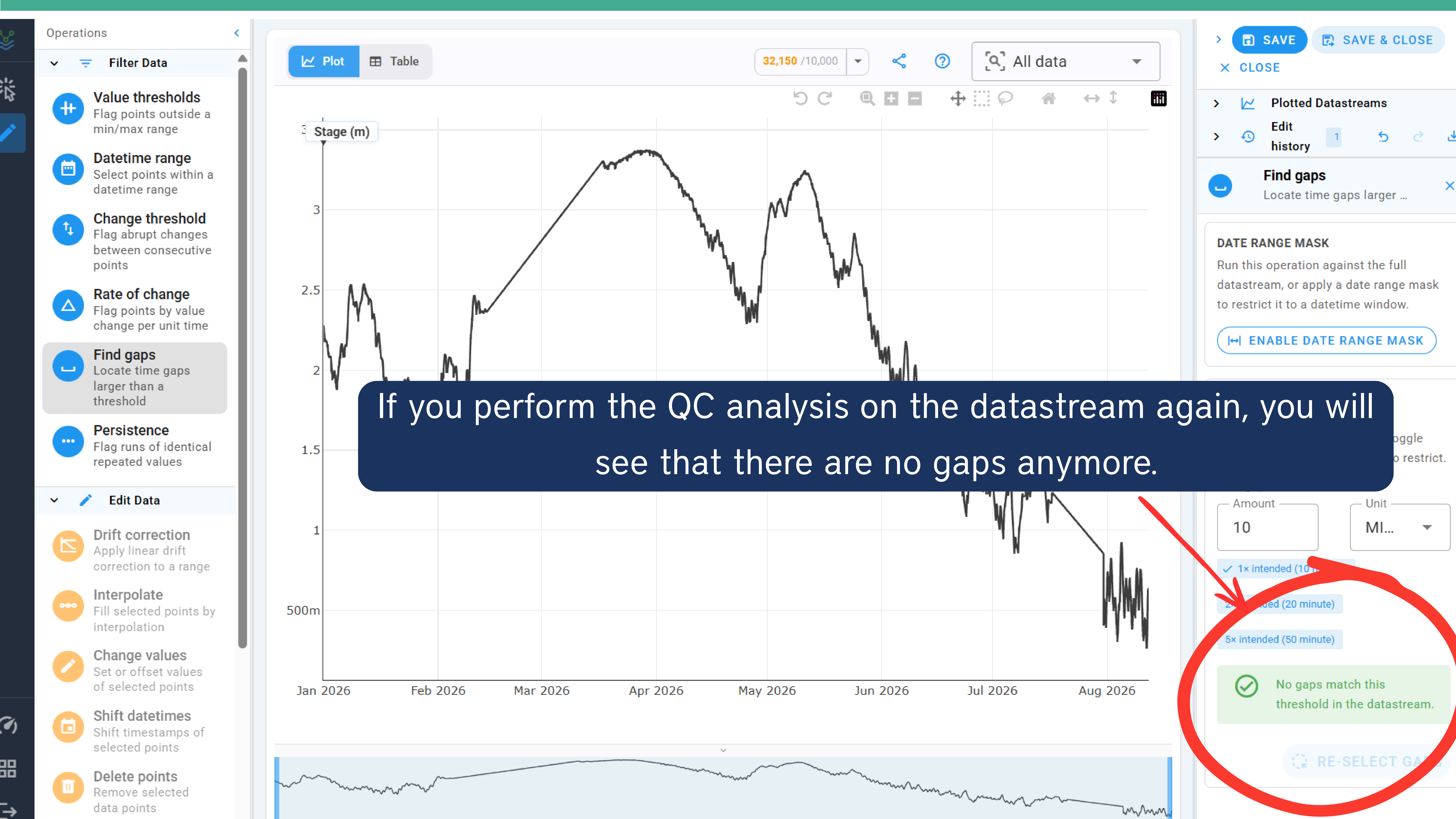
You can reuse this file to repeat the same QC analysis.



Operations
Rate of change
Flag points by value change per unit time
Find gaps
Locate time gaps larger than a threshold
Persistence
Flag runs of identical repeated values
Edit Data
Drift correction
Apply linear drift correction to a range
Interpolate
Fill selected points by interpolation
Change values
Set or offset values of selected points
Shift datetimes
Shift timestamps of selected points
Delete points
Remove selected data points
Add Data
Qualifying comments
Attach qualifier flags to selected points
Add points
Insert new data points manually
Fill gaps
Generate points in
Plot
Table
10892 points selected
43,042 /10,000
All data
Stage (m)
Jan 2026
Feb 2026
Mar 2026
Apr 2026
May 2026
Jun 2026
Jul 2026
Aug 2026
LIN_INT
SAVE
SAVE & CLOSE
CLOSE
Plotted Datastreams
57 ms
61 ms
You can Save the QC results.
Pick one or more qualifier flags to apply.
Qualifiers
NEW QUALIFIER
Already applied
LIN_INT
APPLY



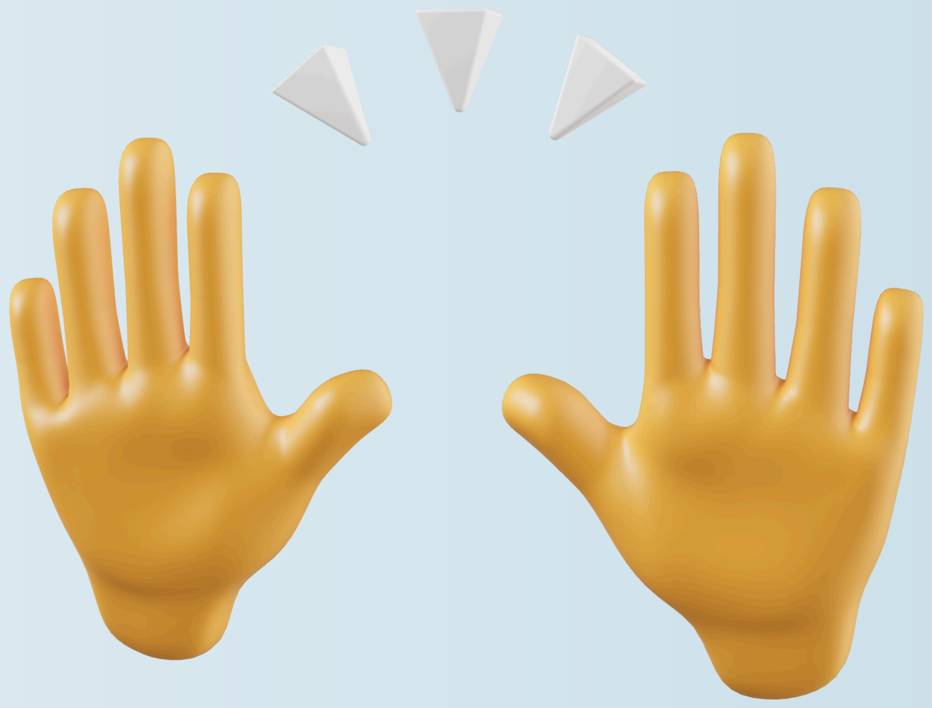
Once submitted, you will receive a message confirming that the QC observations have been successfully submitted and are now available in your datastream.
Submit QC observations?
2 edits pending
This will overwrite existing server observations in the submitted time range (replace mode). This action cannot be undone.
CANCEL
SUBMIT
Operations
Filter Data
Value thresholds
Flag points outside a min/max range
Datetime range
Select points within a datetime range
Change threshold
Flag abrupt changes between consecutive points
Rate of change
Flag points by value change per unit time
Find gaps
Locate time gaps larger than a threshold
Persistence
Flag runs of identical repeated values
Edit Data
Drift correction
Apply linear drift correction to a range
Interpolate
Fill selected points by interpolation
Change values
Set or offset values of selected points
Shift datetimes
Shift timestamps of selected points
Delete points
Remove selected data points
Plot
Table
10892 points selected
32,150 /10,000
All data
SAVE
CLOSE
Plotted Datastreams
2.5
2
1.5
1
500m
Jan 2026
Feb 2026
Mar 2026
Apr 2026
May 2026
Jun 2026
Jul 2026
Aug 2026


Operations
Filter Data
Value thresholds
Flag points outside a min/max range
Datetime range
Select points within a datetime range
Change threshold
Flag abrupt changes between consecutive points
Rate of change
Flag points by value change per unit time
Find gaps
Locate time gaps larger than a threshold
Persistence
Flag runs of identical repeated values
Edit Data
Drift correction
Apply linear drift correction to a range
Interpolate
Fill selected points by interpolation
Change values
Set or offset values of selected points
Shift datetimes
Shift timestamps of selected points
Delete points
Remove selected data points
Plot
Table
32,150 /10,000
All data
Stage (m)
3
2.5
2
1.5
1
500m
Jan 2026
Feb 2026
Mar 2026
Apr 2026
May 2026
Jun 2026
Jul 2026
Aug 2026
SAVE
SAVE & CLOSE
CLOSE
Plotted Datastreams
Edit history
1
Find gaps
Locate time gaps larger ...
DATE RANGE MASK
Run this operation against the full datastream, or apply a date range mask to restrict it to a datetime window.
ENABLE DATE RANGE MASK
Amount
10
Unit
5× intended (50 minute)
No gaps match this threshold in the datastream.
If you perform the QC analysis on the datastream again, you will see that there are no gaps anymore.

Before QC:

After QC:

Good job! 🎉 😎



You have successfully automated data loading and performed data quality analysis!

See you in the next session!

Now, take a break!

Please be back at 15:35